



RIZAL TECHNOLOGICAL UNIVERSITY
M. Eusebio Avenue, Maybunga, Pasig City

**FOVB-AIOT: FOUR-IN-ONE VITAL SIGN SENSOR WITH BMI CALCULATION
USING AI AND IOT FOR HEALTH RISK PREDICTION
IN RTU PASIG CLINIC**

An Undergraduate Thesis
Presented to the Faculty of the
College of Engineering, Architecture and Technology

In Partial Fulfillment
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Bachelor of Science in Computer Engineering

By

Alcantara, Mar Kevin P.
Berongoy, Bernie C.
Gamboa, Khin Andrei R.
Llona, Reymart G.
Oavenada, Erick John A.
Relevo, Paul Andrew A.
Sagadraca, Yuri Lorenz C.

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APPROVAL SHEET

The thesis entitled "FOVB-AIoT: Four-in-One Vital Sign Sensor with BMI Calculation Using AI and IoT for Health Risk Prediction in the RTU Pasig Clinic," prepared and submitted by Mar Kevin P. Alcantara, Bernie C. Berongoy, Khin Andrei R. Gamboa, Reymart G. Llona, Erick John A. Oavenada, Paul Andrew A. Relevo, and Yuri Lorenz C. Sagadraca, in partial fulfillment of the requirements for the Bachelor of Science in Computer Engineering, has been examined and is hereby recommended for oral examination.

Date

ENGR. EZEKIEL C. NEQUIT
Thesis Adviser

PANEL OF EXAMINERS

APPROVED by the Oral Examination Committee with a grade of _____.

ENGR. RONEL D. PAGLOMUTAN
Chairman

ASSOC. PROF. CORAZON S. ASPURIA
Member

ENGR. KENT BRYAN P. PADUA
Member

ACCEPTED as partial fulfillment to the requirements for the degree, Bachelor of Science in Computer Engineering.

March 07, 2026
Date

DR. WILFREDO D. TAMAJO
Dean, College of Engineering

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B.C.B.
E.J.A.O
K.A.R.G
M.K.P.A
P.A.A.R.
R.G.L.
Y.L.C.S.

ABSTRACT

This study presents the development of the **FOVB-AIoT**, a kiosk-based health monitoring system that integrates a Four-in-One Vital Sign Sensor with Body Mass Index (BMI) calculation using Artificial Intelligence (AI) and the Internet of Things (IoT) for Health Risk Prediction. Designed to improve efficiency and accuracy over traditional manual screening methods, the FOVb-AIoT platform measures key physiological parameters, including body temperature, heart rate, respiratory rate, and blood pressure, while calculating BMI

The system integrates sensor modules, microcontrollers, AI algorithms, and IoT technology. Vital signs are measured sequentially, ensuring reliable readings. An RFID-based registration system allows quick user access, and all collected health data are transmitted to a web-based IoT database for real-time access by RTU Clinic personnel. The integrated AI module analyzes the vital sign data to detect unusual patterns and generate a Comprehensive Risk Score, indicating potential health risks.

The FOVb-AIoT operates reliably and efficiently, improving the speed and convenience of health screening. The AI module successfully analyzed the data, provided health recommendations, and supported early medical intervention.

Keywords: *FOVB-AIoT, RTU, Risk Score, IoT-based, Database*

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LIST OF ACRONYMS

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AI – Artificial Intelligence
APEs – Annual Physical Examinations
Bio-Sync – Biometric Synchronization Engine
BMI – Body Mass Index
BP – Blood Pressure
BSCPE – Bachelor of Science in Computer Engineering
CSS – Cascading Style Sheets
DOH – Department of Health
FOVB-AIoT – Four-in-One Vital Sign Sensor with BMI Calculation using AI and IoT for Health Risk Prediction in RTU Pasig Clinic
HR – Heart Rate
HRP – Health Risk Prediction
HTML – HyperText Markup Language
IoT – Internet of Things
ISO/IEC – International Organization for Standardization / International Electrotechnical Commission
JS – JavaScript
LLM – Large Language Model
LiDAR – Light Detection and Ranging
PREP – Personalized Results and Educational Protocol
RFID – Radio Frequency Identification
RR – Respiratory Rate
RTU – Rizal Technological University
SD – Standard Deviation
SENTINEL-IoT – Secure End-to-end Network for Telemetered Evaluation and Logging via IoT
TF Luna – Time-of-Flight
UCARE – University Clinic Assessment Record Enclave
UI – User Interface
VI – Verbal Interpretation
VITAL-PRO – Vital Sign Protocol for Rapid Operation
WHO – World Health Organization
WM – Weighted Mean
XGBoost – eXtreme Gradient Boosting
YOLO – You Only Look Once

CHAPTER I

THE PROBLEM AND ITS BACKGROUND

Introduction

The assessment of vital signs, including blood pressure, heart and respiratory rate, oxygen saturation, body temperature and Body Mass Index (BMI), plays an important role in determining the overall health condition of an individual. These parameters indicate whether the body functions within normal ranges or exhibits early warnings of potential illnesses. Abnormalities such as unusual temperature readings, irregular heart rate, abnormal respiratory rate, abnormal blood pressure, and unhealthy BMI levels may signal health risks that require immediate attention. Regular monitoring of vital signs supports early detection and prevention of health problems, especially in institutional settings where large populations must be assessed efficiently [26].

Among the major health indicators, blood pressure remains a critical marker of cardiovascular health, while BMI categories are widely used to identify whether a person is underweight, normal weight, overweight, or obese [2], [6]. Respiratory rate also serves as an important physiological indicator because deviations from normal ranges may point to underlying respiratory or metabolic conditions [10]. These benchmarks help guide the interpretation of health data and strengthen the role of vital-sign monitoring in preventive care. The integration of Artificial Intelligence (AI)

. and the Internet of Things (IoT) in healthcare enhances accuracy, efficiency, and predictive capability, allowing faster and more reliable medical assessments. These technologies work together to analyze data from different health parameters, enabling more precise interpretation and prediction of potential health risks [16]. AI algorithms process readings from multiple sensors and compare them with standard health ranges, while IoT systems support data transmission and accessibility across connected devices. This integration supports timely insights that contribute to the early detection of abnormalities and promote more proactive healthcare practices. AI- and IoT-based systems also reduce manual workload while improving early detection and response to health conditions, making healthcare more efficient and data-driven [14], [16].

In the Philippines, Annual Physical Examinations (APEs) are commonly conducted in educational institutions to monitor student health and support preventive care [26]. However, these procedures often involve long queues and extended processing times, causing inconvenience and delays for students. In RTU Pasig Branch, students frequently experience prolonged waiting periods during enrollment and when obtaining medical certificates required for registration and examinations. These inefficiencies not only create frustration but also contribute to administrative delays. Such challenges highlight the need for innovative solutions that streamline health-screening processes, reduce waiting time, and improve overall efficiency in university clinics.

The proposed system is intended to provide a more efficient and user-friendly approach to health assessment. It integrates multiple vital-sign measurements with intelligent data analysis to predict potential health risks and offer preventive recommendations. By automating the evaluation process and delivering results in accessible formats, the system aims to reduce manual errors, save time, and promote early awareness of health conditions among users.

Background of the Study

Health assessment plays a vital role in promoting early disease prevention and maintaining overall well-being. In environments where healthcare facilities face high patient volumes and limited manpower, timely and accurate evaluation of an individual's health condition remains essential. Vital signs and BMI serve as key indicators of physiological stability and help identify potential health risks before serious symptoms develop [2], [6], [10].

Clinical standards provide the basis for interpreting these indicators. The American Heart Association provides updated classifications for cardiovascular assessment, particularly for blood pressure [2]. The Centers for Disease Control and Prevention defines standardized BMI categories that support consistent health classification across populations [6]. In addition, age-specific respiratory-rate reference values provide a basis for interpreting respiratory health indicators [10]. Together, these standards help strengthen the interpretation of measured health data

and support the design of automated monitoring systems.

Despite the importance of these health indicators, traditional methods of measuring and interpreting them are often time-consuming, dependent on manual input, and susceptible to human error. In many settings, different devices are used separately for temperature, blood pressure, oxygen saturation, and anthropometric measurements. This can slow down the screening process and may affect consistency in recording and interpretation.

To address these limitations, recent developments in AI and IoT have paved the way for automated health-assessment systems capable of collecting, processing, and analyzing physiological data in real time. These technologies support the simultaneous or integrated measurement of multiple health indicators such as body temperature, heart rate, respiratory rate, blood pressure, oxygen saturation, and BMI while reducing the burden of manual operation [14], [16]. Intelligent health kiosks supported by AI can interpret gathered data, summarize results, and provide predictive insights into possible health risks based on established thresholds. Such integration promotes faster, more consistent, and more data-driven health evaluation.

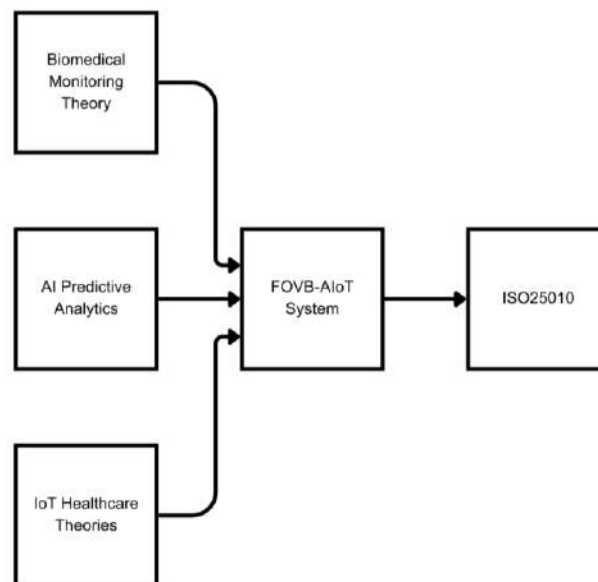
Within the context of RTU Pasig Clinic, the need for a more efficient health-screening system is evident. A kiosk-based platform that combines multiple sensors, AI-based analysis, and IoT connectivity offers a practical approach to improving health monitoring. Such a system can help reduce waiting time, improve record

management, and provide timely health insights that may support early intervention and preventive care.

Theoretical Framework

Figure 1.1

Theoretical Framework



This study is grounded in Biomedical Monitoring Theory, Internet of Things Healthcare Systems Theory, Artificial Intelligence Predictive Analytics Theory, and Software Quality Evaluation Theory, which together support the development and evaluation of the proposed FOVB-AIoT system.

Biomedical Monitoring Theory explains that vital signs such as heart rate, body temperature, blood pressure, respiratory rate, oxygen saturation, and body mass

index are essential physiological indicators used to assess an individual's health condition. Sensor-based health-monitoring systems enable continuous and automated data acquisition, thereby improving diagnostic consistency and reducing dependence on manual measurement. In the context of this study, biomedical monitoring theory supports the integration of multiple sensors within the FOVB-AIoT system to provide objective health evaluation and reduce human error.

The framework is further supported by Internet of Things Healthcare Systems Theory. IoT healthcare systems explain how interconnected devices and communication networks enable real-time data transmission, remote monitoring, and centralized data processing [12]. IoT architecture facilitates continuous health-data collection and seamless communication between sensors and processing units, supporting automated health monitoring. In the proposed system, IoT technology enables sensor components to transmit physiological data to the processing platform, ensuring real-time monitoring and more efficient data storage and access.

Artificial Intelligence Predictive Analytics Theory also serves as a major foundation of the study. This theory explains that machine-learning algorithms analyze structured data to detect patterns and generate predictive insights. Predictive models transform raw biomedical data into actionable health-risk assessments, thereby enabling early detection of possible medical conditions [14], [16]. In this study, the FOVB-AIoT system applies this principle through the XGBoost algorithm, which

processes collected vital signs and generates predictive health-risk scores to support proactive health decision-making.

System evaluation is anchored on software quality principles defined in the ISO/IEC 25010 standard, which establishes quality characteristics such as reliability, efficiency, usability, and security as benchmarks for software performance [11]. ISO/IEC 25010 provides a structured framework for evaluating whether a system meets functional and user requirements, ensuring dependable and user-friendly operation. In this study, the ISO/IEC 25010 model serves as the evaluative basis for assessing the reliability, efficiency, usability, and overall quality of the FOVB-AIoT system in automated health monitoring.

Collectively, Biomedical Monitoring Theory supports physiological data acquisition, IoT Healthcare Systems Theory enables real-time connectivity, AI Predictive Analytics Theory facilitates intelligent risk assessment, and ISO/IEC 25010 ensures structured quality evaluation. These integrated theoretical foundations justify the development and assessment of the proposed automated vital-sign monitoring system.

Conceptual Framework

Figure 1.2

System Design Paradigm

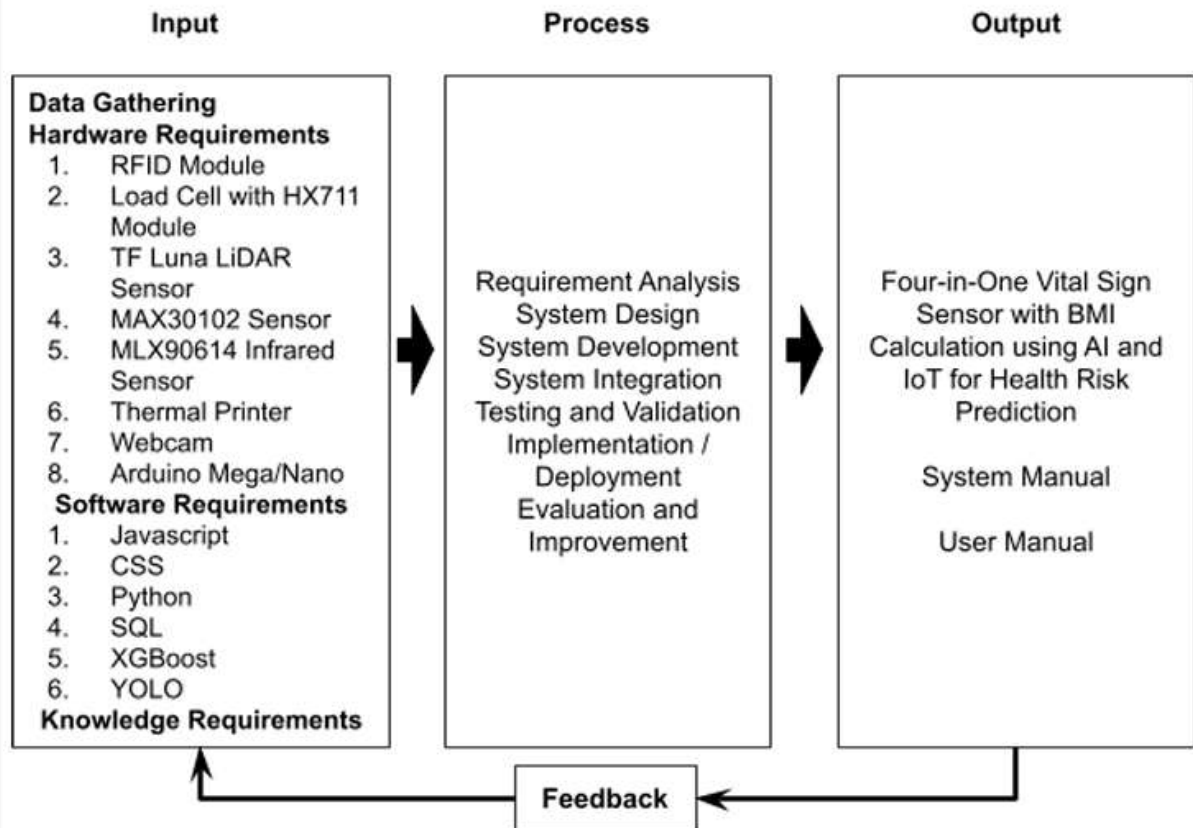


Figure 1.2 illustrates the Input–Process–Output flow of the FOVB-AIoT system.

The System Design Paradigm serves as the framework that guides how the system's components interact and function together. It defines the structure, process, and data flow from input to output, ensuring that the system operates efficiently and achieves its intended purpose.

Input

In this phase, the system identifies the necessary components for the input stage. It collects all required data from the sensors during each testing session.

Hardware Requirements

The system utilizes a Dell Mini PC (Intel i5 10th Gen) as the central processing unit responsible for data processing, storage, and AI-based health risk analysis, while communicating with the Arduino and database to enable real-time monitoring, record-keeping, and web-based reporting. A Fortress CF160TM 15.6-inch capacitive touchscreen monitor serves as the primary user interface, allowing users to interact with the system and conveniently view real-time readings and AI-generated results. The Arduino Mega functions as the main microcontroller that receives and processes data from all integrated sensors and transmits the results to the mini PC and display unit, with its multiple input/output pins supporting simultaneous component integration. For anthropometric measurements, a load cell with HX711 module measures weight by converting mechanical force into electrical signals for BMI computation, while the TF Luna LiDAR sensor determines height using laser pulse reflection technology for fast and precise readings. Body temperature is measured using the MLX90614 infrared sensor, which detects infrared radiation without physical contact to ensure hygienic operation. The RFID module verifies user registration through card tapping, allowing registered users to proceed directly to measurement while prompting new users to input personal information that is stored in the database. The MAX30102

sensor measures heart rate, respiratory rate, and oxygen saturation (SpO₂) using photoplethysmography technology for real-time, non-invasive monitoring. Blood pressure measurement is performed using a digital BP device that displays systolic and diastolic readings; these readings are captured by a webcam and processed through object detection techniques to extract cardiovascular data for system integration. A thermal printer generates printed copies of health results and recommendations after each session. The webcam serves as the primary visual sensor for object detection, ensuring accurate capture of blood pressure readings and preventing misuse by identifying unnecessary objects. The Arduino Nano operates as the microcontroller for the customized BP enclosure, controlling the LCD screen, LED lighting, and user input button, while receiving processed data from the mini PC to display detected results and manage illumination and system timing. The LCD screen provides real-time visualization of blood pressure readings and system status, LEDs ensure consistent internal lighting for accurate camera detection, and a button allows users to initiate BP measurement. Additionally, a speaker delivers voice guidance and audio feedback throughout the registration and measurement process, enhancing usability and accessibility within the kiosk-based healthcare system.

Software Requirements

The system relies on various software tools to handle health monitoring and process data from the physical sensors to the main software. Arduino IDE is used to write and upload code to the microcontrollers that gather hardware readings. For the user interface, the web application is built using React JS, HTML, CSS, and JavaScript to provide a responsive design in which vital signs, BMI results, and alerts update instantly without page reloading. On the backend, Python processes the incoming sensor data and performs calculations, while the Flask framework acts as the web server connecting the hardware, database, and frontend interface.

For advanced data analysis, the system integrates artificial intelligence. The XGBoost algorithm serves as the health-risk prediction engine, analyzing vital signs to classify health risks and provide early warnings. In addition, a YOLOv11 AI object-detection model reads and extracts blood-pressure numbers from digital monitors. To maintain BMI compliance, the system also gathers visual data from weight-scale and wearable-detection components to help ensure accurate results. The training data for this model is managed through Roboflow, and training is done using Google Colab. Lastly, Tailscale provides a secure virtual private network connection that supports encrypted communication between devices and allows safe remote access for maintenance without exposing the system to the public.

Knowledge Requirements

The study requires knowledge in programming, electronics, and web development to integrate both hardware and software components effectively. It involves skills in Arduino for sensor interfacing and data processing, as well as backend and frontend development for managing, displaying, and analyzing collected data. Understanding of data communication and AI integration is also important to ensure accurate data transfer, real-time monitoring, and reliable health prediction.

Data Gathering

The Data Gathering phase is crucial for the study's General Objective 3, which focuses on conducting a baseline survey to compare the clinic's current manual process with the implemented FOVB-AIoT system. This process involves two main stages: initial data collection through surveys during the developing phase to establish a performance benchmark of the current manual assessment in the RTU Pasig Clinic, and a second, more comprehensive data-gathering stage during the release phase. The post-implementation stage involves conducting further surveys and logging system-performance data from user interactions to formally assess the FOVB-AIoT in terms of time consumption, usability, efficiency, accuracy, and data integrity. The collected data from both stages will then be compared to evaluate the system's overall effectiveness and impact on clinic workflow.

Process

In this phase, the study outlines the VITAL-PRO followed in developing the system, from initial planning and design to development, testing, and final deployment, to ensure that the project is well organized, functional, and effective in achieving its goals.

1. **Requirement Analysis** – In this stage, the team identifies what the system must achieve, such as measuring vital signs, computing BMI, predicting health risks, and printing results. All user needs and clinic requirements are gathered to guide the entire development.
2. **System Design** – The system's architecture, sensor layout, wiring connections, database structure, and user-interface designs are created. This provides a clear blueprint of how the system will look and function before building it.
3. **System Development** – Hardware components are assembled and programmed, while the software, including AI prediction, UI design, and database functions, is coded. This phase turns the plans into working features.
4. **System Integration** – All modules such as sensors, RFID, AI models, data storage, display interface, and printing system are connected and combined into one unified system. This ensures every component communicates properly and works together.

5. **Testing and Validation** – The system undergoes multiple tests to check accuracy, responsiveness, stability, and usability. Sensor readings, BMI calculations, AI predictions, and printer outputs are validated to ensure correct performance. Issues found during testing are fixed immediately.
6. **Implementation/Deployment** – The completed system is installed and used in the RTU Pasig Clinic. Students, staff, nurses, and doctors interact with the kiosk during actual operations, allowing the system to function in real clinic conditions.
7. **Evaluation and Improvement** – Feedback from users and performance logs are analyzed to identify areas for enhancement. Adjustments are made to improve sensor accuracy, interface clarity, speed, and overall reliability.

Output

In the output stage, the system displays all analyzed health results on the Fortress CF160TM 15.6-inch capacitive touchscreen monitor, allowing users to view their measured vital signs, including body temperature, heart rate, respiratory rate, blood pressure, and BMI, together with the AI-generated health-risk prediction that provides an overview of their current health status. To ensure measurement accuracy and proper system utilization, the device integrates three AI-assisted cameras: two serve as a validation mechanism by detecting unnecessary accessories or footwear before the user proceeds to the measurement platform, while the third camera supports blood-pressure acquisition through object detection to assist in proper

positioning and reliable measurement capture. After the assessment is completed, the thermal printer automatically produces the PREP containing the recorded measurements, AI analysis, and personalized preventive recommendations. At the same time, all collected data are transmitted to UCARE, enabling RTU Clinic healthcare personnel to monitor and review results in real time.

Statement of the Problem

Taking vital signs is crucial for identifying health issues and checking changes in the body that may indicate potential problems. Parameters such as heart rate, respiratory rate, body temperature, blood pressure, and BMI are measured manually using various devices. This method is time-consuming, inconvenient, and not suitable for those who want a quick health check. Most conventional devices merely record results and do not aid in predicting early health risks. In response to these limitations, this study aims to address the following questions:

1. How can the system accurately and efficiently measure multiple vital signs to minimize the time spent on manual assessment?
2. How can the system use a Juan AI model specifically (XGBOOST, Gemini 2.0 and GPT-OSS-120b) to analyze vital-sign data and automatically identify early indicators of potential health risks?
3. How can the developed system improve the process of vital-sign measurement compared to the traditional manual method, particularly in terms of efficiency

and convenience, in detecting potential health risks?

3.1. How does the manual method compare with the AI-based system in terms of time, convenience, and efficiency?

3.2. How effective is the AI system in analyzing vital-sign data for identifying potential health risks?

4. How can the system be evaluated using the ISO/IEC 25010 software quality model to determine its performance in terms of reliability, efficiency, and usability?

Objective of the Study

The study aims to develop the FOVB-AIoT, a kiosk-based system capable of measuring multiple vital signs, analyzing results through artificial intelligence, and providing predictive insights about potential health risks in a single, automated, and user-friendly platform.

Specific Objectives

1. To design and develop a system that accurately and efficiently measures multiple vital signs, including body temperature, heart rate, respiratory rate, blood pressure, and BMI, reducing the time required for manual assessment while providing readings comparable in accuracy to standard clinical devices.
2. To develop Juan AI that will analyze vital-sign data and predict health risks based on detected patterns and irregularities.

3. To conduct a baseline survey comparing the clinic's current manual process with the implemented FOVB-AIoT system, serving as a basis for evaluating improvements in efficiency and convenience through the VITAL-PRO.
4. To evaluate the FOVB-AIoT system using the ISO/IEC 25010 software quality model, assessing its quality in terms of reliability, efficiency, and usability.

Research Hypothesis

1. Hypothesis : Time Consumption

Null Hypothesis (H_0):

There is no significant difference in time consumption between the traditional method and the FOVB-AIoT system.

Alternative Hypothesis (H_1):

There is a significant difference in time consumption between the traditional method and the FOVB-AIoT system.

2. Hypothesis: Efficiency

Null Hypothesis (H_0):

There is no significant difference in efficiency between the traditional method and the FOVB-AIoT system.

Alternative Hypothesis (H_1):

There is a significant difference in efficiency between the traditional method and the FOVB-AIoT system.

Scope and Delimitation

The study develops a FOVB-AIoT. It aims to create a kiosk-based system that is efficient, accurate, and user-friendly for measuring vital signs and analyzing potential health risks.

Specifically, this study will:

The scope of the study includes measuring multiple vital signs, namely body temperature, heart rate, oxygen level, respiratory rate, blood pressure, and BMI, and generating predictive health risk insights while storing the collected data in the UCARE database for processing and record-keeping. The system uses Juan AI to analyze data, detect abnormalities, and generate predictive insights. The results are displayed on a touchscreen monitor, with feedback provided through a voice assistant.

Additionally, the system utilizes IoT technology through a web-based platform that presents real-time sensor data, enabling seamless communication between the sensing unit and the UCARE database. It also incorporates AI-assisted cameras for object detection. Two cameras identify predefined classes such as bags, shoes, watches, and other accessories to prompt users to remove them before measurement, while another camera supports blood pressure acquisition through object detection.

There are some limitations to this study, including the following:

The system is designed for on demand use only and is not intended for continuous monitoring of vital signs. The study does not aim to produce a medical-grade device; therefore, it should not be used to replace professional diagnosis or medical consultation.

The camera-based object detection is limited to predefined classes and environmental conditions, which may affect detection accuracy. For wearable detection, the AI model is limited to identifying commonly used wearable items in a school environment, specifically caps, bags, watches, and identification (ID) cards with lace. It should be noted that not all types of wearable items can be detected, as the system focuses only on items that are frequently used in school settings.

For footwear detection, the AI system is limited to detecting shoes and socks only. Similar to wearable detection, the system does not detect all types of footwear but focuses only on those commonly worn by students within the school environment.

The AI risk factor assessment in the system is limited to ten parameters: sex, age, age group, body mass index (BMI), body temperature, heart rate, oxygen saturation (SpO_2), respiratory rate, systolic blood pressure, and diastolic blood pressure. The risk prediction generated by the Juan AI is based solely on these parameters and does not include other medical variables that may influence health conditions.

The blood pressure camera support depends on proper positioning and lighting conditions to ensure accurate readings. The height measurement function is limited to a maximum of 7 feet, which aligns with the average height range of Filipino individuals to maintain measurement accuracy.

Testing and validation of the system will be conducted only at the RTU Pasig Clinic, and large-scale hospital integration is beyond the scope of this study. The Juan AI prediction model will be trained using available datasets and may require further refinement to achieve medical-level precision.

Despite these limitations, the study highlights how combining Artificial Intelligence (AI) and the Internet of Things (IoT) in a kiosk-based multi-sensor system can provide a quick, smart, and accessible approach to monitoring vital signs and predicting potential health risks.

Significance of the Study

The study is significant because it introduces a kiosk-based system capable of measuring four vital signs: heart rate, respiratory rate, body temperature, and blood pressure, as well as calculating BMI using height and weight. By combining these measurements into a single, automated session, the **FOVB-AIoT** system simplifies data collection and ensures organized, accurate results. It also utilizes an AI algorithm to analyze the data gathered and predict potential health risks. Users receive the

PREP protocol through a receipt, enabling easy access to important health information for personal checking and early prevention.

The groups that will benefit from this study are as follows:

1. **RTU students** will have a quicker and easier way to check their vital signs, calculate their BMI, and receive alerts about potential health risks, helping them become more aware of their health status and make early lifestyle adjustments.
2. **RTU faculty members** will benefit from the system as it allows them to conveniently monitor their health within the campus. Regular access to vital sign checking can help them manage stress and workload more effectively, supporting their overall well-being and productivity in teaching.
3. **RTU staff** will also gain from the system's accessibility and ease of use, enabling them to check their vital signs anytime within the university premises. This promotes wellness and encourages a healthier working environment, helping them maintain good health while performing their daily tasks.
4. **Future researchers and students**, particularly those studying health and technology, can utilize this project as a valuable reference and starting point for their own research and innovative ideas.

Definition of Terms

Bio-Sync - (Biometric Synchronization Engine) This refers to the proprietary hardware-software integration framework that manages the simultaneous and time-

aligned data collection from all sensors in the FOVB-AIoT system. It coordinates the readings from the MAX30102, MLX90614, Blood Pressure, Load Cell, and LiDAR sensors, ensuring all vital sign data and BMI measurements are synchronized to a single user profile and timestamp for coherent analysis by the Juan AI.

FOVB-AIoT - The interactive health monitoring kiosk developed for this project, featuring the Four-in-One Vital Sign Sensor with BMI Calculation and integrated AI and IoT. It uses Juan AI, the MAX30102 Finger Pulse Rate Sensor, and other modules to measure and display vital signs, provide health risk predictions, and deliver feedback to users via receipt.

HRP - (Health Risk Prediction) The specific AI-generated output that summarizes the user's potential health risks based on analyzed vital sign data and BMI. It provides a categorical risk level to aid in early awareness.

Juan AI - is the artificial intelligence system responsible for analyzing sensor data from the Four-in-One Vital Sign Sensor to predict the user's health risk. Based on detected patterns and deviations from normal health parameters, it identifies potential health risks and provides recommendations or suggestions to help users take preventive measures or seek medical attention.

PREP - (Personalized Results and Educational Protocol) - This term can be used to describe the system's comprehensive output delivery method, which includes the on-

screen display, voice guidance, and printed receipt containing the user's results and health recommendations.

Pulse Rate - A sensor used in the Juan Kiosk to measure heart rate, oxygen saturation (SpO_2), and respiratory rate from a user's fingertip. In this study, the sensor used is the **MAX30102**, which provides accurate, real-time readings for display on the kiosk interface.

RTU Pasig Branch - Rizal Technological University is the academic institution where this study is conducted and where the Juan Kiosk is implemented for students and staff to conveniently monitor their health.

SENTINEL-IoT - Secure End-to-end Network for Telemetered Evaluation and Logging via IoT. This is the proprietary name for the complete data pipeline and security architecture of the FOVB-AIoT system. It encompasses the encrypted transmission of synchronized biometric data from the kiosk's Arduino microcontroller to the web server, the secure storage of this data within the UCARE database, and the protocols that govern real-time data access for authorized clinic staff, ensuring full data integrity and user privacy.

UCARE - University Clinic Assessment Record Enclave. The secure, dedicated segment of the IoT-based web database where the historical health records and real-time vital sign data of all registered RTU Pasig users are stored, monitored, and made accessible to authorized clinic staff.

Vital-Pro - (Vital Sign Protocol for Rapid Operation) This is the proprietary name for the standardized, sequential procedure that governs the user interaction and automated data acquisition workflow of the FOVB-AIoT system. The protocol ensures a swift and consistent process from user registration via RFID through the sequential measurement of weight, height, body temperature, pulse rate, and blood pressure, culminating in the AI analysis and result printing. Its design directly addresses the study's objective to minimize assessment time and improve operational efficiency in the clinic.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents the related literature and studies that highlights the importance of integration of sensors, BMI calculations, A.I. and IoT in the field of healthcare. It dives on existing research on the vital sign devices, the application of A.I. in predicting health risks and IoT for real-time health data transmission. These studies provide a foundation for the development of the Four-in-One Vital Sign Sensor with BMI Calculation.

Foreign Literature

Theme 1: Technological Advancements in Biomedical Monitoring Systems

Recent advancements in microcontroller and biomedical sensor technologies have enabled the development of compact, multi-parameter health monitoring systems capable of measuring vital signs such as heart rate, blood pressure, oxygen saturation, respiratory rate, body temperature, height, and weight. These innovations support automated and real-time health assessment in both clinical and non-clinical environments, enhancing accessibility and efficiency in health monitoring.

Ahmed [1] conducted a comparative analysis of medical sensor technologies, emphasizing accuracy, reliability, and cost-effectiveness in

physiological data acquisition. The study highlights the importance of selecting validated sensors to ensure clinically meaningful measurements. Similarly, Chowdhary [2] documented advancements in biomedical sensing, particularly in miniaturization and signal processing, which improve measurement precision. These studies establish that modern biomedical sensors can provide reliable real-time data when properly calibrated and validated.

However, existing literature often evaluates sensors in isolation rather than as part of integrated multi-parameter monitoring systems. This gap underscores the need for systems that unify multiple validated sensors within a single platform to support comprehensive health assessment.

Theme 2: Clinical Standards and Health Classification Benchmarks

Accurate interpretation of health data requires adherence to established clinical standards. The American Heart Association provides updated classifications for heart rate and blood pressure, serving as medical benchmarks for cardiovascular assessment [3]. Additionally, the Centers for Disease Control and Prevention defines standardized BMI categories, ensuring consistent health classification across populations [4]. Harten and Anderson [5] further established age-specific respiratory rate reference values, which are essential for interpreting respiratory health indicators.

These standards provide a foundation for automated monitoring systems by establishing medically accepted thresholds for data interpretation. Lim et al. [6] further

support the use of region-specific BMI classifications, ensuring appropriate health assessment across populations.

Despite the availability of standardized benchmarks, many automated systems fail to incorporate localized health classifications. This limitation highlights the need for monitoring solutions that align both global medical standards and local health metrics.

Theme 3: IoT and AI Integration in Healthcare Systems

Internet of Things (IoT) technology has expanded healthcare monitoring capabilities through real-time connectivity and remote data transmission. Boonsong [7] evaluated IoT system architectures, focusing on latency, scalability, and reliability. Jawad et al. [8] identified interoperability and data security as significant challenges in IoT healthcare implementation, emphasizing the need for secure and efficient system design.

Artificial Intelligence (AI) further enhances monitoring capabilities by enabling predictive analytics and automated data interpretation. Kumar and Patel [9] demonstrated that AI algorithms can detect early abnormalities and reduce manual workload, improving health assessment efficiency. Li et al. [10] reported improved diagnostic accuracy when AI processes multi-modal sensor data, highlighting AI's role in enhancing data-driven decision-making.

In computer vision applications, YOLO-based object detection techniques have shown high accuracy in extracting numerical readings from medical device displays, supporting camera-based digitization approaches [11], [12]. These findings confirm the feasibility of integrating AI-driven display recognition within automated monitoring systems.

Despite these advancements, most IoT and AI healthcare studies focus on individual components rather than fully integrated solutions. The absence of unified systems combining sensor validation, AI analytics, and IoT connectivity presents an opportunity for innovation in automated health monitoring platforms.

Theme 4: Sensor Validation and Measurement Reliability

Reliable health monitoring requires rigorous validation and calibration procedures. Malhotra [13] emphasized calibration and noise reduction techniques to enhance measurement accuracy. Studies involving load cells, LiDAR sensors, infrared thermometers, and digital blood pressure monitors consistently recommend repeated measurements and averaging methods to minimize variability [14]–[17].

García et al. [18] reinforced that multiple trials improve statistical confidence and measurement stability, supporting standardized validation protocols. These practices ensure that biomedical devices produce consistent and reliable results suitable for health assessment.

However, validation studies often examine devices independently rather than within integrated systems. This gap underscores the need for comprehensive validation frameworks that evaluate multi-parameter monitoring systems as unified platforms.

Theme 5: Ethical Governance and Data Privacy in Digital Health

Ethical considerations remain essential in AI-driven healthcare systems. The Belmont Report establishes principles of respect for persons, beneficence, and justice in human subject research. The World Health Organization further outlines governance guidelines for AI in healthcare, emphasizing transparency, accountability, and fairness [19].

Data privacy is equally critical in digital health applications. Cabrera and Lee [20] highlighted the necessity of encryption and anonymization to protect sensitive health information. Compliance with ethical and legal frameworks ensures responsible handling of health data in automated monitoring systems.

Despite established ethical guidelines, challenges remain in balancing data accessibility and privacy. Future systems must prioritize secure data management while enabling meaningful health analytics.

Theme 6: Probability Sampling in Quantitative Research

Probability sampling ensures that each member of the population has an equal chance of being selected, allowing researchers to obtain representative and unbiased results. Cluster sampling is commonly used in educational research where respondents are naturally grouped into classes or institutional units [21]. This approach simplifies data collection while maintaining statistical reliability.

Theme 7: Respondent Selection and Sample Size

Selecting appropriate respondents and determining sufficient sample size are essential in quantitative research. An adequate number of respondents is necessary to ensure statistical reliability and meaningful interpretation of results [21]. Clearly defined respondent groups improve the reliability of comparative analysis in quantitative studies [21].

Theme 8: Survey Questionnaires in System Evaluation

Survey questionnaires are widely used to measure user perceptions and system performance. Likert-scale questionnaires allow researchers to quantify user opinions and analyze perceptions systematically [22]. In system evaluation, usability, reliability, and efficiency are commonly assessed using structured questionnaires [23].

Theme 9: Likert Scale Interpretation and Weighted Mean Analysis

Likert-scale analysis is commonly used in survey research to interpret user perceptions. Likert scales convert subjective responses into measurable numerical values that can be statistically analyzed [22]. Weighted mean analysis allows researchers to summarize survey responses and provide clear interpretations of respondent perceptions [23].

Theme 10: Data Gathering Methods in Research

Effective data gathering is essential for ensuring reliable research outcomes. Systematic data collection procedures help organize information and support accurate analysis [23]. Combining multiple data collection methods strengthens the validity of research findings.

Synthesis and Identified Gap (Foreign Literature)

The reviewed foreign studies demonstrate significant progress in sensor validation, AI integration, and IoT-based healthcare systems. However, most research focuses on individual components rather than a fully integrated, kiosk-based multi-parameter system that combines validated sensors, AI-driven display digitization, predictive analytics, and standardized health classifications within a single platform. This gap supports the need for an integrated system that unifies these technologies into a practical and accessible solution.

Theme 11: Advanced AI Validation, Clinical Alignment, and Real-Time Data Extraction

To ensure predictive health models are actually reliable, recent research points out the need to move away from standalone algorithms. Kabir et al. [41] showed that blending traditional machine learning setups with Large Language Models (LLMs) through voting fusion greatly boosts prediction accuracy. But relying on just one language model can lead to clinical hallucinations. To fix this, Ali et al. [40] pushed for a multi-agent consensus framework—essentially a "council of AIs"—proving that having multiple AI models check each other drastically cuts down on individual biases.

Beyond just getting the AI to agree, using AI in healthcare means it must strictly follow medical ethics. If general-purpose LLMs are deployed without rules, they pose huge risks. Chen et al. [42] explained how setting strict system prompts to align language models ensures their outputs stay factual and within safe boundaries. Building on this, Wang et al. [43] stressed the need for Expert-in-the-Loop (EITL) pipelines, which embed human medical expertise into the AI's logic to filter out conclusions that might be mathematically right but medically wrong. Liu et al. [44] ultimately found that when LLMs are guided to mimic a doctor's diagnostic steps, it heavily improves the safety of automated health screening.

Of course, before any of this AI validation can happen, the system first needs to accurately capture the physical data in real time. This is where efficient computer

vision models come into play. The introduction of architectures like the Single Shot MultiBox Detector (SSD) [46] and the YOLO (You Only Look Once) framework [45] completely changed real-time object detection. They struck the perfect balance between speed and accuracy, which is exactly what allows modern systems to extract medical vital signs right at the edge without lagging.

Theme 12: Advantages of Lightweight Language Models in Resource-Constrained IoT Healthcare

Even though large language models (LLMs) are incredibly smart, they demand a massive amount of memory and computing power. This makes it practically impossible to run them locally on constrained IoT devices like a standard Mini PC. Looking into edge AI deployment, Sciammarelli [47] pointed out that these hardware bottlenecks are exactly why developers are shifting toward Small Language Models (SLMs) or lightweight, open-source options. By using quantized, compressed models, a system can still handle natural language processing right on the CPU, completely avoiding the need for expensive GPU setups.

On top of the hardware limitations, relying purely on cloud-based LLMs brings up serious concerns about latency and patient privacy—two things you simply cannot compromise in medical IoT. Mou and Lou [48] found that running context-adapted lightweight models locally allows the system to offer instant clinical support while keeping sensitive patient data entirely on the device. This edge-based execution not

only kills the lag time but also keeps vital sign data safe from outside interception during the initial validation step. For systems that eventually need to generate complex medical reports, researchers suggest a distributed approach: use the local lightweight model for fast, secure validation, and then offload the heavy text-generation tasks to a larger cloud-based LLM via an API [49]. This setup keeps the whole healthcare platform fast, secure, and perfectly suited for limited hardware.

Local Literature

Theme 1: National Healthcare Policies and Preventive Health Monitoring

Preventive healthcare and technology-driven health solutions are national priorities in the Philippines. The Department of Health emphasizes health monitoring initiatives in educational institutions to promote early detection and preventive care [24]. This policy aligns with the deployment of automated monitoring systems that support regular health assessment and preventive interventions.

The Department of Science and Technology – Philippine Council for Health Research and Development highlights AI-driven innovations as viable solutions for improving healthcare accessibility and efficiency [25]. These initiatives demonstrate institutional support for technology-based health systems, reinforcing the feasibility of automated monitoring platforms in local settings.

Additionally, the National Nutrition Council emphasizes the importance of regular vital sign monitoring as a preventive health strategy [26]. Monitoring physiological

indicators enables early identification of potential health risks, supporting evidence-based health interventions and community health improvement.

Collectively, these policies establish a strong institutional foundation for automated health monitoring systems that enhance accessibility, efficiency, and preventive care in educational and community environments.

Theme 2: Local Health Standards and Objective Health Assessment

Accurate health evaluation requires adherence to localized health classification standards. The Philippine Statistics Authority provides BMI classifications tailored to the Filipino population, ensuring culturally relevant and statistically appropriate health interpretation [27].

Quianzon-Manuel et al. [28] reported inconsistencies in self-reported height and weight data, highlighting the limitations of subjective health reporting. These findings justify the implementation of automated measurement systems that reduce human error and enhance data reliability.

Theme 3: Local Innovations in Health Monitoring Systems

Tolentino et al. [29] developed a smart health screening system for rural communities, demonstrating the feasibility of deploying automated monitoring solutions in resource-limited environments.

Dela Cruz et al. [30] introduced an all-in-one vital sign monitoring prototype integrating multiple sensors, supporting real-time data acquisition and system feasibility in local settings.

Theme 4: Validation and Statistical Methods in Biomedical Systems

Marquez et al. [31] emphasized calibration protocols in load cell systems to enhance accuracy. Lopez [32], Torres and Ramos [33], Martinez [34], and Reyes and Domingo [35] provided statistical validation techniques such as paired t-tests, absolute error, and percent error.

De Guzman [36] and Santos and Rivera [37] supported the use of weighted mean and standard deviation for analyzing survey data and measurement consistency.

Theme 5: Institutional Health Monitoring and Sampling Methodologies

The Department of Health recognizes schools as clustered populations suitable for structured health monitoring studies [24]. Cluster sampling in educational environments supports efficient and representative data collection [21].

Theme 6: Ethical and Legal Compliance in Digital Health Systems

The Data Privacy Act of 2012 ensures lawful collection and protection of personal health data [38]. Ethical compliance requires transparency, informed consent, and responsible data management.

Theme 7: AI Object Detection and Trial Evaluation in Health Assessment

Systems

YOLO-based models enable fast and accurate object detection. Redmon and Farhadi [11] improved detection performance through deep learning techniques, while Liu et al. [12] introduced SSD for efficient object detection.

Szeliski [39] emphasized that repeated trials and statistical evaluation are essential for validating computer vision systems.

Synthesis of Related Literature

This emphasizes the concepts from literature and explores the ideas that are connected or related to AI and IoT for Vital sign reading or HRP.

Vital Sign System Synthesis

[illegible]

Legends:

- A: IoT-Based Vital Signs Monitoring System for Preventive Healthcare (Jawad et al.)
- B: IoT Approach to Remote Monitoring of Heart Rate, SpO₂, and Temperature (Costrada et al., 2022)
- C: Using Heart Rate and SpO₂ Pulse Oximeter with AI and IoT (Kakumanu et al., 2021)
- D: A Wireless Body Temperature and Oxygen Saturation Monitoring system (Hoseinzadeh et. al, 2022)
- E: Development of Complex System for Remote Heart Rate Monitoring (Kramov & Bauzha, 2020)
- F: Real-Time Intelligent Healthcare Monitoring with Deep Learning (Panganiban et al., 2020)
- G: Validity of Self-Reported Height, Weight, and BMI (Quianzon-Manuel et al., 2023)
- H: Environmental Factors Affecting BMI Using Machine Learning (Mendoza et al., 2024)
- I: Smart Health Screening System for Rural Communities (Tolentino et al., 2021)
- J: HealthWatch: An All-in-One Health Monitoring System (Dela Cruz et al., 2024)
- K: Deep Learning-Based IoT System for Remote Monitoring (Islam et al., 2023).
- L: Proposed Study – FOVB-AIoT : Four-in-One Vital Sign Sensor with BMI Calculation using AI and IoT for Health Risk Prediction in RTU Pasig Clinic.

The systems reviewed in the related literature were evaluated based on specific parameters that highlight their core functionalities and technological scope. The parameters used for comparison include SpO₂ and Heart Rate (HR) Monitoring, Body Temperature Measurement, Blood Pressure Reading, BMI Calculation (Height and Weight), IoT/Web-Based Connectivity, Artificial Intelligence (AI) Integration, Receipt Result or Notification, Voice Guidance, Real-Time Data Access, Kiosk Deployment, HRP, and RFID-Based User Registration. In the comparative analysis, a check mark (✓) denotes the presence of a particular feature, while a cross mark (×) indicates its absence.

Based on the analysis, Systems A, B, and C each possess five (5) key functionalities, System D also has five (5), System E includes four (4), System F provides six (6), Systems G and H each contain three (3), Systems I and J demonstrate eight (8) due to the inclusion of blood pressure monitoring, and System K offers five (5) key functionalities.

The proposed System L – FOVB-AIoT : Four-in-One Vital Sign Sensor with BMI Calculation using AI and IoT for Health Risk Prediction in RTU Pasig Clinic for HRP – integrates all the listed parameters, including the newly added blood pressure reading and RFID-based user registration, into a single comprehensive platform. It features a kiosk-based deployment equipped with AI-driven health analysis, RFID-based registration for secure user identification, real-time IoT data transmission, voice-

guided operation, and automated notifications for health results. This integration addresses the limitations observed in previous systems by providing a more complete, intelligent, and user-friendly health checking solution suitable for preventive and institutional healthcare applications.

RESEARCH METHODOLOGY

This chapter presents the sampling plan, respondent descriptions, statistical analysis of data, instruments used, and data collection procedures applied in the study. These research methods were utilized to develop the FOVB-AIoT.

Research Design

This study adopts the Agile Development Life Cycle as the research design because it supports a flexible, iterative, and user-centered approach suitable for developing a kiosk-based vital sign monitoring and AI prediction system. Agile emphasizes continuous improvement through repeated cycles of planning, designing, building, testing, and refining. Instead of completing the entire system in a single linear process, features are gradually developed, evaluated, and improved based on real user feedback. This ensures that every version of the system becomes less time consuming, accurate, efficient, reliable and data secured, aligning with the needs of clinic staff, students, and healthcare workflows. Through this dynamic structure, the researchers are able to adjust features such as sensor calibration, AI model behavior, and UI design as the project progresses, resulting in a system that evolves and improves throughout development.

Figure 3.1

Prototyping Development Agile Cycle



Figure 3.1 illustrates an iterative development cycle where each phase Planning, Designing, Developing, Testing & Survey, Release, and Feedback & Iteration connects in a continuous loop. The process begins with planning and moves sequentially through the stages until the system is released. After deployment, feedback from users leads back to another planning phase, creating a cycle of continuous improvement. This loop ensures that each prototype becomes more refined and meets the requirements of actual clinic operations.

Planning – This stage focuses on preparing the project by identifying what needs to be done, what resources are required, and when tasks should be completed. The researchers create a roadmap that includes the schedule, budget, and tools needed.

Simple planning techniques such as timelines, diagrams, or sketches are used to make sure the project is realistic and achievable.

Designing – In this phase, the researchers translate the requirements into a clear design of how the system will look and function. The design serves as the blueprint for the prototype, showing both the appearance of the interface and how the features will work. This step bridges the gap between what the users need and how the system will be built.

Developing – The actual building of the system takes place. The researchers start coding, connecting hardware components, and developing the Bio-Sync to integrate the sensor data streams based on the design. Since errors are common, this stage involves a cycle of building, checking, and improving until a usable prototype is created.

Testing & Survey – Once a working prototype is ready, it is tested to check if it meets the expected performance. Surveys and feedback from respondents in the RTU Pasig Branch are also gathered to assess whether the system is accurate, reliable, and easy to use. Any issues found during this stage are fixed through revisions.

Release – Once a functional version meets the required standards, it is deployed in the RTU Pasig Branch clinic for real-world evaluation. This version allows users to interact with the system in its intended environment, helping the researchers gather

practical feedback about its reliability, convenience, and effectiveness. Each release is considered a working version that may still be improved in later iterations.

Feedback and Iteration – After deployment, feedback from users and system performance data are analyzed to identify areas needing enhancement. This input informs the next planning cycle, allowing the system to evolve over time. Updates may include refining the AI predictions, optimizing sensor accuracy, or improving the user interface. Through this loop of continuous improvement, the system becomes more efficient, user-friendly, and aligned with real healthcare needs.

Locale of the Study

The study will be implemented at the RTU – Pasig Branch clinic, where the system will be tested and deployed. Students will participate in health assessments to evaluate the system. This location was selected to enable collaboration with clinic personnel and assess the system's practicality and effectiveness in real health monitoring.

Figure 3.2

Location of the Deployment



Probability Sampling

The use of probability sampling, particularly cluster sampling, is widely supported in quantitative and institutional research because it ensures that each member of the population has a known chance of selection, thereby increasing the representativeness and generalizability of the study findings. Probability sampling techniques are recommended when researchers aim to make valid inferences about a target population, as they minimize sampling bias and enhance the statistical reliability of the collected data [9]. In educational and institutional environments, cluster sampling is especially appropriate because populations are often naturally grouped into categories such as schools, departments, or academic roles. Studies

have shown that cluster sampling allows researchers to efficiently collect data from populations that are already organized into identifiable groups while maintaining methodological rigor and representative sampling [9]. Furthermore, cluster-based sampling designs are commonly used in public health and educational research because they streamline data collection processes and allow researchers to evaluate interventions within real-world institutional settings where populations are structured into clusters [8]. These methodological advantages justify the use of cluster probability sampling in institutional health studies such as the evaluation of the FOVB-AIoT kiosk system.

Description of Respondents

The respondents of the study consisted of 72 third-year Bachelor of Science in Computer Engineering (BSCPE) students from Rizal Technological University – Pasig Campus. The participants were grouped according to their respective class blocks: PCEIT-601A (35 students) and PCEIT-601E (37 students), as presented in Table 3.2. Cluster sampling was employed in selecting the respondents, wherein intact sections were chosen to represent the population. Cluster sampling is commonly used in educational research where participants are naturally grouped into classes or academic sections, allowing researchers to efficiently collect data while maintaining representation of the target population [9].

The use of cluster sampling allowed the researchers to efficiently gather data and systematically compare the ratings of the two groups regarding the manual or traditional approach and the system-based approach in obtaining vital signs and Body Mass Index (BMI). This method facilitated a clearer comparison of perceptions in terms of efficiency and time convenience. In quantitative survey research, sample sizes between 30 and 100 respondents are considered adequate for evaluating system performance, usability, and user perception, particularly in technology-based system assessments [8].

All participants underwent actual testing and evaluation of the system to assess its performance and to provide feedback based on their direct experience using the questionnaire consisting of indicators on time convenience, efficiency, usability, reliability, and security and data integrity. Selecting respondents from defined academic clusters ensured consistency in comparison and strengthened the reliability of the evaluation results. Furthermore, methodological studies emphasize that samples exceeding 50 respondents are generally sufficient for statistical analysis in system usability and technology evaluation studies, enabling meaningful interpretation of survey-based performance indicators [9].

Table 3.1

Survey Respondents

Block	Frequency	Percentage (%)
PCEIT-601A	35	48.61%
PCEIT-601E	37	51.39%
Total	72	100%

Research Instrument

The research instruments for this study will be surveys, prototype testing, and data analysis. These instruments will measure user satisfaction, system reliability, and areas for potential improvement. The findings derived from these instruments will provide valuable insights for the successful implementation of the AI and IoT-based vital sign system with BMI calculation.

Survey Questionnaire

The survey questionnaire was designed to evaluate the overall performance and effectiveness of the FOVB-AIoT with BMI Calculation using AI and IoT for Health Risk Prediction (HRP). The instrument employed a 4-point Likert Scale ranging from 4 – Very Satisfied to 1 – Very Dissatisfied and was structured to quantitatively assess key system features. The use of Likert-scale questionnaires is widely recommended in technology and system evaluation studies because it enables researchers to measure user perceptions, usability, and system performance in a structured and

statistically analyzable manner [13]. Two sets of questionnaires were utilized in the study: the first questionnaire evaluated the manual or traditional method of obtaining vital signs and Body Mass Index (BMI), while the second questionnaire assessed the developed FOVB-AIoT system after actual user interaction and testing.

The system evaluation questionnaire included categories such as Time Consuming, Efficiency, Usability, Reliability, and Security and Data Integrity to measure different aspects of system performance, which are commonly used indicators in information system and healthcare technology assessments [23]. However, for comparative analysis between the traditional method and the developed system, only the categories of Time Consuming and Efficiency were used to determine whether the developed system improves the speed of measurement and streamlines the overall process of obtaining vital signs and BMI.

The data gathered from the respondents were processed through weighted mean calculations to evaluate the overall performance of the system. Table 3.2 presents the weighted mean ranges alongside their corresponding verbal interpretations, creating a clear framework for measuring user satisfaction in all testing criteria. Furthermore, the researchers utilized Cronbach's Alpha to guarantee the internal reliability of the survey instrument. This statistical approach confirms that the questions accurately target the intended constructs by measuring the degree of relatedness among grouped items.

Table 3.2

Ranges of Weighted Mean

Scale	Weighted Mean	Verbal Interpretation
4	3.26 – 4.00	Very Satisfied
3	2.51 – 3.25	Satisfied
2	1.76 – 2.50	Dissatisfied
1	1.00 – 1.75	Very Dissatisfied

Table 3.2 presents the weighted mean scales and their corresponding verbal interpretations used in analyzing the responses from the survey questionnaire. This table serves as the basis for interpreting the computed weighted means of the system's evaluated parameters, such as accuracy, usability, efficiency, and reliability. The researchers adopted a 4-point Likert scale method, where numerical ranges are assigned equivalent descriptive interpretations ranging from 4 (Very Satisfied) to 1 (Very Dissatisfied). The use of weighted mean interpretation in Likert-scale questionnaires is widely applied in survey-based research because it allows researchers to summarize respondents' perceptions and evaluate system performance indicators in a clear and systematic manner [13]. Furthermore, the use of a 4-point Likert scale without a neutral option encourages respondents to provide a definite evaluation of the system, which improves the reliability and clarity of results in usability and technology evaluation studies [8].

Table 3.3

Interpretation of Cronbach's Alpha values

Cronbach's Alpha Values	Verbal Interpretation
0.90 and above	Excellent
0.80 – 0.89	Good
0.70 – 0.79	Acceptable
0.60 – 0.69	Questionable
0.50 – 0.59	Poor
Below 0.50	Unacceptable

Table 3.3 outlines the standard interpretation scale for Cronbach's Alpha coefficients, which serves as the primary benchmark for evaluating the survey's internal consistency. According to established research parameters, an alpha coefficient reaching or exceeding 0.70 is deemed acceptable, confirming the overall reliability of the data-gathering instrument [23]. By applying this specific grading scale to each of the evaluation categories of the system, the researchers can systematically verify that the questionnaire yields dependable and structurally valid results prior to conducting further statistical analysis.

Data Gathering

The proponents will employ a systematic process to collect and organize data that will address the objectives of this study. This will include identifying reliable data sources, designing appropriate data collection methods, and structuring the gathered

information for analysis to ensure an effective evaluation of both technical performance and user experience. A systematic approach to data collection is essential in research because it ensures that the gathered information is accurate, organized, and suitable for statistical analysis and interpretation [8].

To gather the necessary data, the proponents will utilize three key methods: administering surveys to respondents, conducting direct system observations during prototype testing, and reviewing related literature to establish the study's context. The use of multiple data collection methods strengthens the validity of research findings by allowing researchers to obtain information from different sources and perspectives [8]. These methods will enable the researchers to validate the system's functionality and assess its effectiveness in promoting healthcare and wellness within the Rizal Technological University – Pasig Campus community.

Statistical Treatment Data

The data gathered will be analyzed and interpreted by the statistical methodologies enumerated below.

Weighted Mean

To get the weighted mean of each question parameters of the survey, the proponents use the formula of:

$$WM \text{ of each question} = \frac{\sum (Response \cdot Rating)}{Total \text{ of Respondents}} \quad (1)$$

Equation (1) weighted mean (WM) is employed to compute the Verbal Interpretation (VI) according to specific parameters, as outlined in Table 3.1. Table 3.1 presents the rating scale for weighted mean values, which serves as the basis for classifying verbal interpretations. It is used to determine the overall tendency of the data gathered from the prototype measurements as well.

$$\text{Weighted Mean} = \frac{\text{Sum of All Rating Scores}}{\text{No. of Rating Scores}} \quad (2)$$

Standard Deviation

Equation (2) standard deviation measures the consistency and variability of the readings collected from the system, determining how close or spread out the data points are from the mean value.

$$SD = \sqrt{\frac{\sum(X_i - \bar{X})}{n - 1}} \quad (3)$$

Where:

X_i = individual measurement

$\bar{X}$ = mean of all measurements

n = total number of measurements

Difference per Pair (d_i)

Equation (3) difference per pair (d_i) presents the numerical difference between two corresponding measurements taken from the prototype system and the reference

or standard device for the same respondent or trial. This value is necessary for conducting a paired t-test because it reflects how much the prototype's output deviates from the established standard for each individual observation.

$$d_i = X_i - Y_i \quad (4)$$

Where:

X_i = Score of the FOVB-AIoT System (Proposed System)

Y_i = Score of the Traditional Manual Method (Baseline)

T-Test

Equation (4) paired samples T-test will be employed to determine whether a statistically significant difference exists between the Traditional Manual Method (Baseline) and the FOVB-AI System (Proposed System). This study adopts a paired approach because the same group of $n = \{\text{no. student}\}$ student respondents will evaluate both methods. This allows for a direct comparison of their experiences in terms of Time Consumption and System Efficiency using a 4-point Likert scale (1-4).

The formula to be used is:

$$t = \frac{\bar{d}}{SD_d / \sqrt{n}} \quad (5)$$

Where:

$\bar{d}$ = mean of the paired differences

SD_d = standard deviation of the differences

n = number of paired observations (respondents)

Absolute Error

Equation (5) absolute error measures the magnitude of the difference between the prototype's measurement and the true or reference value, without considering whether the difference is positive or negative. It provides a straightforward indication of how far a single reading deviates from the accurate or standard measurement.

$$\text{Absolute Error} = |\text{Measured Value} - \text{True Value}| \quad (6)$$

Error (%Error)

Equation (6) percent Error expresses the **relative size of the measurement error** compared to the true or reference value. It allows comparison of accuracy across different measurement types (e.g., BPM, °C, mmHg) by converting the error into a percentage.

$$\%Error = \frac{|X_m - X_t|}{X_t} \cdot 100\% \quad (7)$$

X_m = measured value (prototype)

X_t = true/reference value

Accuracy

Equation (7) accuracy determines how close the prototype's measured value is to the true value, expressed as a percentage. It reflects the overall correctness of the device's readings. The formula subtracts the relative error from 1 (or 100%) to

represent the proportion of correctness.

$$Accuracy = \left(1 - \frac{|Measured - True|}{True}\right) \cdot 100\% \quad (8)$$

Cronbach's Alpha

To establish the internal consistency of the survey instrument used for the FOVB-AIoT system evaluation, Cronbach's Alpha was employed as the primary statistical tool. This specific metric evaluates the degree of relatedness among a group of survey items, verifying that the questions uniformly assess the intended underlying construct. It is widely used in survey-based research, particularly in Likert-scale questionnaires, to validate the reliability of the instrument.

The Cronbach's Alpha coefficient is computed the following formula:

$$\alpha = \frac{k}{k-1} \left(\frac{\sum \sigma_i^2}{\sigma_t^2} \right) \quad (9)$$

Where:

k = number of items

σ_i^2 = variance of each item

σ_t^2 = total variance of the sum of all items

Metrics to validate

To ensure the accuracy and reliability of the system, three trials will be conducted for each hardware component. Repeated measurements are a standard approach for evaluating measurement repeatability and reducing the effect of random variation in experimental results. The results will be validated using the Mean, Standard Deviation, and Percentage Error. The Mean will determine the average performance of the system across trials, the Standard Deviation will measure the consistency or repeatability of the readings, and the Percentage Error will compare the system's results with those of standard reference devices to assess accuracy. These metrics are commonly used in repeated-measurement and calibration practice to determine whether a system produces stable and acceptably accurate measurements. In calibration work, replicate comparisons are used to assess repeatability and measurement stability [10], [19], [20].

System Testing

This section will present the hardware and software testing that will be conducted based on the system's functional features. Each hardware component and software module will be carefully tested to verify proper operation and to ensure that every function will perform according to its intended purpose. The testing will aim to identify and correct any issues that may affect performance, reliability, or accuracy. The following test cases validate the VITAL-PRO user interaction sequence:

Table 3.4

Software Testing

Test Data/ Action	Expected Result	Trial 1	Trial 2	Trial 3	Ave. Value	Error (%)	Accuracy (%)	Remark
Touch start button	The user will proceed to home screen	1	1	1	1	0%	100%	Passed
User account registration creation	User successfully created account	1	1	1	1	0%	100%	Passed
User tap the ID	The system will validate the user	1	1	1	1	0%	100%	Passed

User start the button for BMI Phase	The system will display the weight, height, and the computed BMI	1	1	1	1	0%	100%	Passed
User touch the button for body temperature	The body temperature will display	1	1	1	1	0%	100%	Passed
User put the left index finger in the pulse oximeter	The system will display the heart rate, respiratory rate and oxygen saturation	1	1	1	1	0%	100%	Passed
The assistant nurse touch the button to start the measurement of blood pressure	The system will display the result of blood pressure with diastolic and systolic	1	1	1	1	0%	100%	Passed
User touch the button for AI Result	The system will calculate the prediction of the health risk	1	1	1	1	0%	100%	Passed
User touch the button to start printing the result	The system will print the results of the calculation of health risk prediction and the vital signs	1	1	1	1	0%	100%	Passed
User touch the button to email the result	The system will email the results of the of health risk prediction and the vital signs	1	1	1	1	0%	100%	Passed

Table 3.4 Software testing presents the unit testing of the system, where the test data represent the different software features tested, and the results indicate that all functions performed properly.

Hardware Component Testing and Validation

The hardware components undergo validation and accuracy testing to ensure that the system's measurements of vital signs and BMI are consistent with established

clinical standards. The readings from the system's integrated sensors are compared directly against trusted reference devices commonly used in healthcare settings. This ensures that measurements for weight, height, body temperature, heart rate, and blood pressure are accurate and reliable before being processed by the AI for HRP.

Table 3.5

Hardware Components Testing

Manual Devices	System Components	Expected Values	Trial 1	Trial 2	Trial 3	Remarks
Digital Weight Scale	Load cells and HX711	52.55kg	52.89kg	52.90kg	52.71kg	Passed
Tape Measuring	Tf luna lidar	165.5 cm	165.59cm	164.59cm	164.90cm	Passed
Non Contact Temp Gun	MLX90614	36.9°C	36.41°C	36.43°C	36.51°C	Passed
Digital Blood Pressure	Digital Blood Pressure	117 SYS 74 DIA	114 SYS 69 DIA	125 SYS 72 DIA	105 SYS 64 DIA	Passed
Oximeter (Heart Rate)	Heart Rate (MAX30102 Sensor)	80BPM	85BPM	79BPM	83BPM	Passed
Oximeter (Respiratory Rate)	Respiratory Rate (MAX30102 Sensor)	20 breaths per minute	23 breaths per minute	18 breaths per minute	21 breaths per minute	Passed
Oximeter (Oxygen Rate)	Oxygen Saturation (MAX30102 Sensor)	100%	98%	100%	100%	Passed

Table 3.5 presents the comparative results between the system's measured readings and those obtained from the corresponding standard reference devices across trials. For each parameter, the mean value of the three trials was computed, followed by the calculation of the absolute error, percent error, and overall percent accuracy to evaluate the system's measurement reliability and performance.

Ethical Considerations

This study collects sensitive health data using a four-in-one vital sign sensor with BMI calculation and AI-based health risk prediction, requiring strict ethical adherence. Privacy and confidentiality are prioritized, ensuring all personal and physio

logical data, including vital signs, BMI, and predicted risks, are protected and used solely for research purposes. Participants are fully informed about the study and provide consent before any measurements. The system ensures accuracy and reliability, with the device intended only as a supportive health-monitoring tool, not a medical diagnosis. Physical safety is maintained through safe components, and AI outputs are presented in clear, neutral language to protect psychological well-being.

AI-specific ethical considerations include preventing algorithmic bias and ensuring fairness. Limitations of the AI model are acknowledged, and predictions are not overgeneralized. Transparency is upheld by explaining system operations, data usage, and prediction scope to participants. Data handling follows the Data Privacy Act of 2012 (RA 10173), with secure storage, limited retention, anonymization where possible, and deletion after research. The study complies with institutional ethical requirements, respects participants' dignity, emphasizes voluntary participation, and allows withdrawal at any time without penalty.

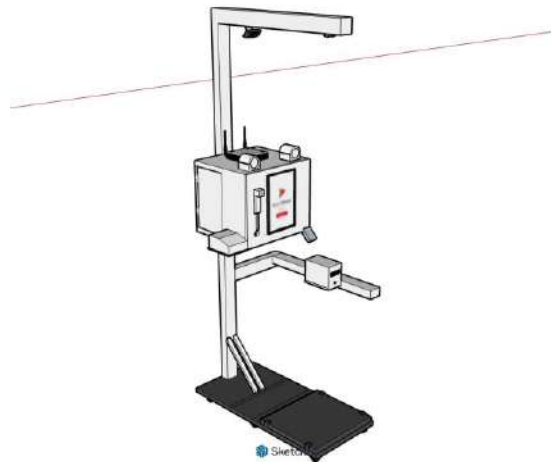
CHAPTER IV
THE PROJECT DESIGN

This chapter of the FOVB-AIoT system. It discusses the results related to the system's hardware and software implementation, including the accuracy and efficiency of vital sign measurements, AI object detection (BMI compliance and Extraction of result from Digital Blood Pressure), AI-based health risk prediction, and overall system performance. The chapter also presents system flow diagrams, technical specifications, and testing outcomes. Furthermore, it analyzes the evaluation results gathered from respondents using the ISO 25010 software quality model in terms of time consuming, reliability, efficiency, usability and data & security providing a comprehensive assessment of the system's effectiveness in improving the vital sign measurement process compared to traditional manual methods.

Project Overview

Figure 4.1

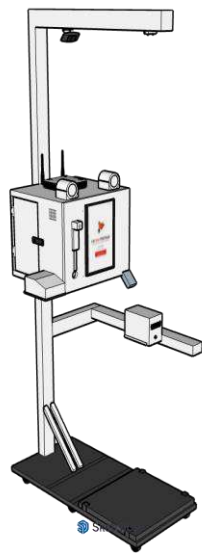
Project 3D Design



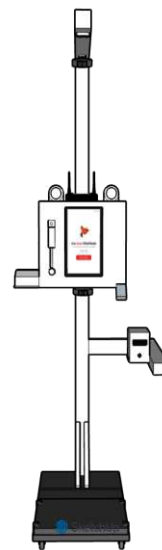
The FOVB-AIoT is a kiosk-based health monitoring system developed to automate the measurement and analysis of multiple vital signs in a fast and efficient manner. The system integrates various sensors to measure body temperature, heart rate, respiratory rate, blood pressure, oxygen saturation, height, and weight, enabling automatic Body Mass Index (BMI) computation. Collected data are processed using Artificial Intelligence through *Juan AI* to detect abnormalities and predict potential health risks. The results are displayed in real time on a touchscreen interface, securely stored in the UCARE database, and printed as a Personalized Results and Educational Protocol (PREP).

Figure 4.2

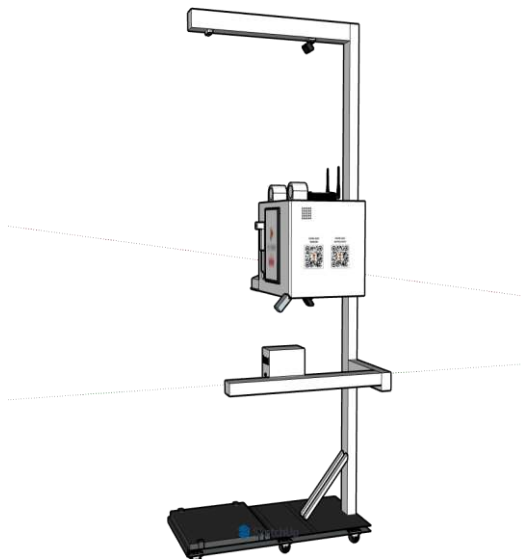
3D Design Layout



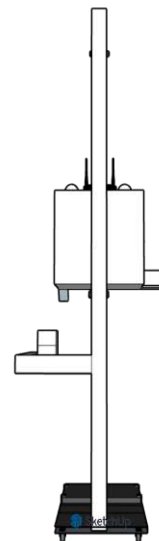
Isometric View



Front View



Side View



Back View

The 3D design layout with dimensions illustrates the physical structure and spatial configuration of the FOVB-AIoT kiosk. It presents the overall height, width, and depth of the system, ensuring that all components are ergonomically positioned for user accessibility and comfort. The design considers proper placement of the touchscreen display, sensors, height measurement module, and input interfaces to allow smooth user interaction during the vital sign measurement process. The dimensions are optimized to accommodate users of varying heights while maintaining stability, safety, and efficient use of space within the clinic environment. This layout serves as a reference for the actual construction of the kiosk, ensuring accurate assembly and alignment of all hardware components.

Figure 4.3

3D Design Layout with Dimensions

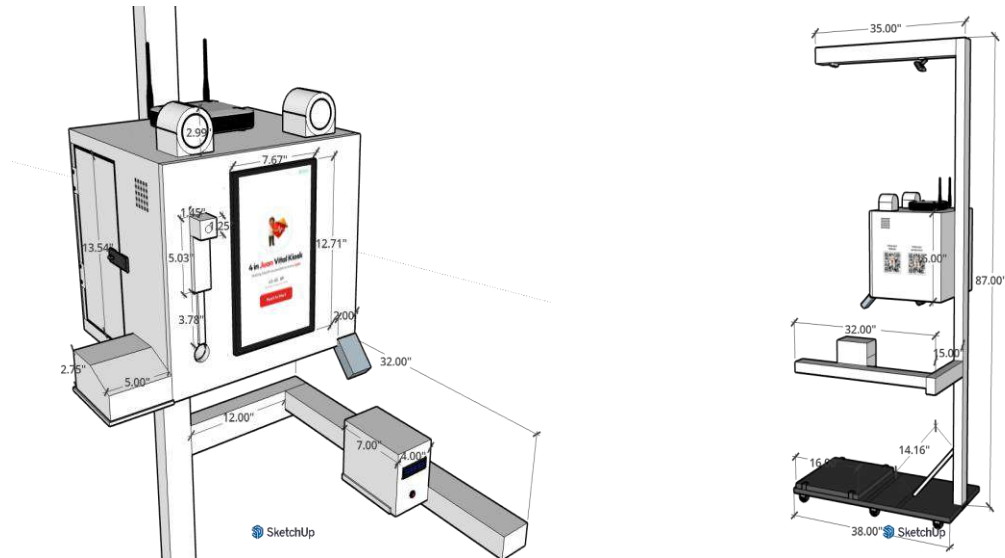


Figure 4.3 illustrates the 3D design layout of the FOVB-AIoT kiosk with specified dimensions, showing the overall height, width, and depth of the system. The figure highlights the proportional spacing and structural arrangement of the kiosk components to ensure stability, ergonomic user interaction, and efficient placement within the clinic environment. The indicated dimensions serve as a reference for accurate fabrication and assembly, ensuring that all hardware components are properly aligned and accessible during operation.

Figure 4.4

3D Design Layout with Labels

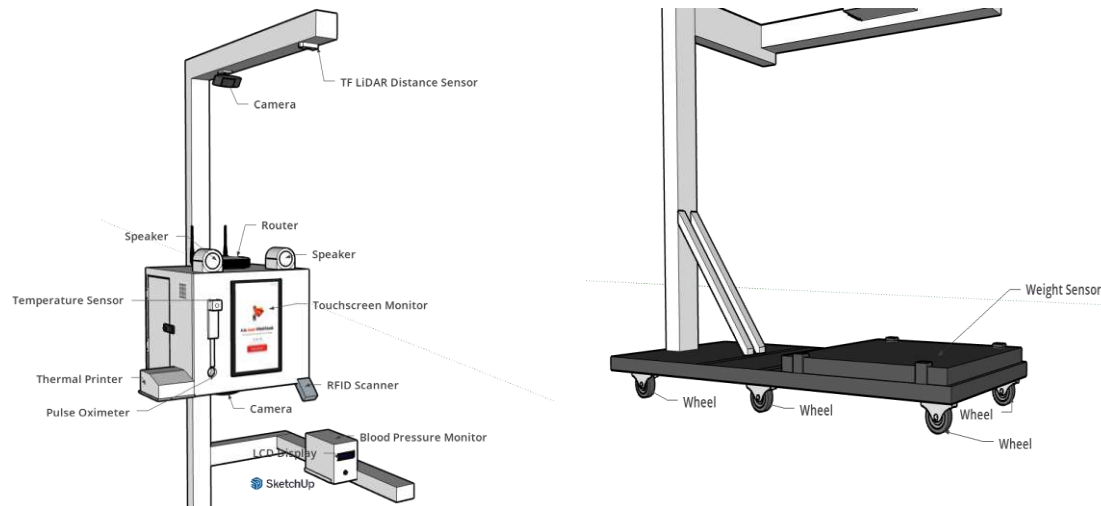


Figure 4.4 shows a 3D design layout of a mobile, integrated health monitoring system with clearly labeled components. The structure features an overhead frame equipped with a camera and TF LiDAR distance sensor for user detection, while the central unit contains devices such as a touchscreen monitor, temperature sensor, pulse oximeter, blood pressure monitor, RFID scanner, thermal printer, speakers, and a router for interaction and data processing. The base platform includes wheels for mobility and a weight sensor for measuring the user's weight, demonstrating a compact and multifunctional system designed for automated health screening.

Figure 4.5

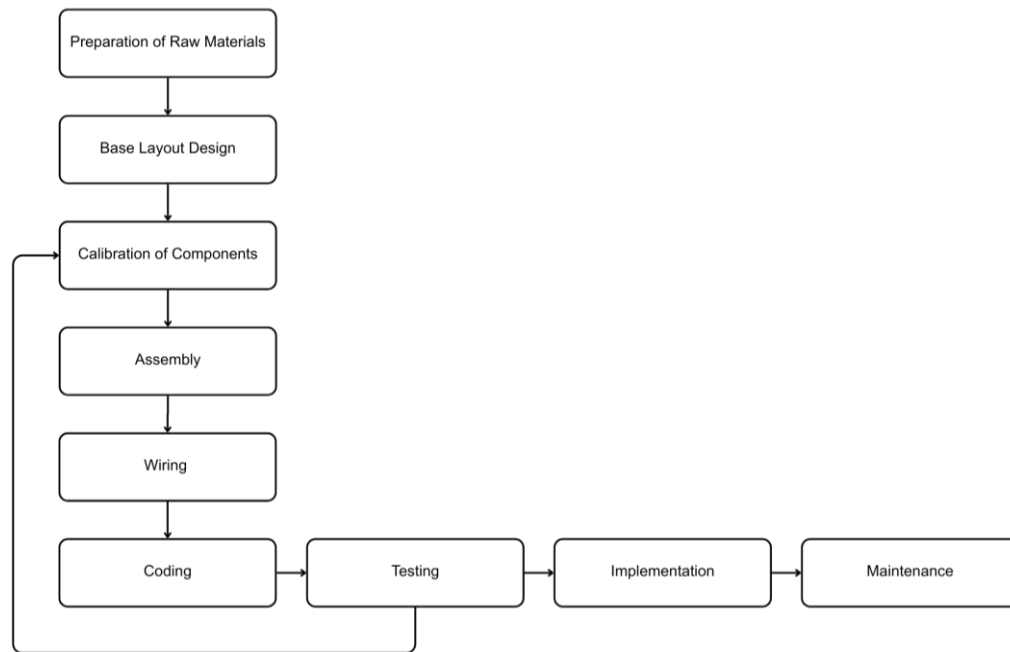
Actual Layout



Figure 4.5 shows the actual layout of the developed FOVB-AIoT kiosk after assembly and implementation. It presents the real physical appearance and placement of all hardware components, including the touchscreen display, sensor modules, RFID reader, and thermal printer. This figure demonstrates how the finalized system aligns with the proposed design layout, confirming proper installation, accessibility, and readiness for actual use in the clinic environment.

Figure 4.6

Process Diagram



Preparation of Raw Materials - This stage focused on gathering all the required hardware components such as sensors, microcontrollers, display units, electronic modules, and mechanical materials. The necessary software tools, libraries, and development environments were also installed to support system development.

Base Layout Design - During this phase, the overall physical and system layout of the kiosk was planned. The placement of sensors, display, wiring paths, and internal components was carefully arranged to ensure user comfort, accessibility, and smooth system operation.

Calibration of Components - Before assembling the system, each sensor and measurement module was individually calibrated. This was done to ensure accurate and reliable readings for temperature, pulse rate, SpO₂, blood pressure, height, and weight by comparing them with standard reference instruments.

Assembly - After calibration, all components were installed into the kiosk enclosure based on the finalized design layout. Proper alignment and secure mounting were ensured to maintain the stability and durability of the system.

Wiring - The electrical connections between sensors, microcontrollers, power supplies, and peripheral devices were carefully completed. Proper wiring was essential to ensure safe operation, accurate data transfer, and consistent system performance.

Coding - Software development was carried out to make the system functional. This included programming the microcontroller, developing the user interface, integrating AI models, setting up database connections, and configuring IoT communication.

Testing (Iterative Loop) - The system was tested multiple times to check its functionality, accuracy, and ease of use. When issues such as sensor inaccuracies, wiring problems, or software bugs were identified, the system was adjusted by returning to the necessary earlier stages. This process was repeated until the system performed as expected.

Implementation - Once testing was successfully completed, the FOVB-AIoT system was installed and used at the RTU Pasig Clinic for actual operation and user interaction.

Maintenance - After deployment, regular monitoring, recalibration, software updates, and hardware inspections were performed to ensure the system remained accurate, reliable, and efficient over time.

Figure 4.7

Block Diagram

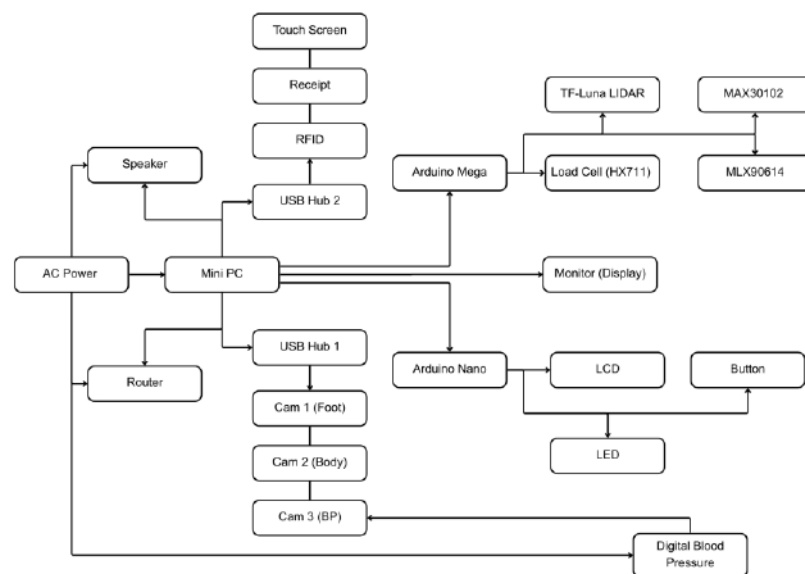
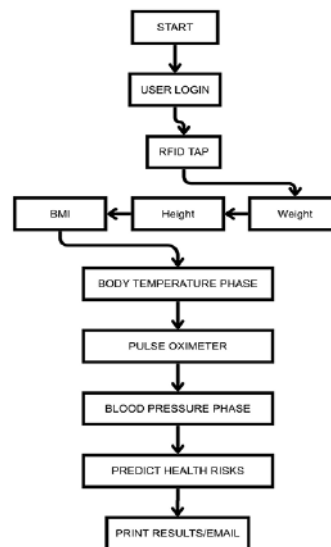


Figure 4.7 presents the block diagram of the FOVB-AIoT system, illustrating the overall interaction between hardware, software, and network components. The diagram shows how input data from various sensors including the temperature sensor, pulse and SpO₂ sensor, blood pressure sensor, load cell, and LiDAR height sensor are collected and processed by the Arduino Mega microcontroller. The processed data

are transmitted to the mini PC, where further computations, AI-based health risk prediction, and data management are performed. The system outputs are displayed through the touchscreen interface, printed via the thermal printer, and securely transmitted to the UCARE database through IoT connectivity for real-time monitoring and record storage. This block diagram provides a clear representation of the system's functional flow and data processing architecture.

Figure 4.8

Screen Transition Diagram

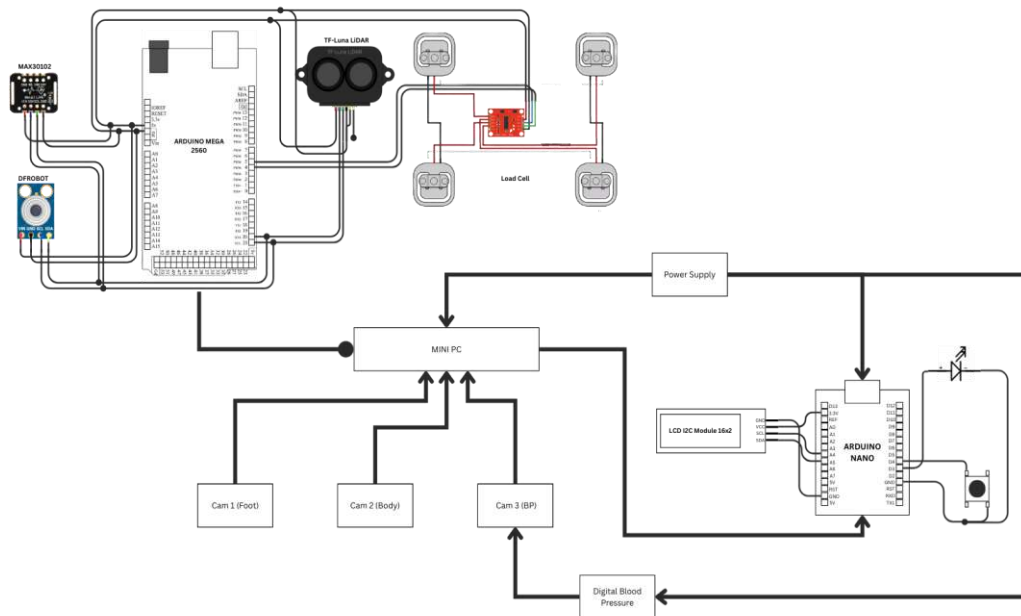


The Transition Diagram systematically outlines the operational flow of the FOVB-AIoT system from beginning to end, illustrating how a user progresses through the automated health assessment process. The cycle starts in the Standby state, waiting for a user to initiate the process. Upon interaction, the system moves into the

Login/Registration state, where the user is authenticated either through an RFID card tap or manual data input for new users. The process then transitions to a critical checkpoint, the Pre-Measurement Clearance state, utilizing the AI model to verify that the user has properly positioned themselves and removed all footwear or wearables that could compromise the accuracy of the sensors. Once clearance is granted, the system enters the Measurement phase to automatically capture all vital signs and Body Mass Index data. This collected health information is then passed to the AI Analysis state for the computation of the Comprehensive Risk Score and the generation of personalized recommendations. Finally, the flow reaches the Results Display state, where the user is presented with their assessment via email or a printed receipt, concluding the session as the system gracefully returns to the Standby state for the next use.

Figure 4.9

Schematic Diagram



The schematic diagram illustrates the detailed electrical connections and circuitry of the FOVB-AIoT system. It shows how the sensors, including the MAX30102 pulse sensor, MLX90614 temperature sensor, blood pressure module, load cell with HX711, LiDAR height sensor, and RFID module, are interfaced with the Arduino Mega microcontroller. The diagram also presents the power supply distribution, signal pathways, and communication interfaces used for data transmission between components. This schematic serves as a technical reference for wiring, troubleshooting, and system replication, ensuring proper electrical connectivity, safe operation, and reliable data acquisition throughout the system.

Figure 4.10

Flow Chart (Registration)

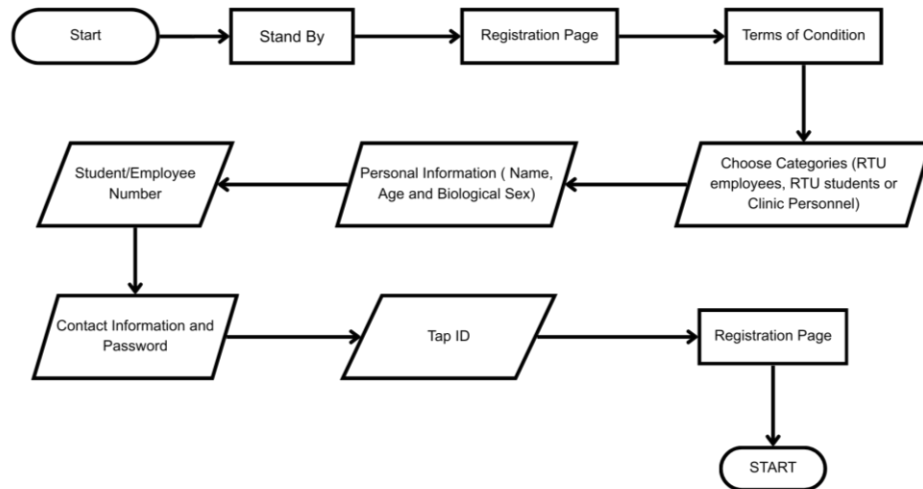


Figure 4.10 illustrates the system's registration workflow, which initiates from a standby state and directs users to the main page to accept the terms and conditions. Subsequently, registrants must select their specific user category such as RTU student, employee, or clinic personnel and input requisite data, including personal demographics, institutional identification numbers, and contact details with a secure password. The procedure finalizes when the user taps their physical ID for hardware authentication, prompting the system to loop back to the initial start state to complete the registration cycle.

Figure 4.11

Flow Chart (Log In)

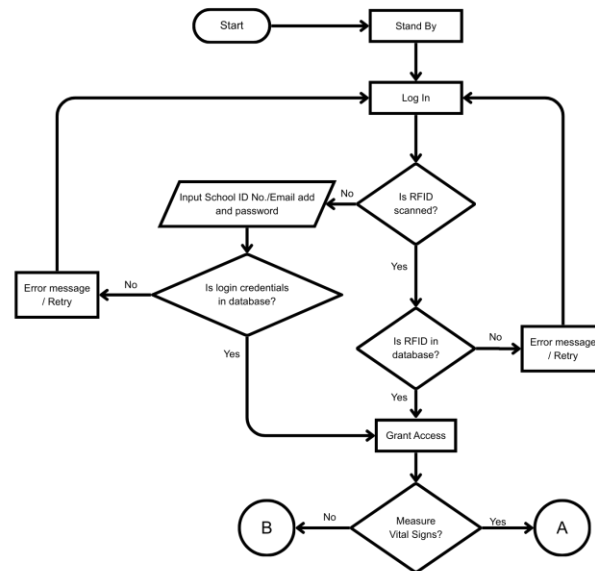


Figure 4.11 shows the steps for the system's login process, which gives users two options: scanning their RFID card or manually entering their School ID or email along with their password. If the system matches the scanned RFID or the typed credentials with the database, it successfully grants access. However, if the RFID is not registered or the entered details are incorrect, the system displays an error message and returns the user to the login screen to try again.

Flow Chart (Measurement)

This section presents the overall measurement workflow of the system, which includes both the All Vital Signs and Specific Vital Signs flow charts. It illustrates the step-by-step process of conducting health measurements, from system preparation

and user guidance to the actual recording of vital signs and validation of results. The flow charts demonstrate how the system ensures accuracy and proper sequencing whether the user selects a complete vital sign assessment or measures specific parameters only.

Figure 4.12

Flow Chart (Measurement with All Vital Signs)

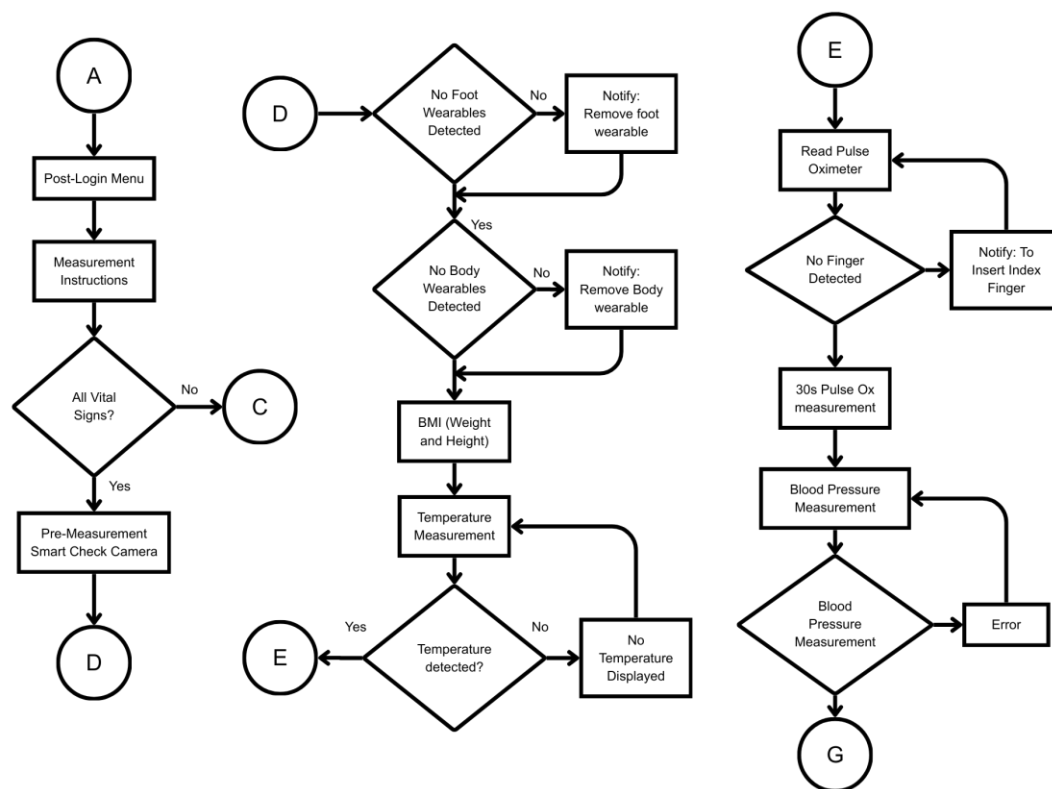


Figure 4.12 presents the complete workflow for measuring all vital signs within the system. After login and measurement instructions, the system performs pre-

checks such as detecting wearables and ensuring proper setup before proceeding with BMI, temperature, pulse oximeter, and blood pressure measurements. Notifications are provided when errors or incorrect placements are detected, and the process ends once all vital signs are successfully recorded.

Figure 4.13

Flow Chart (Measurement with Specific Vital Signs)

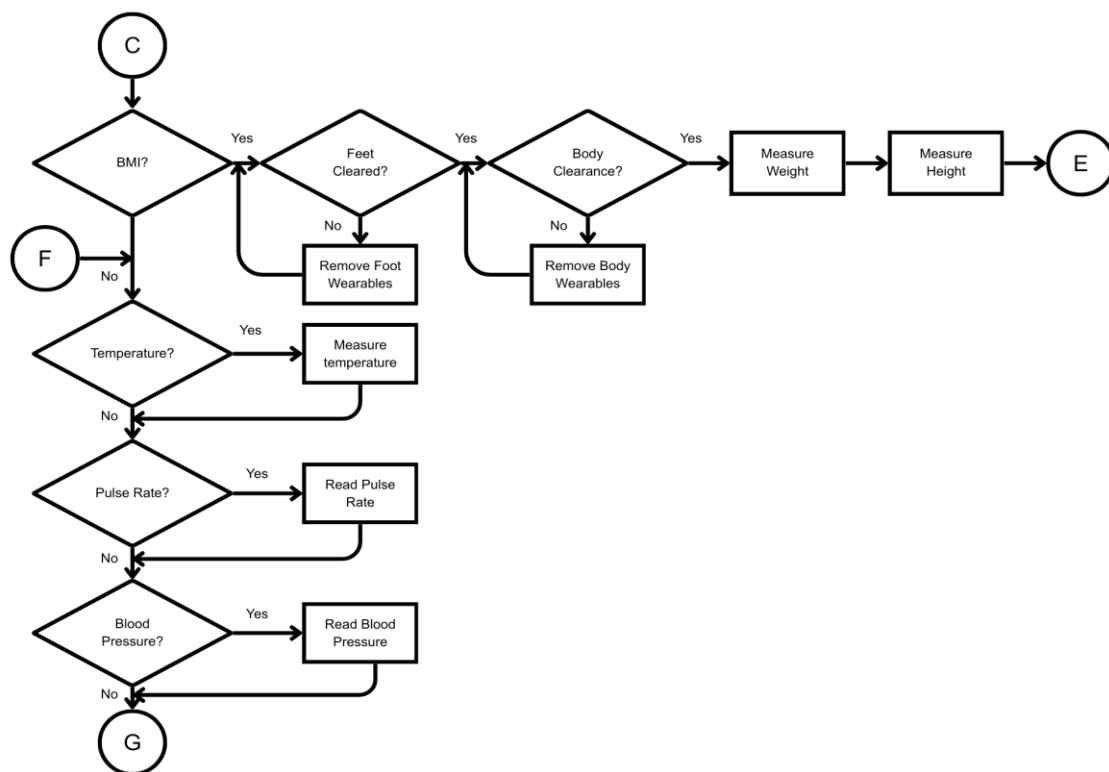


Figure 4.13 upon successful login, the user accesses the main menu and selects either all vital signs or specific parameters for measurement. The system provides instructions to remove footwear and unnecessary body wearables and to stand correctly within the camera's view. An AI based pre-measurement clearance

check is conducted in a loop to detect any footwear or obstructive wearables; if detected, the user is notified and the verification process repeats until clearance is achieved. Once approved, the system measures the selected vital signs height, weight, temperature, pulse rate, and blood pressure and subsequently performs AI-based analysis to generate a health risk prediction.

Figure 4.14

Flow Chart (Final Results)

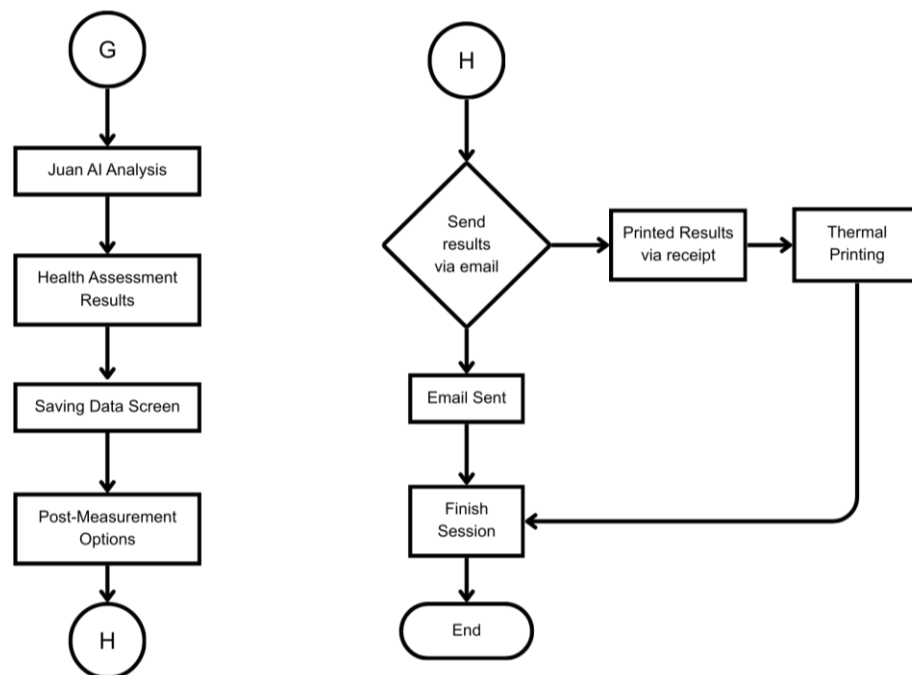


Figure 4.14 presents the final results workflow of the system, beginning with the Juan AI Analysis and the generation of Health Assessment Results. After the results are produced, the system saves the data and displays the post-measurement

options for the user. The user can then choose to receive the results either via email or through a printed receipt using thermal printing, after which the session is finalized and the process ends.

Flow Chart (Dashboards)

This section presents the workflow of the system dashboards, covering the navigation and functionalities available to different user roles such as admin, doctor, nurse, and patient. It illustrates how users access their respective overview pages and navigate through modules including patient management, health overview, measurement history, and account settings. The flow chart highlights the structured flow of dashboard interactions from login to session termination.

Figure 4.15

Flow Chart (Admin)

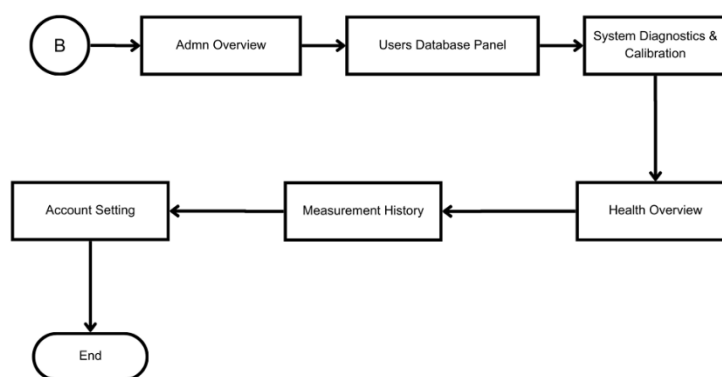


Figure 4.15 presents the workflow of the Admin Dashboard, starting from system access at Start and proceeding to the Doctor Overview page. The admin can

then navigate to Assigned Patients / All Users and Health Overview to monitor patient information and assessment results. The process also includes reviewing Measurement History and managing Account Setting, before ending the session at End.

Figure 4.16

Flow Chart (Doctor)

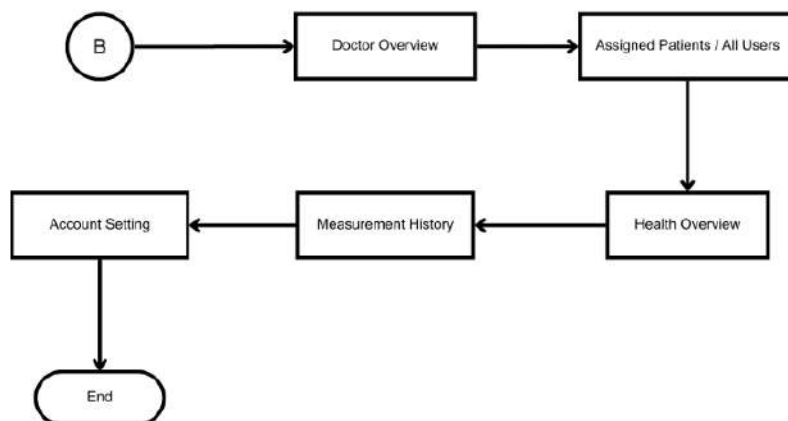


Figure 4.16 presents the workflow of the Doctor module, beginning at Start and proceeding to the Doctor Overview dashboard. The doctor navigates through Assigned Patients / All Users, Health Overview, and Measurement History to review patient records and assessment data. The process then moves to Account Setting for profile management before ending the session at End.

Figure 4.17

Flow Chart (Nurse)

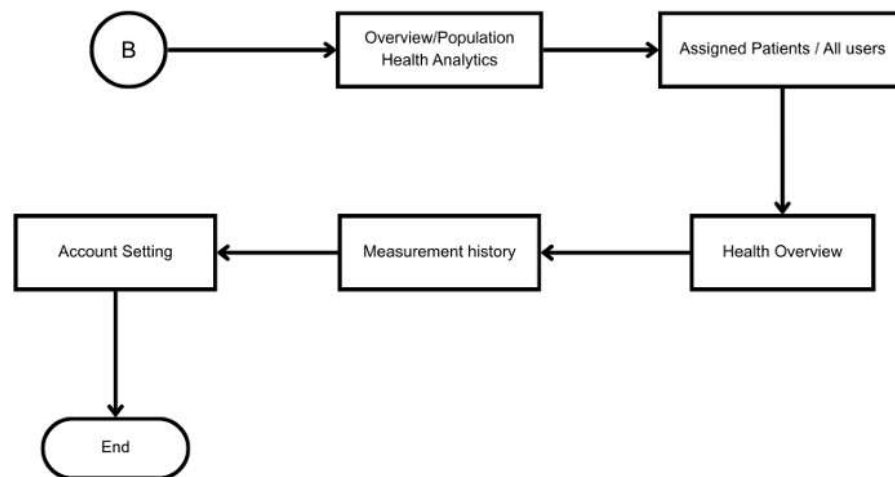


Figure 4.17 presents the workflow of the Nurse module, beginning at Start and proceeding to the Nurse Overview / Population Health Analytics dashboard. The nurse navigates through Assigned Patients / All Users, Health Overview, and Measurement History to review patient information and vital sign records. The process then moves to Account Setting for profile management before ending the session at End.

Figure 4.18

Flow Chart (Patient)

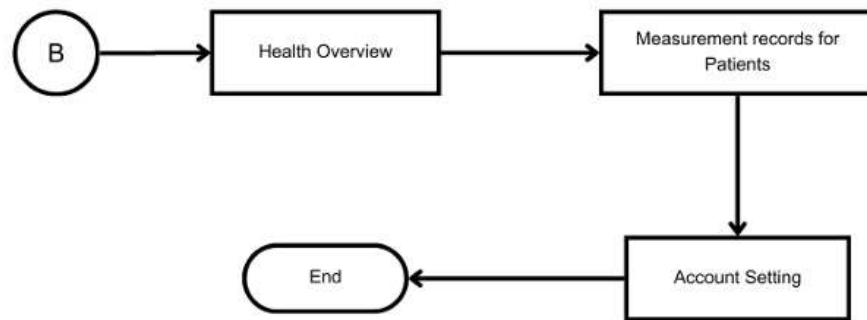
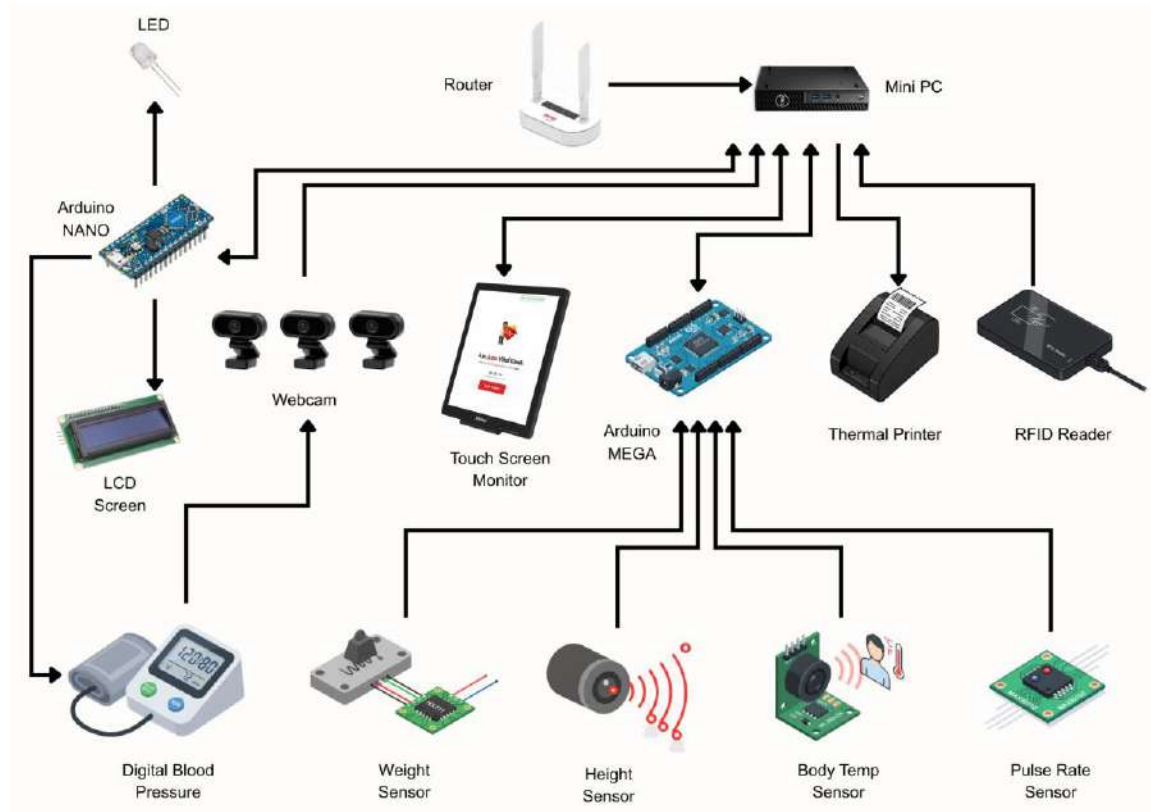


Figure 4.18 presents the workflow of the Patient/User module, beginning at Start and proceeding to the Patient/User Health Overview dashboard. The user can navigate to Measurement History to review previous vital sign records and assessment results. The process then continues to Account Setting for profile management before ending the session at End.

Figure 4.19

System Architecture

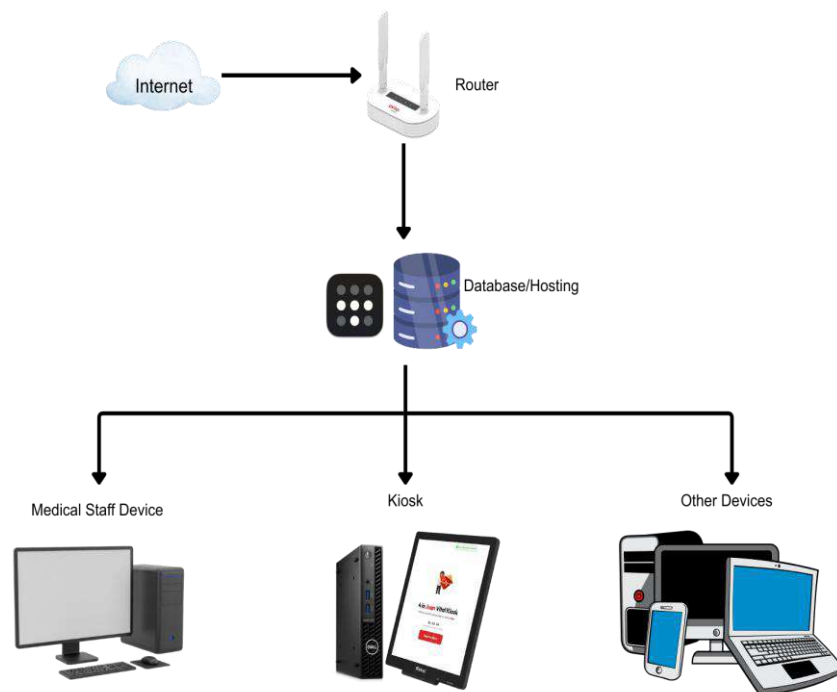


The system architecture of the FOVB-AIoT defines the structured interaction between hardware components, software modules, and network services that enable automated health assessment. The architecture is composed of three main layers: the sensing layer, processing layer, and application layer. The sensing layer consists of multiple biomedical and physical sensors connected to the Arduino Mega for real-time data acquisition. The processing layer includes the mini-PC, which handles data processing, AI-based health risk prediction through *Juan AI*, and system control using

Python and Flask. The application layer provides a web-based user interface developed with React JS and facilitates secure data storage and transmission through the UCARE database using IoT connectivity. This layered architecture ensures efficient data flow, scalability, reliability, and secure access to health information for both users and authorized clinic personnel.

Figure 4.20

Network Architecture



The network architecture of the FOVB AIoT system securely transmits sensor data from the Arduino Mega to a local minicomputer for processing. The Flask backend then analyzes this data and artificial intelligence predictions before sending

it to the UCARE database. To support distinct clinic functions, the architecture operates in a standalone kiosk mode for continuous patient measurements and a remote mode for administration. These two variations are securely bridged over the internet using Tailscale. This encrypted virtual private network overlay allows all users to access the web application from outside locations, with permissions and health record visibility strictly determined by a role based access control system. This ensures reliable data exchange, centralized management, and privacy compliance without exposing the local network to public vulnerabilities.

AI Models

Powering the FOVB-AIoT system is an integrated AI framework communicating via API to handle data collection and health risk assessment. For data acquisition, a YOLOv11 vision model trained via Roboflow and Google Colab extracts blood pressure readings from digital monitors while simultaneously checking for strict BMI compliance, detecting if the user is wearing footwear or heavy clothing that could skew measurements. After securing these vital signs, the data routes to the Juan AI engine, which maximizes clinical reliability through a multi-agent consensus architecture. XGBoost acts as the core predictive model here, calculating an initial risk score based on numerical deviations from baseline health metrics. Because the kiosk operates on a resource-constrained Windows Mini PC, running massive language models locally is unfeasible due to hardware bottlenecks [47]. To address this, the system utilizes an edge-cloud collaborative architecture for its secondary validation layer. A lightweight

GPT Open-Source Software (GPT OSS) instance runs locally at the edge, providing rapid, secure, on-device validation that protects patient privacy [48]. Meanwhile, the computationally heavy semantic reasoning is offloaded to a Gemini LLM in the cloud via API [49]. Running parallel analyses, these models generate their own risk probabilities so the system can cross-reference the outputs from XGBoost, the edge-based GPT OSS, and the cloud-based Gemini to establish a mathematically verified consensus score [41]. This triangulation effectively filters out biases and potential AI hallucinations before Gemini formats the verified consensus into safe, readable preventative recommendations to guarantee the final health report is medically sound [40].

To guarantee the clinical validity of this finalized data, the system's language models undergo a strict clinical alignment process before communicating with the user. Instead of allowing the AI to generate arbitrary health responses, the reporting engine is constrained by an Expert-in-the-Loop (EITL) framework [43]. Through explicitly designed system prompts based on established healthcare guidelines, the artificial intelligence is essentially trained to evaluate and output the consensus risk scores exactly as a human physician would expect [42]. This clinical alignment ensures that the generated health reports mirror the diagnostic logic, tone, and ethical boundaries of a medical professional, providing accurate preventative insights while strictly avoiding unauthorized medical prescriptions [44].

Concept of Operation

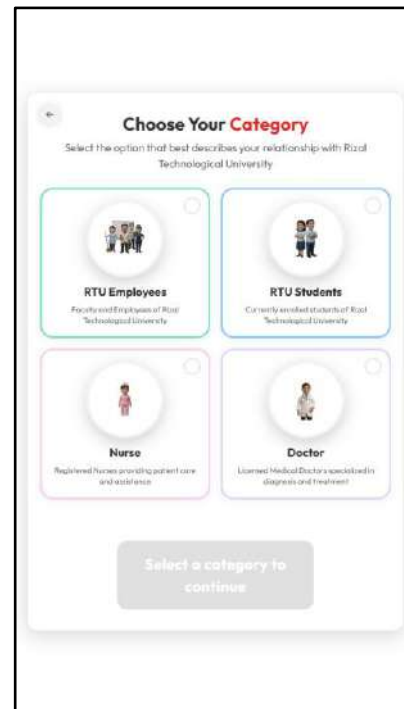
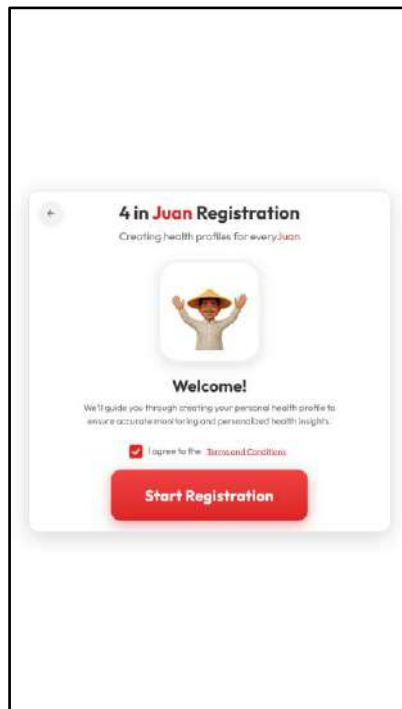
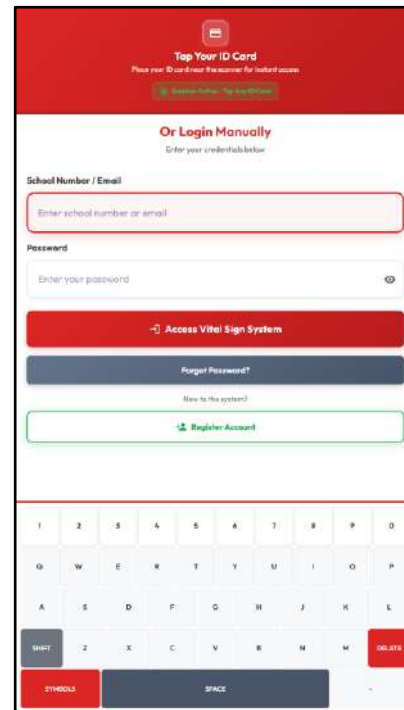
The **FOVB-AIoT system** operates as an automated, kiosk-based health assessment platform designed to provide efficient and guided health screening. Users begin by registering through a touchscreen interface by entering their personal information or tapping their RFID card if already registered, while a voice assistant guides them throughout the process. Before measurements begin, the system performs a pre-clearance check using AI-assisted cameras to detect footwear and body accessories such as bags, watches, and other wearables that may affect measurement accuracy. Once cleared, the system proceeds to the automated health screening phase where IoT-based sensors measure height, weight, body temperature, heart rate, respiratory rate, oxygen saturation, and blood pressure, with the Body Mass Index (BMI) computed automatically. All collected data are processed in real time and stored in the UCARE database, where Juan AI analyzes the readings to identify abnormalities and generate predictive health risk insights. The results are displayed on the touchscreen monitor and generated as a Personalized Results and Educational Protocol (PREP), which can be printed or sent via email, while healthcare personnel at the RTU Clinic can access the records through a monitoring dashboard for efficient review and timely health intervention.

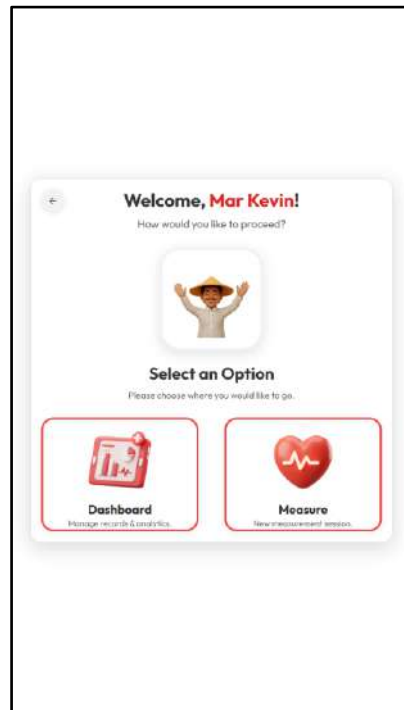
Figure 4.21

Concept of Operation



Kiosk Application Interface





1 Name 2 Age 3 Sex

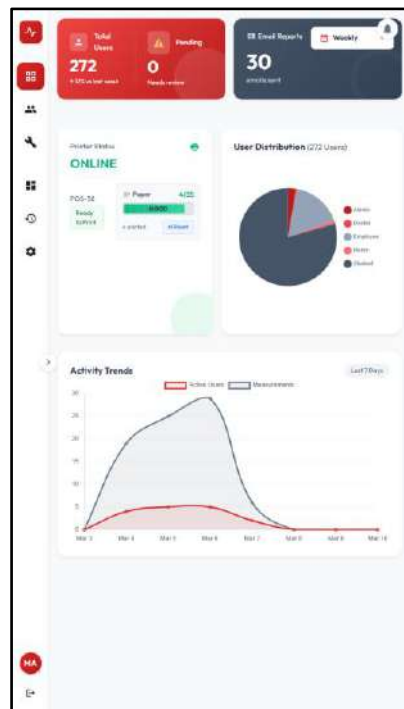
What's your name?
Great to have you here! Let's start with your name.

First Name: Juan Middle Name: Santos

Last Name: Dela Cruz Suffix: None

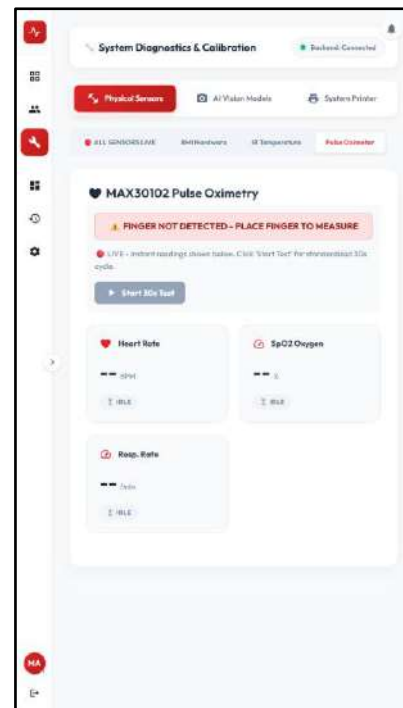
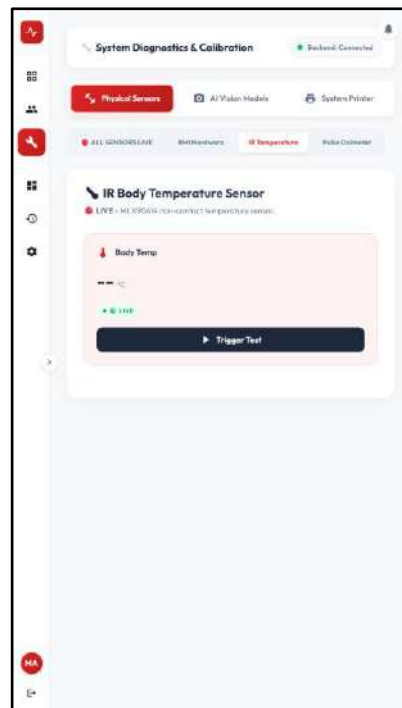
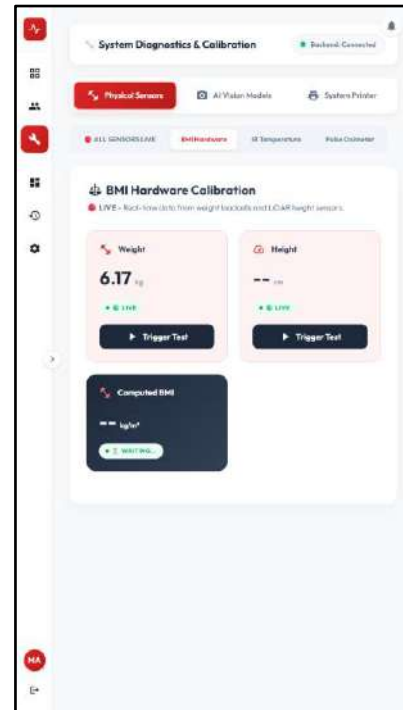
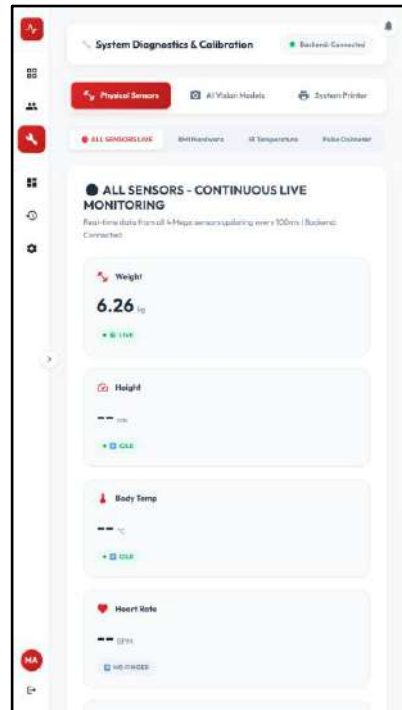
Continue

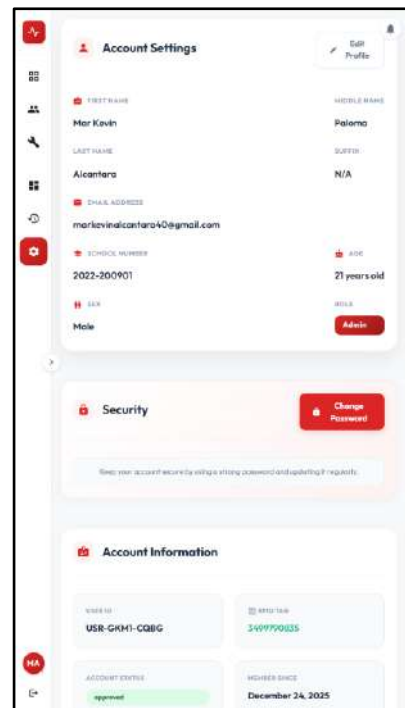
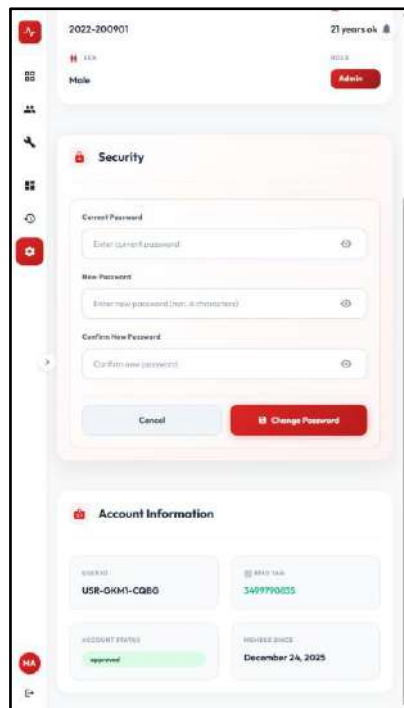
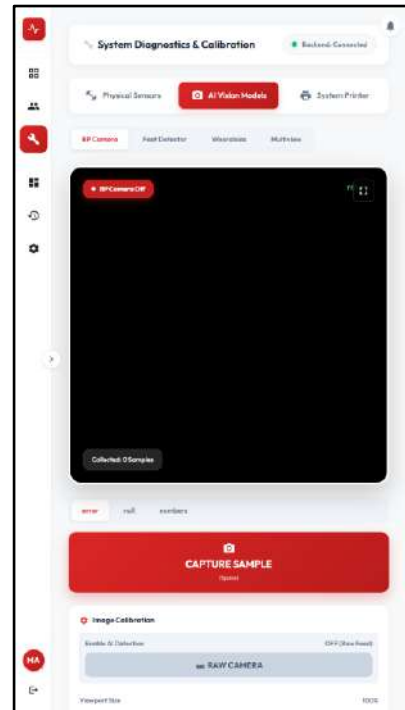
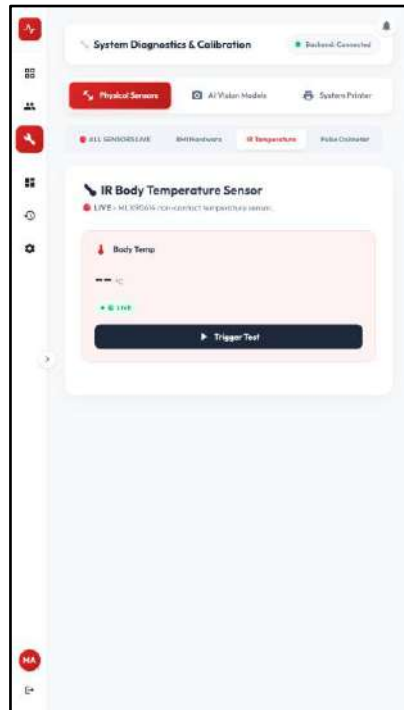
Q W E R T Y U I O P
A S D F G H J K L
SHIFT Z X C V B N M DELETE
SPACE

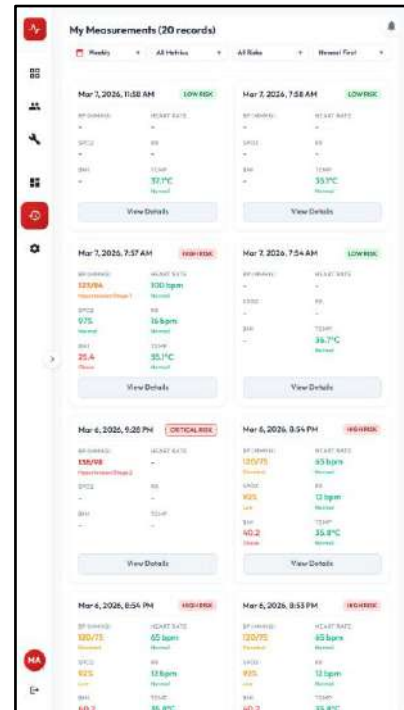
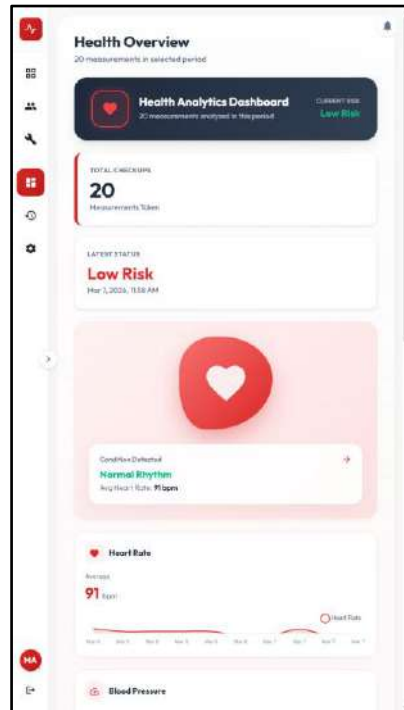


Registered Users Database (272 records)

MA	Mar Kevin Alcantara	Admin	3937-308961	10/06/2020	markevin@cantara10@gmail.com	Approved
NT	Nurse Test	Nurse	1	10/06/2020	Nurse@gmail.com	Approved
DT	Doctor Test	Doctor	2	10/06/2020	Doctor@gmail.com	Approved
ST	Student Test	Student	2	10/09/2020	Student@rtu.edu.ph	Approved







Ready to Begin!
Let's start gathering your vital signs.

NAME: Mer Kevin Alcantara | AGE: 21 years old | AKA DESIGN: Young Adult (18-24) | SEX: Male

ROLE: Admin

Relax & Stay Still
Breathe normally and keep calm during the process.

Ask for Assistance
Our medical staff is here to help if you need anything.

Remove Accessories
Please remove watches, caps, shoes, all socks, and other accessories.

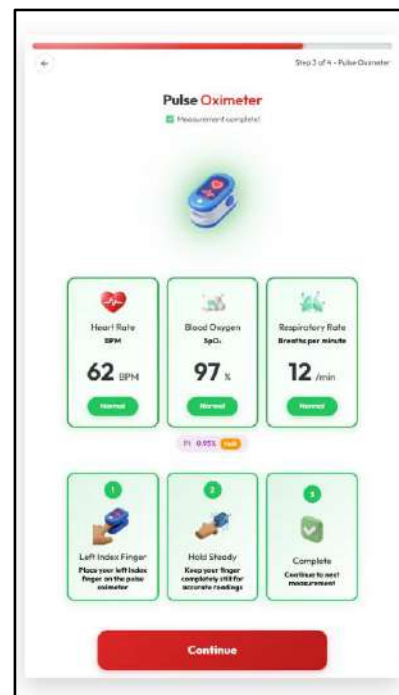
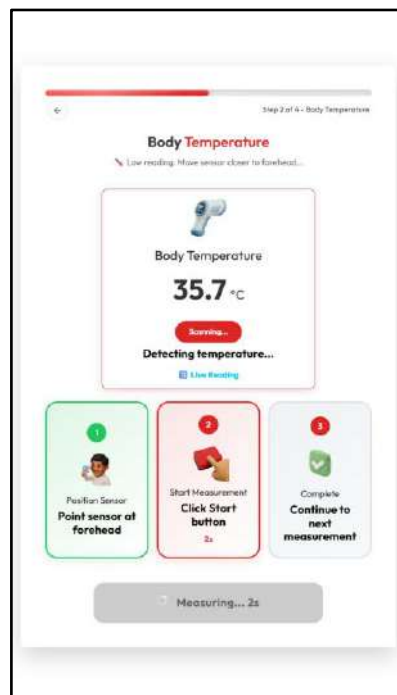
Start Measurements

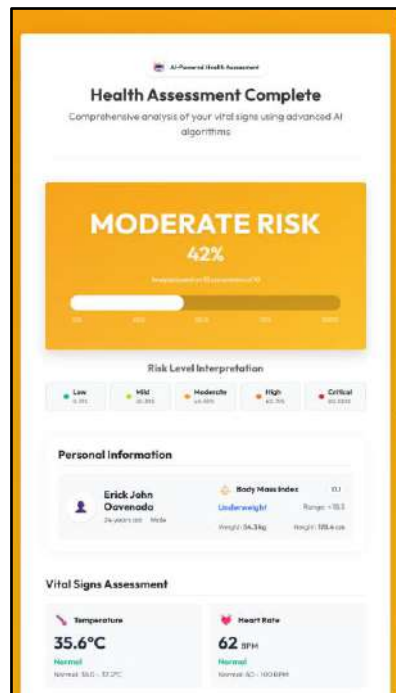
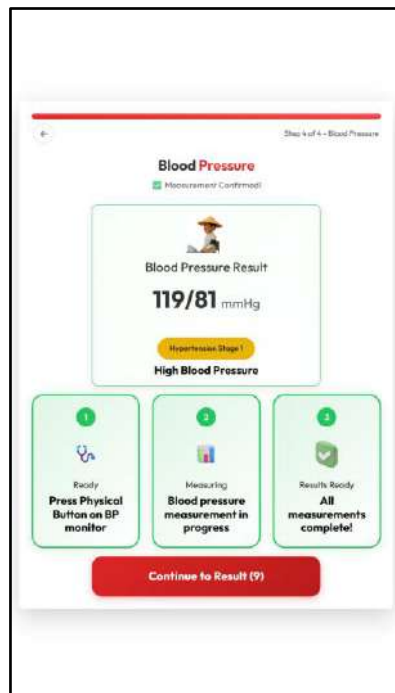
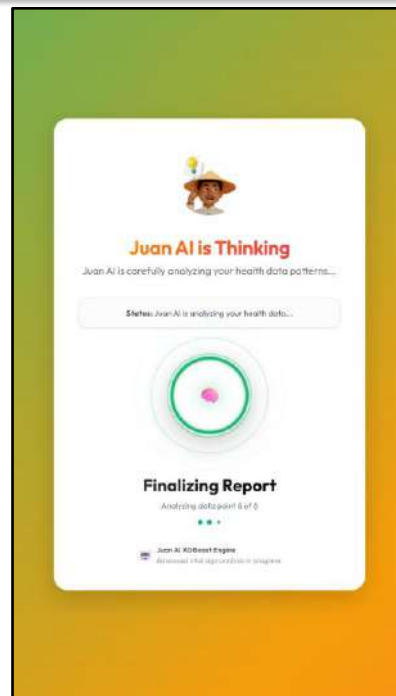
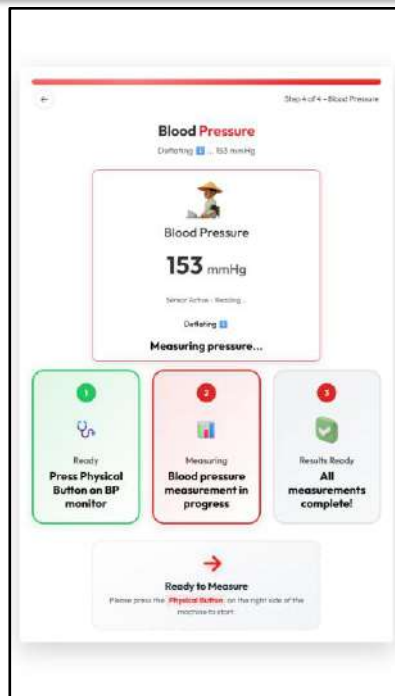
Measurement Checklist
Select the vital signs you want to measure for this session.

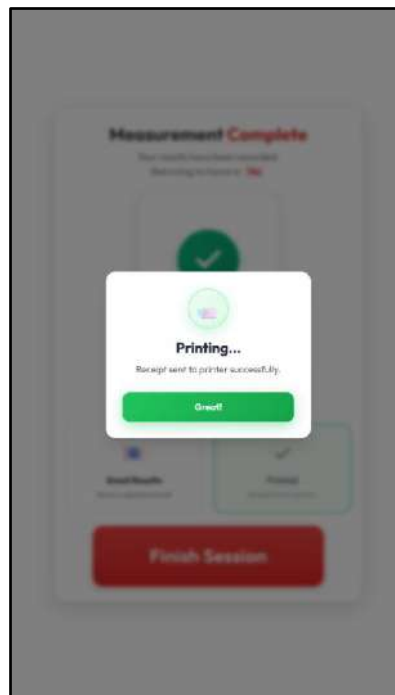
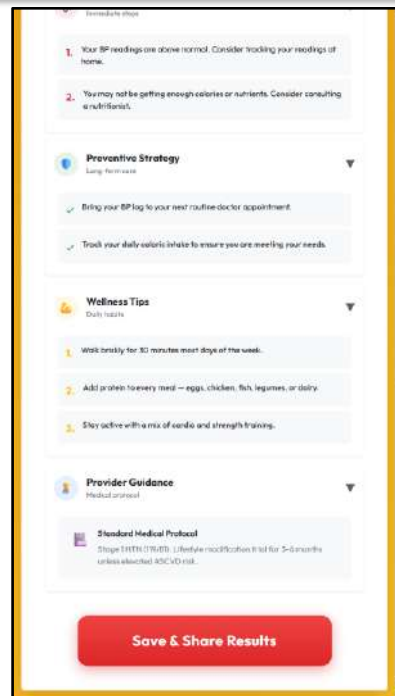
- ☒ **BMI** (Height & Weight)
- ☒ **Body Temperature** (Forehead Sensor)
- ☒ **Pulse Oximeter** (Heart Rate & SpO2)
- ☒ **Blood Pressure** (Cuffless & Oscillometric)

Done (4)

Start Measurement (4)







21 PM

PATIENT INFORMATION
Name: Etos John Gonsales
Age: 24 years
Gender: Male
BMI: 17.1 (Underweight)

ACTUAL STONE
Body Temp: 36.6 C (Normal)
Heart Rate: 67 BPM (Normal)
Systolic: 115 (Normal)
Diastolic: 75 (Normal)
Resp. Rate: 17/min (Normal)
BP: 115/75 mmHg (Hypertension 1)
Age: 24
Height: 54.3 kg
Weight: 170.4 cm

AI RISK ASSESSMENT
Risk Level: 45%
Category: Moderate Risk

DIAGNOSTIC ACTIONS
- Your BP readings are above our normal. Consider tracking your read tags of read.
- You may not be getting enough calories or nutrients. Consider consulting a nutritionist.

PREVENTIVE STRATEGIES
- Bring your BP tag to your next visit for doctor adjustment.
- Track your daily calorie intake to ensure you are meeting your needs.

WELLNESS TIPS
- Walk briskly for 30 minutes 3-4 days of the week.
- Add protein to every meal: egg, chicken, fish, beans, or dairy.
- Stay active with a mix of cardio and strength training.

PROVIDER GUIDANCE
Stage 1 HTN (115/75). Lifestyle modification trial for 3-6 months if no elevated ASCVD risk. Underweight (BMI 17.1). Screen for malnutrition, thyroid disorders, malabsorption, GIC, albumin, LDL.

*** IMPORTANT DISCLAIMER ***
This AI health assessment is for informational purposes only. Consult a doctor for medical advice.

Generated by Juan AI
Report ID: MEAS-20250310-6W63U

Health Report
March 01, 2025 - 12:30 PM

Hi Etos, Bonnie

Here are your results from your recent session at the Vital Signs Kiosk. Keep this record for your personal health tracking.

Overall Health Status: GREEN (25%)

HEALTHY APP
20.6 mmHg
WAGE 11.5 kg WAGE 170.4 cm

HEALTHY APP
115/67 mmHg

HEALTHY APP
92 bpm

HEALTHY APP
15 bpm

HEALTHY APP
97%

HEALTHY APP
36.2%

Overall Analysis & Recommendations

Medical Recommendations
- Your blood pressure is mildly elevated. Monitor it regularly over the coming weeks.

Preventive Strategy
- Check your blood pressure to determine a trend at the same time.

Wellness Tips
- Reduce salt in your cooking and avoid eating extra salt at the table.
- Build healthy habits now—long-term health benefits for lifelong health.

Provider Outlook
Blood pressure (BP) (115/67). Follow-up in 3 months with lifestyle intervention. No pharmacotherapy needed at this time.

Disclaimer: This automated report is for informational purposes only. It is not an official medical diagnosis. Please consult a qualified healthcare provider for professional medical advice.

1 Name 2 Age 3 Sex

What's your name?
Start by having your name. Let's start with your name.

First Name Middle Name
Ezekiel Santos

Last Name Suffix
Nequit None

Continue →

Q W E R T Y U I O P
A S D F G H J K L
SHIFT Z X C V B N M DELETE
SPACE

1 Name 2 Age 3 Sex

How old are you?
Select your birthday to calculate your age.

Select Your Birthday
Scroll to select

Month Day Year
January 1 2005
February 2 2002
March 3 2001
April 4 2000
May 5 1999

Your Age: 25
Born on March 3, 2001

Continue →

← 1 2 3
Name Age Sex

What is your biological sex?
Required for accurate health assessments

Biological Sex

Male

Female

We ask for your biological sex to provide accurate BMI calculations and health assessments. This information will not be shared publicly.

Continue to Account Setup

← 1 2 3
Name Age Sex

What is your biological sex?
Required for accurate health assessments

Biological Sex

Male

Female

We ask for your biological sex to provide accurate BMI calculations and health assessments. This information will not be shared publicly.

Continue to Account Setup →

← 4 5 6
Student ID Contact ID Top

Enter Student Number
Your official RTU student identification number

Student Number

2022-200885

Numbers and hyphens only (e.g., 2002-2009010)

* Your student number will be used for official identification and academic records.

Continue →

1 2 3 4 5 6 7 8 9 0
Q W E R T Y U I O P
A S D F G H J K L
SHIFT Z X C V B N M DELITE
ENTER SPACE

← 4 5 6
Student ID Contact ID Top

Contact Information
We'll use these for important updates and account recovery

Email Address

ezekielnequit@rtu.edu.ph

gmail.com outlook.com yahoo.com yahoo.com gmail.com

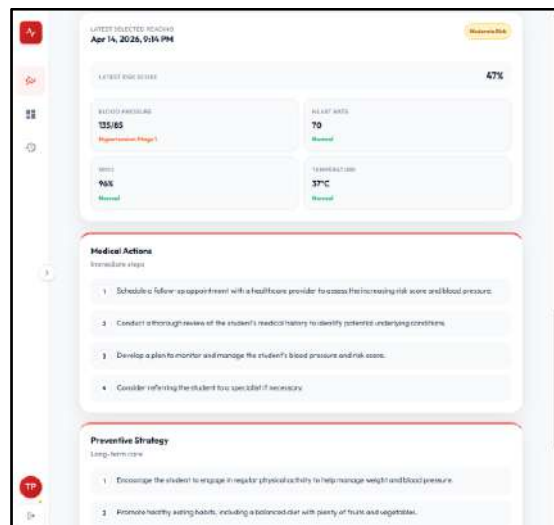
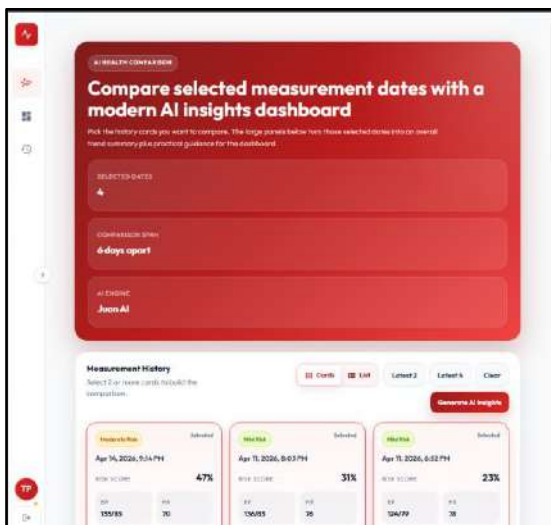
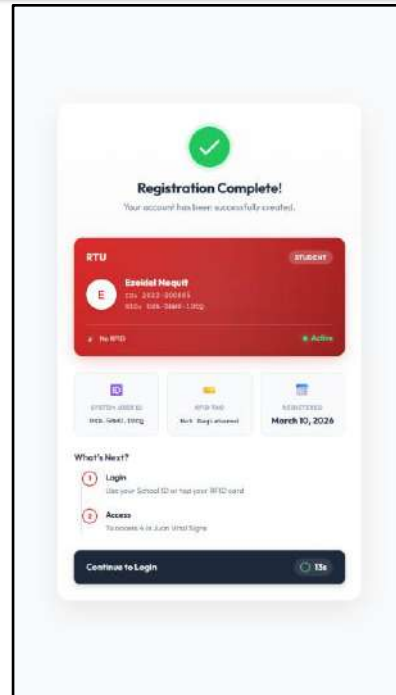
Create Password

4-10 characters

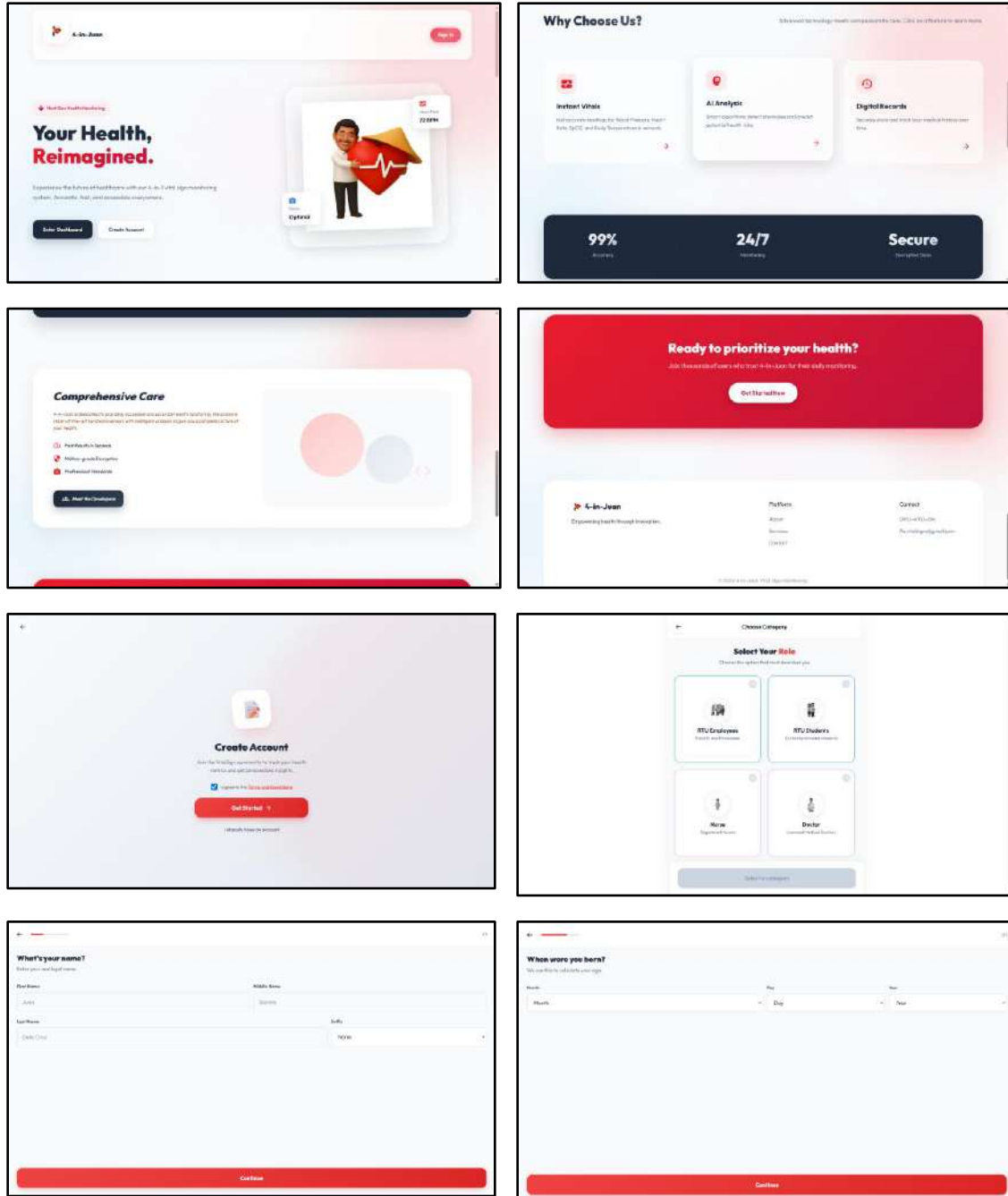
We'll use these details to send you important health updates and account recovery information.

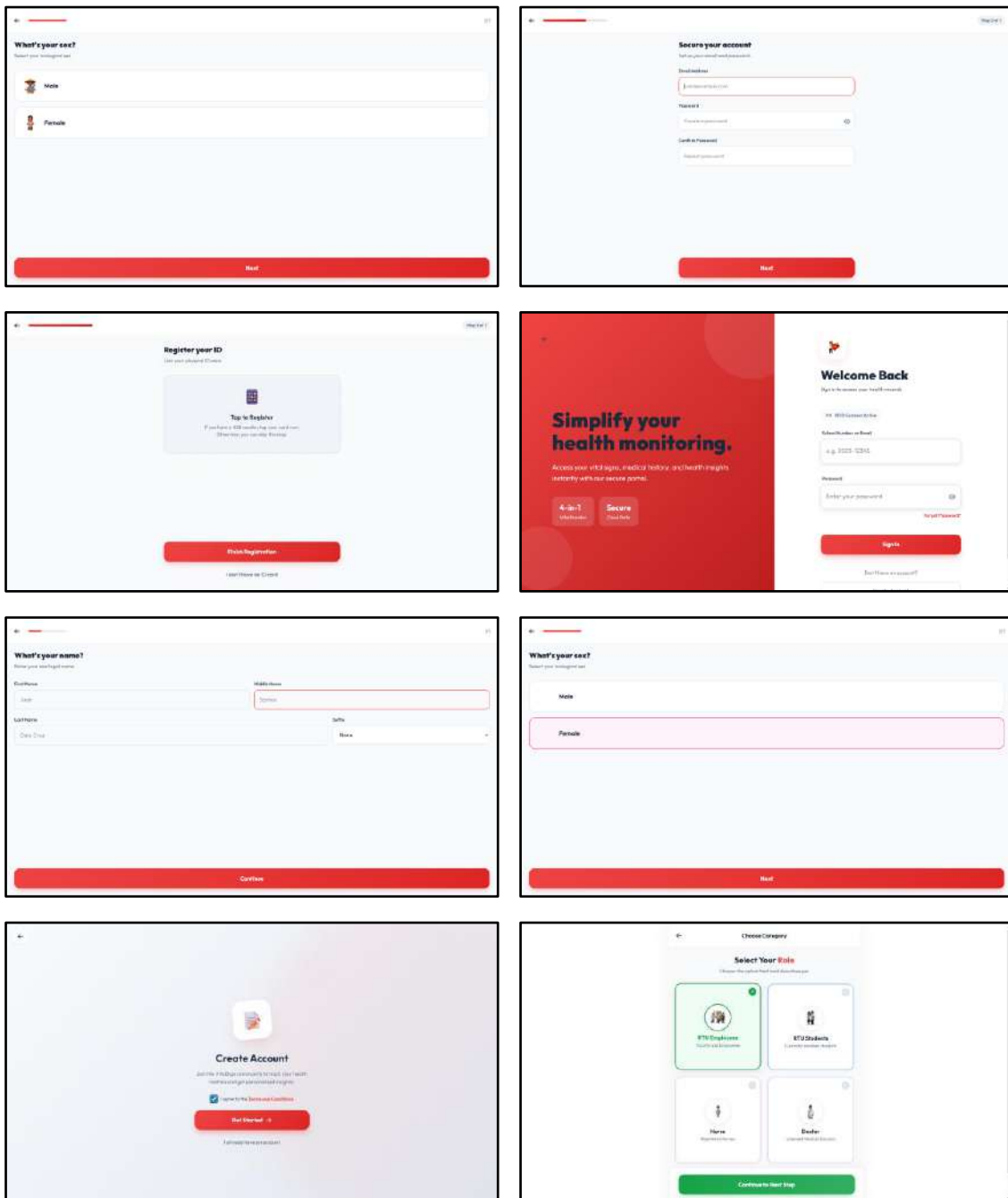
Continue →

1 2 3 4 5 6 7 8 9 0
Q W E R T Y U I O P
A S D F G H J K L
SHIFT Z X C V B N M DELITE
ENTER SPACE



Web Application Interface





What were you born?

My health is important to me

Month: Day: Year:

Next

What's your sex?

My health is important to me

Male

Female

Next

Secure your account

Set up your account with password

Email Address:

Password:

Confirm Password:

Next

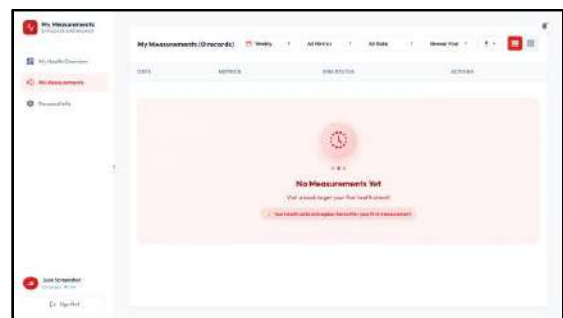
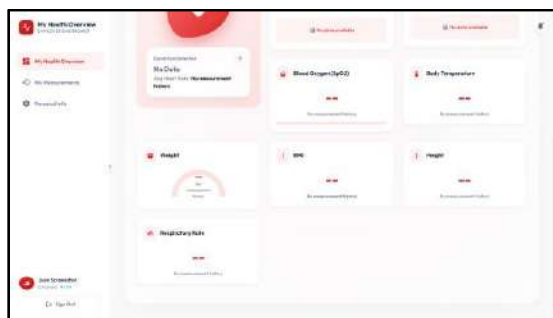
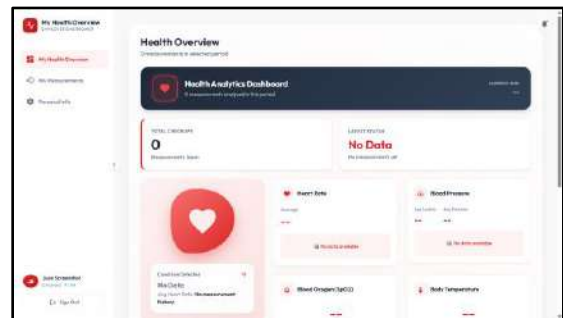
Register your ID

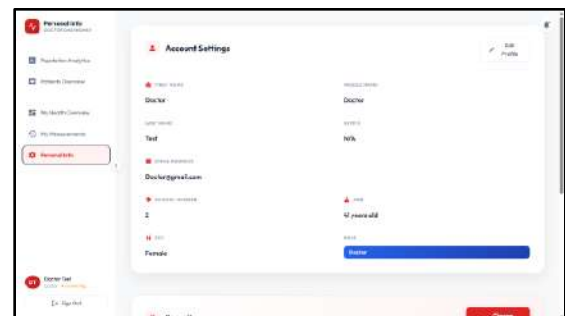
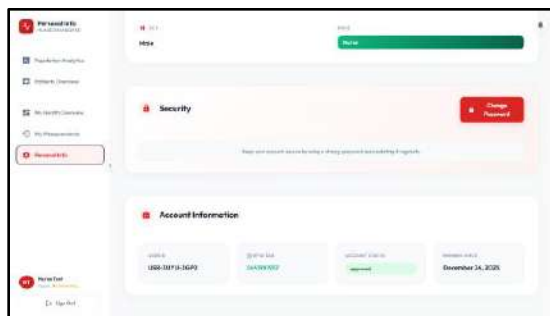
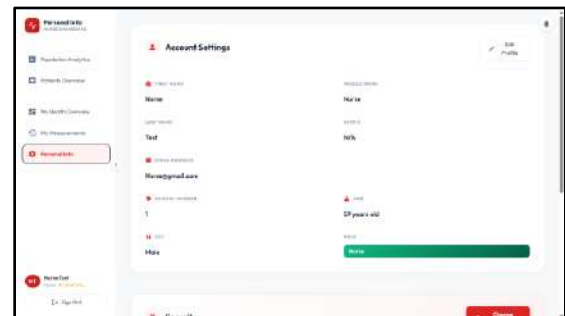
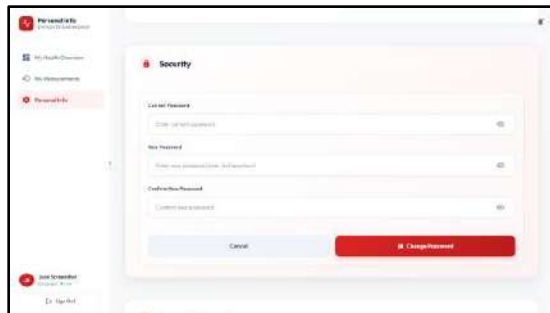
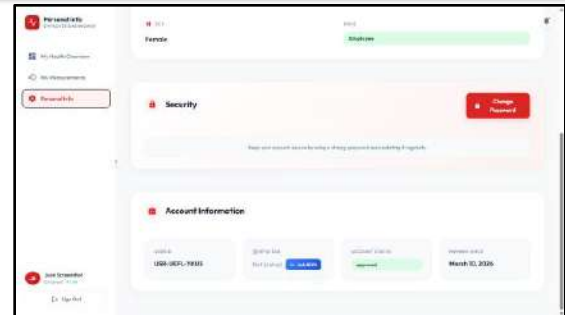
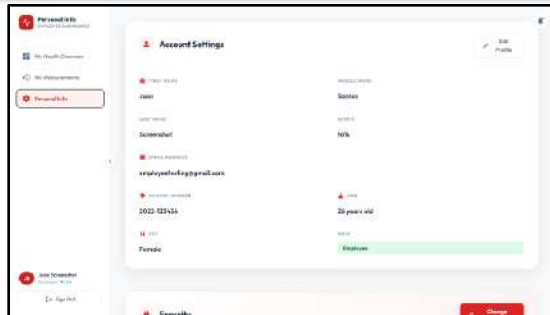
Use your physical ID card

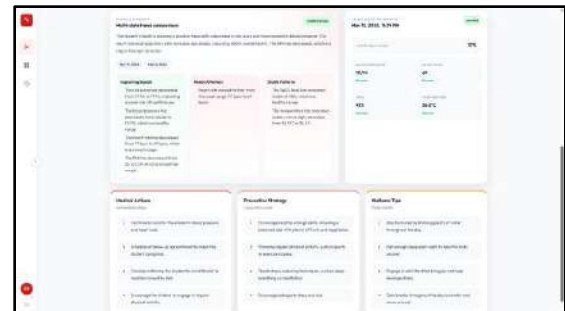
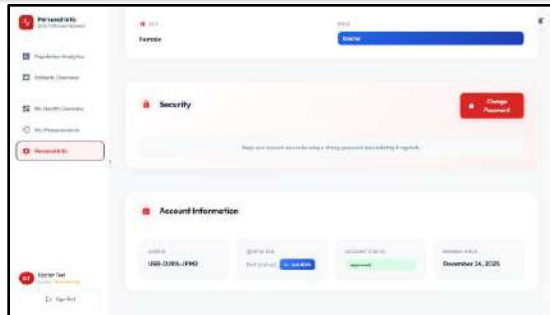
Tap to Register

If you have a QR code, tap your card now. Otherwise, you can skip this step.

Click Registration







Testing Results

In this section, the results of hardware and AI system testing are presented, showing the performance, accuracy, and reliability of the FOVB-AIoT components and models.

Unit Testing

Table 4.1

Result of Hardware Components Testing

Components	Expected Result	Trial 1	Trial 2	Trial 3	Ave. Value	Error %	Accuracy %	Remarks
Arduino Mega	Boot time: 200ms	195ms	198ms	196ms	196.3ms	1.85	98.1	Passed
Arduino Nano	Boot time: 150ms	145ms	148ms	147ms	146.7ms	2.2	97.8	Passed

Table 4.1 shows the results of unit testing for the hardware components of the FOVB-AIoT system. Each component was tested for its expected functionality, including boot time for the Arduino Mega and Nano, with multiple trials conducted to ensure consistency. The average values, error percentages, and calculated accuracy demonstrate that all tested components are functioning correctly and meet the required performance standards.

System Testing

Table 4.2

Load Cells with HX711 Weight Sensor

Person	Expected Values	Trial 1	Trial 2	Trial 3	Ave. Value	Error (%)	Accuracy (%)	Remarks
Person 1	53.5kg	53 kg	53.5 kg	53.5 kg	53.33 kg	0.32%	99.68%	Passed
Person 2	64.5kg	64.5 kg	64.5 kg	64 kg	64.3 kg	0.26%	99.74%	Passed
Person 3	53.5kg	53 kg	53.5 kg	53.5 kg	53.33 kg	0.32%	99.68%	Passed

Table 4.2 is used to validate the performance of the system's load cells by comparing expected weights with measured values across repeated trials, and the results demonstrate that the system achieves very high accuracy (approximately 99.7%) with minimal error, confirming that the weight measurements are consistent, reliable, and suitable for use in the proposed system.

Table 4.3

Tf Luna Lidar Height Sensor

Person	Expected Values	Trial 1	Trial 2	Trial 3	Ave. Value	Error (%)	Accuracy (%)	Remarks
Person 1	175.4 cm	175.4 cm	174.4 cm	175.4 cm	175.07 cm	0.19%	99.81%	Passed
Person 2	165.5 cm	164.4 cm	163.4 cm	163.4 cm	163.73 cm	1.07%	98.93%	Passed
Person 3	175 cm	176.4 cm	175.4 cm	173.4 cm	175.07 cm	0.04%	99.96%	Passed

Table 4.3 presents the comparison between manual tape measurements and the TF-Luna LiDAR sensor readings across multiple trials, showing low error rates and high accuracy (approximately 98.9%–99.9%), which confirms that the sensor provides precise, consistent, and reliable height measurements suitable for integration into the proposed system.

Table 4.4

MLX90614 Temperature Sensor

Person	Expected Values	Trial 1	Trial 2	Trial 3	Ave. Value	Error (%)	Accuracy (%)	Remarks
Person 1	35.6 °C	35.8 °C	35.6 °C	35.4 °C	35.6 °C	0.01%	99.9%	Passed
Person 2	36.5 °C	35.9 °C	36.5 °C	37 °C	36.4 °C	0.08%	99.92%	Passed
Person 3	35.7 °C	35.7 °C	35.7 °C	35.3 °C	35.57 °C	0.36%	99.64%	Passed

Table 4.4 presents the comparison between the non-contact temperature gun and the MLX90614 sensor across multiple trials, showing very minimal error and high accuracy (approximately 99.6%–100%), which confirms that the sensor provides precise, consistent, and reliable body temperature measurements suitable for integration into the proposed system.

Table 4.5

Heart Rate (MAX30102 Sensor)

Person	Expected Values	Trial 1	Trial 2	Trial 3	Ave. Value	Error (%)	Accuracy (%)	Remarks
Person 1	98 BPM	103 BPM	98 BPM	88 BPM	96.33 BPM	1.70%	98.30%	Passed
Person 2	112 BPM	112 BPM	111 BPM	112 BPM	111.67 BPM	0.30%	99.70%	Passed
Person 3	74 BPM	77 BPM	74 BPM	74 BPM	75 BPM	1.35%	98.65%	Passed

Table 4.5 presents the comparison between the oximeter heart rate readings and the MAX30102 sensor measurements across multiple trials, showing generally low error rates and high accuracy for most cases, which indicates that the sensor can reliably measure heart rate, though slight variations may occur due to movement or physiological differences among users.

Table 4.6

Performance Evaluation of Respiratory Rate (MAX30102 Sensor)

Person	Expected Values	Trial 1	Trial 2	Trial 3	Ave. Value	Error (%)	Accuracy (%)	Remarks
Person 1	17 breaths per minute	17 breaths per minute	18 breaths per minute	16 breaths per minute	17 breaths/min	0.00%	100.00%	Passed
Person 2	17 breaths per minute	18 breaths per minute	17 breaths per minute	17 breaths per minute	17.33 breaths/min	1.94%	98.06%	Passed
Person 3	14 breaths per minute	13 breaths per minute	17 breaths per minute	14 breaths per minute	14.67 breaths/min	4.79%	95.21%	Passed

Table 4.6 presents the comparison between oximeter-based respiratory rate measurements and the MAX30102 sensor readings across multiple trials, showing low error percentages and high accuracy (approximately 95.2%–100%), which confirms that the sensor can reliably and consistently measure respiratory rate for integration into the proposed system.

Table 4.7

Performance Evaluation of Oxygen Saturation (MAX30102 Sensor)

Person	Expected Values	Trial 1	Trial 2	Trial 3	Ave. Value	Error (%)	Accuracy (%)	Remarks
Person 1	97%	97%	94%	97%	96%	1.03%	98.97%	Passed
Person 2	98%	97%	98%	98%	97.6%	0.34%	99.66%	Passed
Person 3	97%	97%	96%	97%	96.67 %	0.34%	99.66%	Passed

Table 4.7 This table presents the comparison between oximeter oxygen saturation readings and the MAX30102 sensor measurements across multiple trials, showing minimal error and high accuracy (approximately 99.0%–99.7%), which confirms that the sensor provides reliable, consistent, and accurate SpO₂ measurements suitable for integration into the proposed system.

Table 4.8

AI Object Detection (Displayed Digits of Blood Pressure Device)

Class of Numbers	Trial 1 (%)	Trial 2 (%)	Trial 3 (%)	Ave. Value	Error (%)	Accuracy (%)	Remarks
0 digit	95	98	97	96.7	3.3	96.7	Passed
1 digit	93	97	96	95.3	4.7	95.3	Passed
2 digit	98	95	96	96.3	3.7	96.3	Passed
3 digit	92	98	94	94.7	5.3	94.7	Passed
4 digit	95	96	94	95.0	5.0	95.0	Passed
5 digit	94	95	96	95.0	5.0	95.0	Passed
6 digit	93	94	95	94.0	6.0	94.0	Passed
7 digit	95	94	95	94.7	5.3	94.7	Passed
8 digit	94	96	95	95.0	5.0	95.0	Passed
9 digit	95	96	94	95.0	5.0	95.0	Passed

Note. The AI object detection model for blood pressure was evaluated with a minimum accuracy requirement of above 90%. Accuracy results of 90% or below are not accepted by the system, while results greater than 90% are considered acceptable for reliable detection and system performance.

Table 4.8 presents the results of AI object detection testing for the customized blood pressure system. The table shows the precision of the YOLO-based AI model in detecting individual numerical digits and null states displayed on the digital blood pressure device across multiple trials. The results indicate that the system consistently achieved high accuracy with minimal error, confirming the reliability of the AI-assisted image processing approach in accurately extracting and synchronizing blood pressure readings for automated health assessment.

Table 4.9

Performance Evaluation of Digital Blood Pressure (AI Detection)

Person	Expected Values	Trial 1	Trial 2	Trial 3	Ave. Value	Error (%)	Accuracy (%)	Remarks
Person 1	113 SYS	113 SYS	113 SYS	113 SYS	113 SYS	0.00%	100%	Passed
	64 DIA	64 DIA	64 DIA	64 DIA	64 DIA			
Person 2	120 SYS	120 SYS	120 SYS	120 SYS	120 SYS	0.00%	100%	Passed
	71 DIA	71 DIA	71 DIA	71 DIA	71 DIA			
Person 3	111 SYS	111 SYS	111 SYS	111 SYS	111 SYS	1.54%	98.46%	Passed
	65 DIA	65 DIA	60 DIA	65 DIA	64 DIA			

Table 4.9 presents the comparative validation of the AI-detected blood pressure readings against a standard digital reference device. The results demonstrate the system's exceptional reliability, with two out of three measurements achieving 100% accuracy and 0.00% error for both Systolic and Diastolic readings.

The single deviation observed in Person 3 resulted in a marginal error of 1.54% and an accuracy of 98.46%. This confirms that the integrated AI model is highly effective and consistent in accurately extracting and synchronizing blood pressure data, validating its suitability for continuous, automated health assessment.

Table 4.10

Performance Evaluation of AI Object Detection (Foot Wearables)

Object Class	Confidence Level			Ave. Value (%)	Error (%)	Accuracy (%)	Remarks
	Trial 1 (%)	Trial 2 (%)	Trial 3 (%)				
Barefeet	91	93	92	92.0	8	92	Passed
Foot Wear	94	95	96	95.0	5	95	Passed
Socks	97	95	92	94.7	5.3	94.7	Passed

Note. The AI object detection model for foot wearables was evaluated with a minimum accuracy requirement of above 90%. Accuracy results of 90% or below are not accepted by the system, while results greater than 90% are considered acceptable for reliable detection and system performance.

Table 4.10 presents the results of AI object detection testing for foot wearables. Multiple trials were conducted to evaluate the model's precision in detecting classifications such as bare feet, footwear, and socks. The average values, error percentages, and high accuracy rates achieving 95.0% for footwear, 94.7% for socks, and 92.0% for bare feet show that the AI model reliably identifies these specific conditions with high precision, successfully meeting the expected performance requirements.

Table 4.11

Performance Evaluation of AI Object Detection (Body Wearables)

Objects Class	Trial 1 (%)	Trial 2 (%)	Trial 3 (%)	Ave. Value	Error (%)	Accuracy (%)	Remarks
Bag	94	96	95	95.0	5	95	Passed
Cap	89	91	90	90.0	10	90	Passed
ID	91	92	93	92.0	8	92	Passed
Watch	92	94	93	93.0	7	93	Passed

Note. The AI object detection model for body wearables was evaluated with an accuracy requirement of 90% or greater. Accuracy results below 90% are not accepted by the system, while results 90% or greater are considered acceptable [11], [12] for reliable detection and system performance.

Table 4.11 presents the results of AI object detection testing for body wearables using the YOLOv11 model. Multiple trials were conducted to evaluate the model's detection performance for objects such as bags, caps, IDs, and watches. The average values, error percentages, and accuracy results indicate that the AI model was generally able to identify these objects with acceptable performance.

Table 4.12

Juan AI (Health Risk Prediction)

Classification (Risk Score and Classification)						
Person	Doctor's Classification	AI MODELS			Juan AI's Classification	Remarks
		XGBoost	GEMINI-2.0-flash	GPT-OSS-120B		
Person 1	Low risk	Low Risk 4%	Low Risk 6%	Low Risk 3%	Low Risk	Passed
Person 2	Mild risk	Mild risk 30%	Mild risk 28%	Mild risk 31%	Mild risk	Passed
Person 3	Moderate risk	Moderate risk 46%	Moderate risk 47%	Moderate risk 44%	Moderate risk	Passed
Person 4	High risk	High risk 67%	High risk 70%	High risk 69%	High risk	Passed
Person 5	Priority risk	Priority risk 87%	Priority risk 85%	Priority risk 88%	Priority risk	Passed

Note. The percentage values apply only to the AI risk score because the AI models generate a numerical risk score expressed in percent. The corresponding risk category is then determined based on the threshold ranges presented in Table 4.13. In contrast, the Doctor Expectation is expressed only as a categorical interpretation of the patient's condition and does not have an equivalent percentage value.

Table 4.12 presents the validation results of the Juan AI Risk Assessment module. To evaluate the module's ability to prevent AI bias and hallucination, five user profiles corresponding to established medical baselines (Doctor Expectation) were processed simultaneously by the three integrated AI models: XGBoost, GEMINI-2.0-

Flash, and GPT-OSS-120B. The results demonstrate that the models consistently agreed on the correct classification brackets (Low, Mild, Moderate, High, and Priority Risk). Notably, the numerical risk score generated by the primary machine learning algorithm (XGBoost) and the dual Large Language Models (Gemini and GPT) showed an exceptionally minimal deviation of only 1 to 3 percent. This tight margin confirms the effectiveness of the strict system prompting, proving the LLMs can safely validate the numerical data without generating random variations. Driven by this consensus, the Final Judgement achieved a 100% classification accuracy to the Doctor's Expectation with a 0% error rate. This validates that the 4-in-1 device can accurately and reliably process physiological data into precise health warnings.

Table 4.13

Classification of Risk Score

Classification	Risk score
Low Risk	0-19
Mild Risk	20-39
Moderate Risk	40-59
High Risk	60-79
Priority Risk	80-100

Table 4.13 classifies the computed risk score into five categories: Low Risk (0–19), Mild Risk (20–39), Moderate Risk (40–59), High Risk (60–79), and Priority Risk (80–100). This five-level scheme is based on established clinical triage models, such as the Emergency Severity Index (ESI) and Manchester Triage System (MTS), which

group patients by severity and urgency. Dividing the 0–100 score into fixed intervals makes the results easier to interpret, with lower scores reflecting stable conditions and higher scores indicating greater physiological abnormality and risk of deterioration. The highest range, Priority Risk, represents patients who may require immediate medical attention. This quintile-based approach ensures consistent clinical interpretation, supports objective decision-making, and enhances the practicality of the risk assessment model by allowing healthcare providers to prioritize resources effectively in real-world settings.

Survey Data Gathering

In this section, the researchers conducted a comprehensive analysis and presentation of the system testing results. These findings were essential in evaluating the performance, accuracy, and reliability of the developed FOVB-AIoT system and served as a critical basis for forming the study's conclusions and recommendations.

Table 4.14

Mean Ratings of Time Consumption of the Manual and FOVB-AIoT Vital Sign Taking Process

Questions (Manual)	Mean	Rating
I am satisfied with how quickly I receive my vital signs at the clinic.	2.49	Dissatisfied
I am satisfied with how easily I can obtain my BMI at the clinic.	2.47	Dissatisfied
I am satisfied with how fast the clinic completes the procedure of measuring my vital signs and BMI.	2.42	Dissatisfied
I am satisfied with how quickly I am informed about my health risk after getting my vital signs and BMI.	2.39	Dissatisfied
Overall = 2.44 (Dissatisfied)		
Questions (FOVB-AIoT)	Mean	Rating
The system measures my vital signs quickly.	3.63	Very Satisfied
The system computes my BMI within a short amount of time.	3.68	Very Satisfied
The entire process from measurement to result display is completed quickly.	3.64	Very Satisfied
The system provides my health risk prediction promptly after the measurements.	3.65	Very Satisfied
Overall = 3.65 (Very Satisfied)		

Table 4.14 presents the mean ratings of the respondents on the time consumption of the manual and FOVB-AIoT vital sign taking process, based on the average responses of 72 respondents.

For the manual vital sign taking process, the results show that the overall mean (2.44) is interpreted as *Dissatisfied*, indicating that the respondents were generally not

satisfied with its time efficiency. From highest to lowest, the indicator on the system measuring vital signs quickly obtained the highest mean (2.49), followed by the system computing BMI within a short amount of time (2.47), the entire process from measurement to result display being completed quickly (2.42), and the system providing health risk prediction promptly after the measurements (2.39). These findings suggest that although the measurement of vital signs was perceived as the fastest aspect of the manual process, the prompt generation of health risk prediction was rated lowest, which contributed to the overall dissatisfaction with its time consumption.

For the FOVB-AIoT system, the findings reveal that the overall mean (3.65) is interpreted as Very Satisfied, indicating that the respondents were highly satisfied with its time efficiency. From highest to lowest, the indicator on the system computing BMI within a short amount of time obtained the highest mean (3.68), followed by the system providing health risk prediction promptly after the measurements (3.65), the entire process from measurement to result display being completed quickly (3.64), and the system measuring vital signs quickly (3.63). These results indicate that the respondents consistently perceived the FOVB-AIoT system as fast and responsive across all aspects of the process, demonstrating that it performed better than the manual method in terms of time consumption.

Overall, the findings indicate that the FOVB-AIoT system overall (3.65) demonstrated substantially greater time efficiency compared to the manual method's

(2.44), resulting in higher levels of respondent satisfaction and confirming its effectiveness in minimizing time consumption.

Table 4.15

Mean Ratings of Efficiency of the Manual and FOVB-AIoT Vital Sign Taking Process

Questions (Manual)	Mean	Rating
The system measures my vital signs quickly.	2.35	Dissatisfied
The system computes my BMI within a short amount of time.	2.38	Dissatisfied
The entire process from measurement to result display is completed quickly.	2.32	Dissatisfied
The system provides my health risk prediction promptly after the measurements.	2.38	Dissatisfied
Overall = 2.36 (Dissatisfied)		
Questions (FOVB-AIoT)	Mean	Rating
I am satisfied with the speed of the clinic's vital sign recording process.	3.79	Very Satisfied
I am satisfied with the minimal effort required to obtain my BMI result at the clinic.	3.78	Very Satisfied
I am satisfied with how efficiently I can complete the vital signs and BMI process at the clinic.	3.79	Very Satisfied
I am satisfied with how smoothly the clinic completes the entire process (vital signs and BMI).	3.79	Very Satisfied
Overall = 3.79 (Very Satisfied)		

Table 4.15 presents the mean ratings of the respondents on the efficiency of the manual and FOVB-AIoT vital sign taking process, based on the average responses of 72 respondents. For the manual vital sign taking process, Table 4.15 presents the respondents' assessment of its efficiency in terms of speed and promptness during the conduct of vital sign taking. The findings show that the overall

mean (2.36) is interpreted as *Dissatisfied*, indicating that the respondents were generally not satisfied with the efficiency of the manual process. From highest to lowest, “The system computes my BMI within a short amount of time” (2.38) and “The system provides my health risk prediction promptly after the measurements” (2.38) got the same highest mean, followed by “The system measures my vital signs quickly” (2.35), while “The entire process from measurement to result display is completed quickly” (2.32) obtained the lowest mean. These results indicate that although BMI computation and the prompt provision of health risk prediction were viewed as relatively better aspects of the manual process, the overall procedure from measurement to result display was rated least efficient, contributing to the respondents’ dissatisfaction.

For the FOVB-AIoT system, Table 4.15 presents the respondents’ assessment of its efficiency based on the same average responses of 72 respondents. The results reveal that the overall mean (3.79) is interpreted as *Very Satisfied*, which indicates that the respondents were highly satisfied with the efficiency of the system. From highest to lowest, “The system measures my vital signs quickly” (3.79), “The entire process from measurement to result display is completed quickly” (3.79), and “The system provides my health risk prediction promptly after the measurements” (3.79) got the same highest mean, followed by “The system computes my BMI within a short amount of time” (3.78). These findings suggest that the respondents consistently

perceived the FOVB-AIoT system as highly efficient across all indicators, showing that the system performed much better than the manual process in terms of efficiency.

Overall, the results indicate that the FOVB-AIoT system with an overall mean of (3.79), demonstrated significantly higher efficiency compared to the manual vital sign taking process, which obtained an overall mean of (2.36). The consistently higher mean ratings across all indicators suggest that the system effectively improved the speed and promptness of operations, leading to greater respondent satisfaction and confirming its superiority over the manual method in terms of efficiency.

Table 4.16

Mean Ratings of FOVB-AIoT Vital Sign Taking Process in Terms of Usability

Usability	Mean	Rating
The system is easy to use when starting the vital sign measurement process.	3.79	Very Satisfied
The instructions displayed on the screen are clear and easy to understand.	3.79	Very Satisfied
The touchscreen interface responds accurately to user input.	3.76	Very Satisfied
The displayed vital signs, BMI, and health risk results are easy to read and understand	3.74	Very Satisfied
Overall = 3.77 (Very Satisfied)		

Table 4.16 presents the mean ratings of the respondents on the usability of the FOVB-AIoT vital sign taking process, based on the average responses of 72 respondents. It answers the question of how the respondents assessed the system in

terms of ease of use, clarity of instructions, interface response, and readability of results. The findings show that the overall mean (3.77) is interpreted as *Very Satisfied*, indicating that the respondents were highly satisfied with the usability of the system. From highest to lowest, the indicator on the displayed vital signs, BMI, and health risk results being easy to read and understand got (3.79), while the indicators on the system being easy to use when starting the vital sign measurement process and the instructions displayed on the screen being clear and easy to understand got the same mean (3.79). Meanwhile, the indicator on the touchscreen interface responding accurately to user input got (3.76), which was the lowest mean. These results indicate that the respondents found the system highly usable, particularly in terms of the readability of the displayed results.

Table 4.17

Mean Ratings of FOVB-AIoT Vital Sign Taking Process in Terms of Reliability

Reliability	Mean	Rating
The system consistently provides reliable vital sign measurements.	3.81	Very Satisfied
The BMI results generated by the system are stable during repeated use.	3.75	Very Satisfied
The system operates properly without frequent errors or interruptions.	3.75	Very Satisfied
The system maintains good performance during continuous or repeated operation.	3.81	Very Satisfied
Overall = 3.78 (Very Satisfied)		

Table 4.17 presents the mean ratings of the respondents on the reliability of the FOVB-AIoT vital sign taking process, based on the average responses of 72 respondents. The results show that the overall mean (3.78) is interpreted as *Very Satisfied*, indicating that the respondents were highly satisfied with the reliability of the system. From highest to lowest, the indicators on the system consistently providing reliable vital sign measurements and the system maintaining good performance during continuous or repeated operation both obtained the same highest mean (3.81). Meanwhile, the indicators on the BMI results generated by the system being stable during repeated use and the system operating properly without frequent errors or interruptions both obtained the same mean (3.75). These findings indicate that the respondents perceived the system as highly reliable, particularly in terms of consistent measurement and sustained performance during continuous operation.

Table 4.18

Mean Ratings of FOVB-AIoT Vital Sign Taking Process in Terms of Security and Data Integrity

Security and Data Integrity	Mean	Rating
My personal and health information is protected when using the system.	3.75	Very Satisfied
The system prevents unauthorized access to my vital sign and BMI records. The BMI results generated by the system are stable during repeated use.	3.69	Very Satisfied
The system securely transmits data between the device and the database or server.	3.75	Very Satisfied
The system preserves the accuracy and completeness of my health data.	3.71	Very Satisfied
Overall = 3.73 (Very Satisfied)		

Table 4.18 presents the mean ratings of the respondents on the security and data integrity of the FOVB-AIoT vital sign taking process, based on the average responses of 72 respondents. The results show that the overall mean (3.73) is interpreted as *Very Satisfied*, indicating that the respondents were highly satisfied with the system in terms of security and data integrity. From highest to lowest, the indicators on personal and health information being protected when using the system and the system securely transmitting data between the device and the database or server both obtained the same highest mean (3.75). This was followed by the indicator on the system preserving the accuracy and completeness of health data, which was obtained (3.71). Meanwhile, the indicator on the system preventing unauthorized access to vital signs and BMI records obtained the lowest mean (3.69). These findings

indicate that the respondents perceived the system as secure and reliable in handling health information, particularly in protecting personal data and ensuring secure data transmission.

Table 4.19

Cronbach's Alpha Reliability Results

Category	No. of Items	Cronbach's Alpha	Interpretation
Time Consumption	4	0.86	Good
Efficiency	4	0.86	Good
Usability	4	0.90	Excellent
Reability	4	0.87	Good
Security and Data Integrity	4	0.85	Good

Table 4.19 shows all categories obtained with Cronbach's Alpha values greater than 0.70, indicating that the questionnaire has good internal consistency and reliability. The Usability category obtained the highest reliability value of 0.90, which is interpreted as excellent, while the remaining categories Time Consumption, Efficiency, Reliability, and Security and Data Integrity achieved values within the "Good" range.

These results confirm that the survey instrument used in the study is reliable and that the items within each category consistently measure the intended system performance indicators. Therefore, the collected data are considered valid and suitable for further statistical analysis, such as weighted mean evaluation and

comparative analysis between the traditional method and the developed FOVB-AIoT system.

T-test Implementation

Table 4.20

Descriptive Statistics and Paired Differences (Time Consumption)

Measure	Value
Manual Mean	2.441
System Mean	3.649
Mean of Difference	1.208
Standard Deviation	1.099
Number of Respondents	72
Degree of Freedom	71

T - value	Decision
9.33	Reject H_0

Table 4.20 presents the descriptive statistics and paired differences between the manual approach and the proposed system in terms of Time Consumption. The proposed system obtained a higher mean rating ($M = 3.649$) compared to the manual clinic approach ($M = 2.441$), resulting in a mean difference of (1.208). This suggests that the respondents perceived the proposed system as faster in carrying out the vital sign-taking process. Moreover, the computed t-value (9.33) with 71 degrees of freedom indicates a statistically significant difference between the two approaches,

leading to the rejection of the null hypothesis. The magnitude of the t-value shows strong evidence that the observed improvement in time consumption favored the proposed system rather than occurring by chance. Although the standard deviation of the paired differences (SD = 1.099) suggests some variation in respondents' evaluations, the overall result consistently supports the advantage of the system over the manual method.

Thus, the findings indicate that the proposed system significantly enhanced time consumption, showing that it is considerably faster and more efficient than the traditional manual clinic approach.

Table 4.21

Descriptive Statistics and Paired Differences (Efficiency)

Measure	Value
Manual Mean	2.354
System Mean	3.788
Mean of Difference	1.434
Standard Deviation	1.010
Number of Respondents	72
Degree of Freedom	71
T - value	Decision
12.052	Reject H_0

Table 4.21 presents the descriptive statistics and paired differences between the manual clinic approach and the proposed system in terms of Efficiency. The proposed system registered a higher mean rating ($M = 3.788$) compared to the manual approach ($M = 2.354$), resulting in a mean difference of (1.434). This suggests that the respondents rated the proposed system as more efficient than the traditional method.

Furthermore, the computed t-value (12.052) with 71 degrees of freedom indicates that the difference between the two approaches is statistically significant, leading to the rejection of the null hypothesis. The size of the t-value provides strong support that the improvement in efficiency is attributable to the proposed system. Although the standard deviation of the paired differences ($SD = 1.010$) reflects some variation in the responses, the general pattern consistently favors the system over the manual approach.

Thus, the results establish that the proposed system significantly improved efficiency, demonstrating that it is a more efficient and reliable approach than the traditional manual clinic method.

CHAPTER V

SUMMARY OF FINDINGS, CONCLUSION AND RECOMMENDATION

This chapter presents the summary of findings derived from the testing and evaluation of the proposed system, highlighting the performance, accuracy, and reliability of each integrated sensor. It also provides the conclusion based on the objectives of the study and the results obtained from the experiments. Lastly, this chapter offers recommendations for future improvements and possible enhancements to further develop and optimize the system for wider application.

Summary of Findings

This study focused on the design, development, and evaluation of the **FOVB-AIoT**, a kiosk-based health monitoring system that integrates multiple sensors, artificial intelligence, and Internet of Things (IoT) technologies to automate vital sign measurement and health risk prediction.

1. Measurement of Multiple Vital Signs.

The developed system was able to accurately and efficiently measure multiple vital signs simultaneously, including body temperature, heart rate, respiratory rate, blood pressure, oxygen saturation, height, and weight. Through sensor integration and automated data acquisition, the system minimized the time

required for manual assessment while maintaining consistent measurement performance

2. AI-Based Health Risk Prediction

The implementation of the eXtreme Gradient Boosting (XGBoost) AI model enabled the system to analyze collected vital sign data and automatically identify potential indicators of health risks. The AI model successfully generated health risk predictions based on established health parameters, demonstrating its capability to assist in early health monitoring.

3. Improvement over Traditional Manual Methods

The developed system improved the process of vital sign measurement compared to traditional manual methods by integrating multiple measurements into a single automated platform. This reduced the need for multiple devices and minimized the time and effort required during health screenings.

3.1 Manual and AI System Comparison

Results showed that the AI-based system significantly improved time convenience and operational efficiency compared to the manual method. Automated measurement and data processing reduced the time required for vital sign collection and minimized the possibility of human error during recording and interpretation.

3.2 Effectiveness of the AI System

The AI system effectively analyzed the collected data and produced health risk assessments based on the measured parameters. The integration of object detection and automated data analysis contributed to improved accuracy and consistency in the system's health monitoring process.

4. System Evaluation Using ISO 25010

The developed system was evaluated using the ISO 25010 software quality model, which assessed the system in terms of reliability, efficiency, usability, and security and data integrity. Evaluation results indicated that the system performed well across these criteria, demonstrating that the FOVB-AIoT is reliable, efficient, and user-friendly for health monitoring applications.

Overall, the findings indicate that the FOVB-AIoT system provides an effective and practical solution for automated health screening in institutional settings, offering improved efficiency, convenience, and early health risk identification compared to traditional manual methods.

Conclusions

The researchers confirmed that the developed FOVB-AIoT system is both valid and reliable based on the results of development, testing, and evaluation. The system successfully achieved objectives of the study by automating the measurement of multiple vital signs, integrating artificial intelligence for health risk prediction, and

utilizing IoT technology for data storage and monitoring. Based on the results and findings of the study, the researchers conclude the following:

Through the development and testing of the FOVB-AIoT system, the researchers concluded that automated vital sign measurement can successfully meet established clinical standards while improving efficiency. By comparing the system's performance with traditional manual methods, it was confirmed that the integrated biomedical sensors produce highly reliable and accurate readings for body temperature, heart rate, respiratory rate, blood pressure, and BMI. More importantly, the system directly addressed the primary issue of time consumption. Because the kiosk measures multiple vital signs simultaneously, it significantly accelerates the entire assessment process, proving that a centralized, automated approach is just as valid as, but much faster than, standard clinic practices.

The study further concludes that the system's interpretation of health data was guided by recognized medical standards. In particular, the BMI classification used in the system was aligned with the Asian BMI Standard, while the blood pressure assessment followed the American Heart Association (AHA) blood pressure guidelines. These standards served as the basis for classifying the users' health status and strengthened the medical relevance of the system's output. In addition, the results and interpretation of the system were reviewed and approved by a doctor, which further supports the acceptability and validity of the system as a health assessment tool.

Based on the findings gathered from the conducted evaluation, the researchers determined that the proposed system significantly improves efficiency and reduces time consumption compared to the traditional clinic approach. The results revealed that the system streamlines the overall process, making it more accessible, convenient, and less time-consuming for users. In contrast to the manual clinic procedure, the system enables faster transactions and minimizes delays, resulting in a smoother and more efficient experience. Therefore, it was concluded that the implementation of the system enhances operational performance and offers a more effective solution than the existing approach.

Following the comprehensive evaluation conducted using the ISO/IEC 25010 framework, the researchers determined that the proposed system showed significant improvement in terms of efficiency and time consumption compared to the traditional clinic approach in taking vital signs. The baseline results indicated that the system made the process faster, more streamlined, and more convenient for users. Based on the respondents' feedback, the researchers also concluded that the system achieved a satisfactory level of reliability, usability, security, and data integrity, demonstrating accurate performance, user-friendliness, and effective protection of user information. However, these aspects were not directly compared with the clinic's existing process, as the study did not intend to question or evaluate the clinic's standards but instead focused on assessing and validating the overall quality of the proposed system.

Recommendation

The researchers recommend the following for future researchers and other related studies:

1. Incorporate additional medical sensor features to enhance the comprehensiveness of the system's health assessment functions. In addition, the integration of higher-performance processing hardware is advised to facilitate faster data processing, improve AI model responsiveness, and support more advanced health risk analysis while maintaining optimal system performance.
2. Expanding the AI model training dataset with a wider range of health profiles is recommended to improve prediction accuracy and generalization across different age groups and physical conditions.
3. Future versions of the system may include features such as mobile health applications, to enhance accessibility and usefulness for users and clinic staff.
4. Improving the physical design and ergonomics of the kiosk is recommended to ensure better user comfort, accessibility for individuals with disabilities, and suitability for long-term deployment in various institutional or healthcare environments.
5. Furthermore, future researchers may explore the development of a portable and modular version of the system for mobile clinic applications. A portable and modular design would allow the system to be easily transported and deployed

in remote communities, disaster response areas, and outreach medical programs, thereby expanding access to health monitoring services beyond fixed healthcare facilities.

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APPENDICES

APPENDIX A

Research Instruments

Survey Questionnaire (Initial)



RIZAL TECHNOLOGICAL UNIVERSITY
College of Engineering, Architecture and Technology
M. Eusebio Avenue, Maybunga, Pasig City





RIZAL TECHNOLOGICAL UNIVERSITY
College of Engineering, Architecture and Technology
M. Eusebio Avenue, Maybunga, Pasig City



Assessment Questionnaire

We're developing a system namely **Four-in-One Vital Sign Sensor that uses AI and IoT to help predict health risks** for the RTU Clinic to instantly measure key metrics and help predict health risks. Before we implement this system, we need your help to evaluate the current manual process. Your feedback on the timeliness (Time Consumption) of the current manual process will serve as the baseline for our project.

This baseline is crucial because it lets us prove the value of our new technology later. Please note that our system is designed to **assist** medical professionals, not replace them. Once implemented, we will compare the new results to these initial scores to show the impact and effectiveness of this helpful upgrade. Rest assured, all your answers are completely confidential and will only be used to improve clinic services.

Please fill out the details below so we can properly analyze feedback from different groups:

PART 1: Respondent Profile
Name (Optional): _____ Age: _____ years old
Position (Designation): _____ (e.g., Student, Faculty, Staff, Clinic Personnel, Admin)
I use the vital signs and tests (BMI) service about _____ (e.g., Once a month, Rarely, Weekly)

PART 2: Evaluation of the Current Manual Vital Signs Process
DIRECTIONS:
The following section contains statements about your experience with the current manual process for having your vital signs measured. For each statement, indicate the extent to which you agree or disagree based on your personal experience.

A. Time Consumption – refers to how quickly and efficiently the clinic performs its vital sign measurement, BMI and health risk processes.

Questions	1 Very Dissatisfied	2 Dissatisfied	3 Neutral	4 Satisfied	5 Very Satisfied
I'm satisfied with how quickly I get my vital signs at the clinic.					
I'm satisfied with how easily I can obtain my BMI at the clinic.					
I'm satisfied with how fast the procedure of getting my vital signs and BMI at the clinic.					
I'm satisfied with how quickly I was notified about my health risk after getting the vital signs and BMI.					

B. Efficiency – focuses on how quickly and with minimal user effort/resource expenditure the system delivers vital sign measurement, BMI results, and health risk information.

Questions	1 Very Dissatisfied	2 Dissatisfied	3 Neutral	4 Satisfied	5 Very Satisfied
I am satisfied with the speed of the vital sign-measuring process.					
I am satisfied with the low user effort (few steps) required to get my BMI result.					
I'm satisfied with how efficient is get the vital signs and BMI.					
I am satisfied with how smoothly the clinic completes the entire process (Vital Signs and BMI).					

PART 4: Feedback / Suggestions

What improvements would you suggest to make the clinic's vital signs, BMI, and health risk services faster, easier, or more convenient?

Respondent's Signature: _____ Date of Evaluation: _____

Initial Survey Questionnaire

Survey Questionnaire (System)


RIZAL TECHNOLOGICAL UNIVERSITY
 College of Engineering, Architecture and Technology
 M. Eusebio Avenue, Maybunga, Pasig City

Assessment Questionnaire

We are conducting this survey to evaluate the **FOVB-AIoT: Four-in-One Vital Sign Sensor with BMI Calculation using AI and IoT for Health Risk Prediction**, implemented at the RTU Pasig Clinic. The system is designed to automate the measurement of vital signs, compute Body Mass Index (BMI), and provide AI-assisted health risk predictions through an integrated kiosk-based platform. This survey aims to gather user feedback regarding the system's performance, efficiency, usability, reliability, and data security after its implementation.

The results of this survey will be compared with data obtained from the initial baseline survey of the traditional manual vital sign measurement process conducted at the clinic. This comparison will help determine the effectiveness and impact of the FOVb-AIoT system in improving time efficiency, user experience, and overall service delivery. The system is intended to support healthcare personnel and does not replace professional medical judgment. All responses will be treated with strict confidentiality and will be used solely for research, evaluation, and system enhancement purposes.

Please fill out the details below so we can properly analyze feedback from different groups.

PART 1: Respondent Profile
 Name (Optional): _____ Age: _____ years old
 Position (Designation): _____ (e.g., Student, Faculty, Staff, Clinic Personnel, Admin)
 I use the vital signs and take BMI service about: _____ (e.g., Once a month, Rarely, Weekly)

PART 2: Evaluation of the Current Manual Vital Signs Process
DIRECTIONS:
 The following section contains statements about your experience with the current manual process for having your vital signs measured. For each statement, indicate the extent to which you agree or disagree based on your personal experience.

Scale and Interpretation
 4 – Very Satisfied
 3 – Satisfied
 2 – Dissatisfied
 1 – Very Dissatisfied

A. Time Consumption – refers to how quickly and efficiently the clinic performs its vital sign measurement, BMI and health risk processes.

Questions	1 Very Dissatisfied	2 Dissatisfied	3 Satisfied	4 Very Satisfied
The system measures my vital signs quickly.				
The system computes my BMI within a short amount of time.				


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The entire process from measurement to result display is completed quickly.				
The system efficiently records my vital sign data.				

B. Efficiency – focuses on how quickly and with minimal user effort/resource expenditure the system delivers vital sign measurement, BMI results, and health risk information.

Questions	1 Very Dissatisfied	2 Dissatisfied	3 Satisfied	4 Very Satisfied
The system efficiently records my vital sign data.				
The system requires only a few simple steps to obtain my BMI result.				
The system processes and displays my results without unnecessary delays.				
The system smoothly completes the workflow from measurement to health risk output.				

D. Usability – focuses on how easy and intuitive the system is for users to operate, including clarity of instructions, touchscreen interaction, voice assistance, and overall user experience during vital sign measurement and result viewing, with minimal need for guidance or prior training.

Questions	1 Very Dissatisfied	2 Dissatisfied	3 Satisfied	4 Very Satisfied
The system is easy to use when starting the vital sign measurement process.				
The instructions displayed on the screen are clear and easy to understand.				
The touchscreen interface responds accurately to user input.				


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The displayed vital signs, BMI, and health risk results are easy to read and understand.				
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E. Reliability – focuses on the system's ability to consistently perform vital sign measurements accurately and without failure, ensuring stable operation, dependable data processing, and successful completion of measurement tasks under normal clinic usage conditions.

Questions	1 Very Dissatisfied	2 Dissatisfied	3 Satisfied	4 Very Satisfied
The system consistently provides reliable vital sign measurements.				
The BMI results generated by the system are stable during repeated use.				
The system operates properly without frequent errors or interruptions.				
The system maintains good performance during continuous or repeated operation.				

F. Security and Data Integrity – focuses on how well the system protects user information, ensures authorized access only, and maintains the accuracy, confidentiality, and integrity of collected health data stored in the local database and accessed through the web-based dashboard.

Questions	1 Very Dissatisfied	2 Dissatisfied	3 Satisfied	4 Very Satisfied
My personal and health information is protected when using the system.				
The system prevents unauthorized access to my vital sign and BMI records.				
The system securely transmits data between the device and the database or server.				
The system preserves the accuracy and completeness of				


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my health data.				
-----------------	--	--	--	--

PART 4: Feedback / Suggestions

What improvements would you suggest to make the clinic's vital signs, BMI, and health risk services faster, easier, or more convenient?

Respondent's Signature: _____ Date of Evaluation: _____

Initial Survey Questionnaire Approval

 **RIZAL TECHNOLOGICAL UNIVERSITY**
College of Engineering, Architecture and Technology
M. Eusebio Avenue, Maybunga, Pasig City

October 25, 2025

To: Thesis Evaluation Committee
Rizal Technological University
Department of Computer Engineering

Request for Expert Validation of ISO/IEC 25010-Based Software Quality Survey Instrument for Thesis Project

Dear Validators,

We are final-year students of the Computer Engineering program at Rizal Technological University (RTU), and we respectfully request your expert validation of our survey questionnaire for our thesis project.

Survey Purpose and Baseline Assessment

This initial survey aims to establish a baseline by evaluating the current manual process of checking vital signs (Heart Rate, Oxygen Rate, Respiratory Rate, and Body Temperature) at the RTU clinic. The results of this survey, focused entirely on the traditional method, will later be compared to a second survey conducted after the deployment of our proposed AI and IoT-integrated system.

Purpose of Validation: Alignment with ISO/IEC 25010

To ensure the survey accurately assesses the existing manual process and allows for a valid future comparison, we seek your expert validation that the items align with the ISO/IEC 25010 Software Product Quality Model, specifically: Performance Efficiency and Usability.

Survey Instrument Details

Please note that the attached questionnaire utilizes a 5-point Likert scale with direct coding. Your review is crucial to confirm the clarity of the statements given this scoring system:

Scale: 5 = Very Satisfied to 1 = Very Dissatisfied

Our overall thesis focuses on creating a Four-in-One Vital Sign Sensor that uses AI and IoT to provide real-time analysis and predict possible health risks based on the measured vital signs and BMI. The comparison of the manual process (the survey) versus our system (the future survey) is the core method for demonstrating the system's improvement in efficiency and usability.

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College of Engineering, Architecture and Technology
M. Eusebio Avenue, Maybunga, Pasig City

Request for Expert Review

We formally invite the following faculty members to provide their expert assessment:

- Engr. Trixie Anagay
- Prof. Danilo Marcos
- Engr. Marvin Apus

This evaluation will be formally noted by our Thesis Adviser: Engr. Ezekiel Nequit.

The ISO 25010 Evaluation Questionnaire is attached to this letter. We estimate that the review will take approximately 15 minutes. Your feedback will be highly valuable in ensuring the clarity and appropriateness of our survey instrument for its use in comparing the manual and system-based vital sign measurement processes.

Thank you for supporting the System Shapers group in this crucial final stage of our thesis work.

Respectfully,
System Shapers Group Representatives
Department of Computer Engineering
Rizal Technological University

Validators' Signatures:

 Engr. Trixie Anagay

 Prof. Danilo Marcos

 Engr. Marvin Apus

Noted by:

 Engr. Ezekiel Nequit

System Survey Questionnaire Approval

 **RIZAL TECHNOLOGICAL UNIVERSITY**
College of Engineering, Architecture and Technology
M. Eusebio Avenue, Maybunga, Pasig City

January 23, 2026

To: Thesis Evaluation Committee
Rizal Technological University
Department of Computer Engineering

Request for Expert Validation of ISO/IEC 25010-Based Software Quality Survey Instrument

Dear Validators,

We are 4th-year Computer Engineering students of Rizal Technological University (RTU), and we respectfully request your expert validation of the survey questionnaire developed for our thesis project.

The survey is intended to evaluate the Four-in-One Vital Sign Sensor with BMI Calculation using AI and IoT for Health Risk Prediction (FOVb-AIoT) based on the experiences of respondents who have directly used the system. The questionnaire assesses the system's quality as a kiosk-based solution for automated vital sign measurement, BMI computation, real-time health analysis, and AI-assisted health risk prediction.

To ensure the validity and reliability of the instrument, we seek your expert review to confirm that the survey items are properly aligned with the ISO/IEC 25010 Software Product Quality Model, focusing on the following characteristics and sub-characteristics:

- **Performance Efficiency** (Time Behavior, Resource Utilization)
- **Usability** (Appropriateness Recognizability, Learnability, Operability, User Interface Aesthetics)
- **Reliability** (Maturity, Availability, Fault Tolerance)
- **Security and Data Integrity** (Confidentiality, Integrity, Authorized Access)

The attached questionnaire uses a 5-point Likert scale with direct coding to capture user perceptions regarding system speed, efficiency, ease of use, reliability of results, and protection of personal and health-related data.

The FOVb-AIoT system integrates RFID-based user identification, automated data storage, IoT-based real-time monitoring, AI-driven health risk analysis, and result delivery through SMS, email, or printed receipt. This survey serves as the primary tool for evaluating these system features from the user's perspective.

 **RIZAL TECHNOLOGICAL UNIVERSITY**
College of Engineering, Architecture and Technology
M. Eusebio Avenue, Maybunga, Pasig City

We respectfully invite the following faculty members to provide their expert validation of the survey instrument:

- Engr. Rizal Lasui
- Prof. Corazon Aspuria
- Prof. Felissa Molina

This validation will be formally noted by our Thesis Adviser, Engr. Ezekiel Nequit.

The review is expected to take approximately 15 minutes. Your feedback will be highly valuable in ensuring the clarity, relevance, and appropriateness of the survey instrument for evaluating the FOVb-AIoT system.

Thank you for your time and support.

Respectfully,
System Shapers Group Representatives
Department of Computer Engineering
Rizal Technological University

Validators' Signatures:

 Prof. Corazon Aspuria

 Prof. Felissa Molina

 Engr. Rizal Lasui

Noted by:

 Engr. Ezekiel Nequit

APPENDIX B

Permission and Communication

Deployment's RTU Pasig University Doctor Approval

 **RIZAL TECHNOLOGICAL UNIVERSITY**
College of Engineering, Architecture and Technology
M. Eusebio Avenue, Maybunga, Pasig City 

January 14, 2026

Dr. Lusieto B. Oporto, M.D.
University Physician
RTU Pasig Clinic
Rizal Technological University
Pasig City

*Approved!
Use coordinate of Meen
And for any instructions
of the prototype!*


Lusieto B. Oporto, M.D.
January 14, 2026
1:53 PM

Subject: Request for Permission to Deploy Thesis Prototype in RTU Pasig Clinic

Good day, Dr. Oporto,

We are students from the Computer Engineering Department, respectfully request permission to deploy and conduct testing of our thesis entitled **"Four-in-One Vital Sign Sensor with BMI Calculation using AI and IoT for Health Risk Prediction (FOVB-AIoT)"** at the RTU Pasig Clinic.

The proposed system is a kiosk-based health monitoring solution that measures body temperature, heart rate, respiratory rate, and blood pressure in separate phases to ensure accuracy and reliability. It also calculates the user's Body Mass Index (BMI) using height and weight to provide a comprehensive health profile. The system uses RFID-based registration, allowing registered users to proceed quickly with measurements, while unregistered users securely input basic information such as name, age, sex, school number, and email for database storage.

All collected data are transmitted to an IoT-based web database, enabling authorized healthcare personnel to view results in real time. An AI module analyzes the readings to detect unusual patterns, generate a comprehensive health risk interpretation score, and provide basic health recommendations and preventive tips. Users may also receive their results through email, or printed receipt for personal reference.

The deployment is intended **solely for academic purposes**, system evaluation, and data validation. We assure you that all data collected will be handled with strict confidentiality, used only for research purposes, and will comply with ethical and data privacy guidelines. The deployment will be conducted under proper supervision and will not disrupt normal operations.

 **RIZAL TECHNOLOGICAL UNIVERSITY**
College of Engineering, Architecture and Technology
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In this regard, we humbly seek your approval to deploy and test our thesis prototype at your institution. Your support will greatly contribute to the successful completion of our study.

Thank you very much for your time and consideration. We look forward to your favorable response.

Respectfully yours,
System Shapers


Alcantara, Mar Kevin P.


Borongoy, Bernice C.


Gamboa, Winn Andrei R.


Llorca, Raymond G.


Oavero, Erick John A.


Raleno, Paul Andrew A.


Sagorido, Yari Lorenz C.

Noted by:

Engr. Rizal Laqui
Subject Instructor

Data Privacy Letter and Non Disclosure Agreement

 RIZAL TECHNOLOGICAL UNIVERSITY College of Engineering, Architecture and Technology M. Eusebio Avenue, Maybunga, Pasig City  January 27, 2026 Dr. Joavir A. Villanueva Director Data Protection Office Rizal Technological University Mandaluyong City Subject: Intent to Conduct Research and Request for Data Privacy Clearance Dear Dr. Villanueva, We are fourth-year students of the Bachelor of Science in Computer Engineering program at Rizal Technological University (RTU) Pasig Branch, writing to formally request permission to conduct data gathering for our research and to apply for a Non-Disclosure Agreement (NDA) for our activity. In compliance with the Data Privacy Act of 2012, we submit the following details regarding our proposed activity: Research/Project/Activity Title: FOVB-AIoT: Four-in-One Vital Sign Sensor with BMI Calculation using AI and IoT for Health Risk Prediction in RTU Pasig Clinic. Introduction / Background of the Research: The study aims to deploy the FOVB-AIoT, a cloud-based system designed to assist the RTU Pasig Clinic. This system integrates sensors to measure four vital signs and calculates BMI, utilizing AI to predict potential health risks. This initiative seeks to enhance the efficiency of health assessments for the university population. Objectives: The primary objective of this study is to evaluate the functionality, accuracy, and user acceptability of the proposed system. Specifically, it aims to: - Test the system's ability to accurately measure vital signs and calculate BMI in a real-world setting. - Assess the system's practicality and effectiveness in streamlining health monitoring workflows.	 RIZAL TECHNOLOGICAL UNIVERSITY College of Engineering, Architecture and Technology M. Eusebio Avenue, Maybunga, Pasig City  - Gather feedback from respondents regarding the system's reliability and ease of use. 4. Methodology: This study adopts the Agile Development Life Cycle research design. This iterative approach allows for the continuous improvement of the prototype through repeated cycles of planning, designing, building, testing, and refining based on real user feedback. 5. Target Respondents: Faculty, staff, administrators, and students of RTU Pasig Branch. The respondents will include: Inclusion Criteria: Individuals currently enrolled or employed at RTU Pasig. Exclusion Criteria: Individuals inside the campus premises who are not officially affiliated with RTU (e.g., parents, delivery personnel, visitors). 6. List of Data to be Collected: All respondents will be required to provide informed consent prior to participation. The following data will be collected: Personal Information: Name, age, email address, and student or employee number. Physiological Data: Vital sign readings including body temperature, heart rate, respiratory rate, blood pressure, oxygen saturation, height and weight, as captured by the system sensors. Survey Data: Responses regarding user satisfaction, system reliability, and areas for improvement. 7. Duration and Disposal Duration: The study will conduct a survey of one hundred (100) respondents from February 2 to March 31, 2026. The duration of the data collection period may vary depending on participant availability and clinic operations. Disposal: Physical Receipts: The system generates a thermal receipt for the respondent. Once handed over, the receipt is the respondent's responsibility. Any unclaimed receipts will be immediately shredded by the researchers. Digital Data: Upon study completion, the researchers will dispose of their access rights rather than the data itself by surrendering the local database to the clinic and changing all administrative credentials to be exclusively known by the RTU Pasig clinic.
 RIZAL TECHNOLOGICAL UNIVERSITY College of Engineering, Architecture and Technology M. Eusebio Avenue, Maybunga, Pasig City  personnel (doctors, nurses, clinic staff, etc.), ensuring a full transfer of control where the researchers retain no copies of patient data. Respectfully yours, System Owners group members: Alicante, Mar Kevin P. Berongoy, Bernie G. Gambao, Kian Andrei R. Llorca, Raymond G. Oversat, Erick John A. Relenc, Paul Andrew A. Sagorido, Tati Lorenz G. Noted by: Engr. Ezekiel G. Nequit Thesis Adviser	RIZAL TECHNOLOGICAL UNIVERSITY RTU-GENP-GRPFC-7882 REVIEWED 1 Effective MAY 15, 2021 Control # (DATA-2024-01-29-42) Rizal Technological University Bon Avenue, City of Mandaluyong DATA PROTECTION OFFICE NON-DISCLOSURE AGREEMENT I, BERNIE C. BERONGOY as the representative of the group, declare that it is our responsibility to protect the integrity and confidentiality of the enclosed information. As the intended recipient, we ensure that we will not reproduce, share, or disclose any information to anyone without the consent of the data subject. Furthermore, we affirm that the data will only be used to conduct the thesis titled: "FOVB-AIoT: Four-in-One Vital Sign Sensor with BMI Calculation Using AI and IoT for Health Risk Prediction in RTU Pasig Clinic" We guarantee our commitment to strictly adhere to the Data Privacy Act of 2012, as well as any current and applicable Implementing Rules and Regulations of the National Privacy Commission and other relevant legislation, rules, and publications. We acknowledge the necessity of implementing the required security measures to protect the data, which includes establishing appropriate organizational, physical, and technical security actions. Any unauthorized disclosure, risk, or suspicion regarding such matters must be reported immediately to the Data Protection Office. We categorically guarantee that we will not reveal any generally confidential, highly sensitive data, or private information obtained, either directly or indirectly, from the respondents to any third party. Finally, we understand that we may be held liable for any damages caused by the unauthorized processing of this information.  BERNIE C. BERONGOY Signature over printed name: RTU Computer Engineer-404 0-22-1000-000 Date: JANUARY 29, 2026  Endorsed by:  DR. JOAVIR A. VILLANUEVA Signature over printed name: Director, Data Protection Office Data Privacy Officer

APPENDIX C

Data and Analysis

[illegible]

Assessment Questionnaire

Please evaluate the **FOCUS4E** Plans-Go User Site, together with the Certification, using the following questions. Please indicate your level of agreement with each statement by marking the appropriate box. The following numbered list of items are the key areas of focus for the assessment. The assessment will assess your performance with respect to the technical processes and the educational objectives of the program. The assessment will also assess your performance with respect to safety, quality, reliability, and cost awareness. All responses will be kept strictly confidential and will be used only for research and quality improvement purposes.

PART 1: Participant Profile

Name: _____ Age: _____

PART 2: Reasons for Using CitiLink, Technical Process

DIRECTIONS:
The following scaling system represents degrees of your agreement with the stated reason intended for using the user site again frequently. For each statement, indicate the extent to which you agree or disagree based on your personal experience.

LEGEND:

1 = 4 - Very Satisfied
2 = 3 - Satisfied
3 = 2 - Dissatisfied
4 = 1 - Very Dissatisfied

A 5-Star Certification - How satisfied are you with the User Certification of the correct information on the Questions

Questions	1 Very Satisfied	2 Satisfied	3 Dissatisfied	4 Very Dissatisfied
1. I am satisfied with the information on the User Certification of the correct information on the Questions	☐	☐	☐	☐
2. I am satisfied with the information on the User Certification of the correct information on the Questions	☐	☐	☐	☐
3. I am satisfied with the information on the User Certification of the correct information on the Questions	☐	☐	☐	☐
4. I am satisfied with the information on the User Certification of the correct information on the Questions	☐	☐	☐	☐
5. I am satisfied with the information on the User Certification of the correct information on the Questions	☐	☐	☐	☐

5. Frequency - How satisfied are you with the Efficiency of the CitiLink health-care program

Questions

Questions	1 Very Satisfied	2 Satisfied	3 Dissatisfied	4 Very Dissatisfied
1. I am satisfied with the efficiency of the CitiLink health-care program	☐	☐	☐	☐
2. I am satisfied with the efficiency of the CitiLink health-care program	☐	☐	☐	☐
3. I am satisfied with the efficiency of the CitiLink health-care program	☐	☐	☐	☐
4. I am satisfied with the efficiency of the CitiLink health-care program	☐	☐	☐	☐
5. I am satisfied with the efficiency of the CitiLink health-care program	☐	☐	☐	☐

PART 3: Post Implementation Survey (CitiLink user)

Please indicate the extent to which you agree or disagree based on your personal experience in the CitiLink user or the system.

A 5-Star Certification - How satisfied are you with the User Certification of the correct information on the Questions

Questions	1 Very Satisfied	2 Satisfied	3 Dissatisfied	4 Very Dissatisfied
1. I am satisfied with the information on the User Certification of the correct information on the Questions	☐	☐	☐	☐
2. I am satisfied with the information on the User Certification of the correct information on the Questions	☐	☐	☐	☐
3. I am satisfied with the information on the User Certification of the correct information on the Questions	☐	☐	☐	☐
4. I am satisfied with the information on the User Certification of the correct information on the Questions	☐	☐	☐	☐
5. I am satisfied with the information on the User Certification of the correct information on the Questions	☐	☐	☐	☐

[illegible][illegible]

[illegible][illegible][illegible]

[illegible]

Appendix C Table 1 Manual Time Consuming

MANUAL																				
TIME CONSUMING																				
	Q1				Q2				Q3				Q4				MEAN			
LIKELY-SCALE	4	3	2	1	4	3	2	1	4	3	2	1	4	3	2	1	4	3	2	1
Total: 72	11	28	18	15	10	27	22	13	8	29	20	15	10	27	16	19	2.49	2.47	2.42	2.39

Appendix C Table 2 Manual Efficiency

MANUAL																				
EFFICIENCY																				
	Q1				Q2				Q3				Q4				MEAN			
LIKELY-SCALE	4	3	2	1	4	3	2	1	4	3	2	1	4	3	2	1	4	3	2	1
Total: 72	7	27	22	16	7	28	22	15	7	28	18	19	9	25	22	16	2.35	2.38	2.32	2.38

Appendix C Table 3 System Time Consuming

SYSTEM																				
TIME CONSUMING																				
	Q1				Q2				Q3				Q4				MEAN			
LIKELY-SCALE	4	3	2	1	4	3	2	1	4	3	2	1	4	3	2	1	4	3	2	1
				1														3.6	3.6	3.6
Total: 72	51	6	4	1	54	13	5	0	51	17	3	1	51	18	2	1	3.63	8	4	5

Appendix C Table 4 System Efficiency

SYSTEM																				
EFFICIENCY																				
	Q1				Q2				Q3				Q4				MEAN			
LIKELY-SCALE	4	3	2	1	4	3	2	1	4	3	2	1	4	3	2	1	4	3	2	1
				5														3.7	3.7	3.7
Total: 72	7	15	0	0	56	16	0	0	58	13	1	0	57	15	0	0	9	8	9	9

Appendix C Table 5 System Usability

SYSTEM																								
USABILITY																								
Q1					Q2					Q3					Q4					MEAN				
LIKELY-SCALE	4	3	2	1	4	3	2	1	4	3	2	1	4	3	2	1	4	3	2	1				
																	3.7	3.7	3.7	3.7				
Total:	72	57	15	0	0	57	15	0	0	55	17	0	0	53	19	0	0	9	9	6	4			

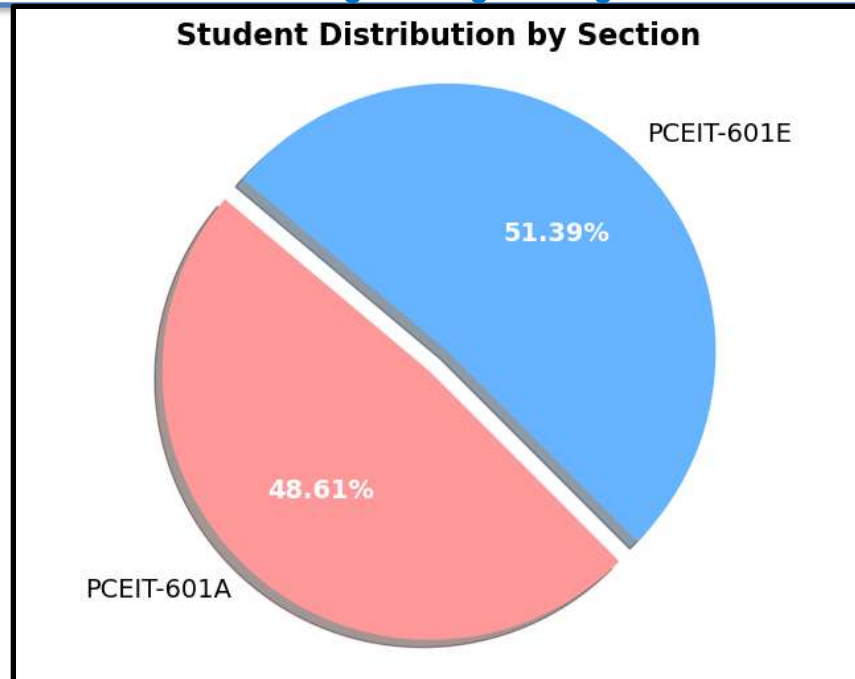
Appendix C Table 6 System Reliability

SYSTEM																					
RELIABILITY																					
		Q1				Q2				Q3				Q4				MEAN			
LIKELY-SCALE	4	3	2	1	4	3	2	1	4	3	2	1	4	3	2	1	4	3	2	1	
																	3.8	3.7	3.7	3.8	
Total:	72	58	14	0	0	54	18	0	0	54	18	0	0	58	14	0	0	1	5	5	1

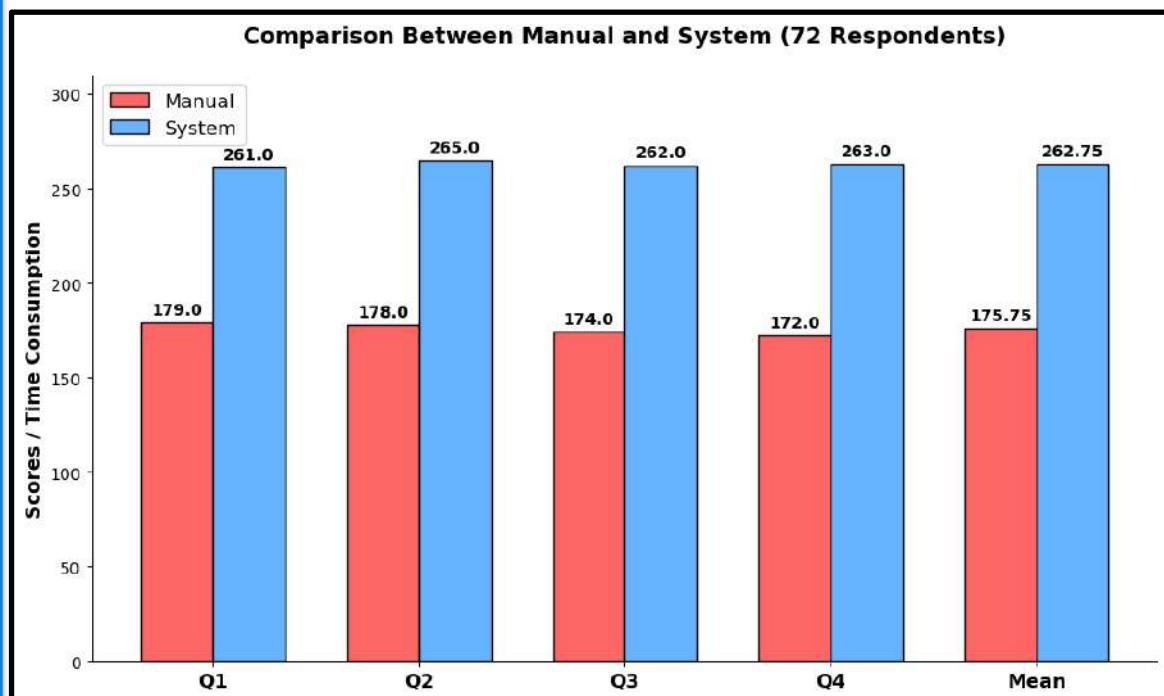
QR Code Linking to the Computation Spreadsheet



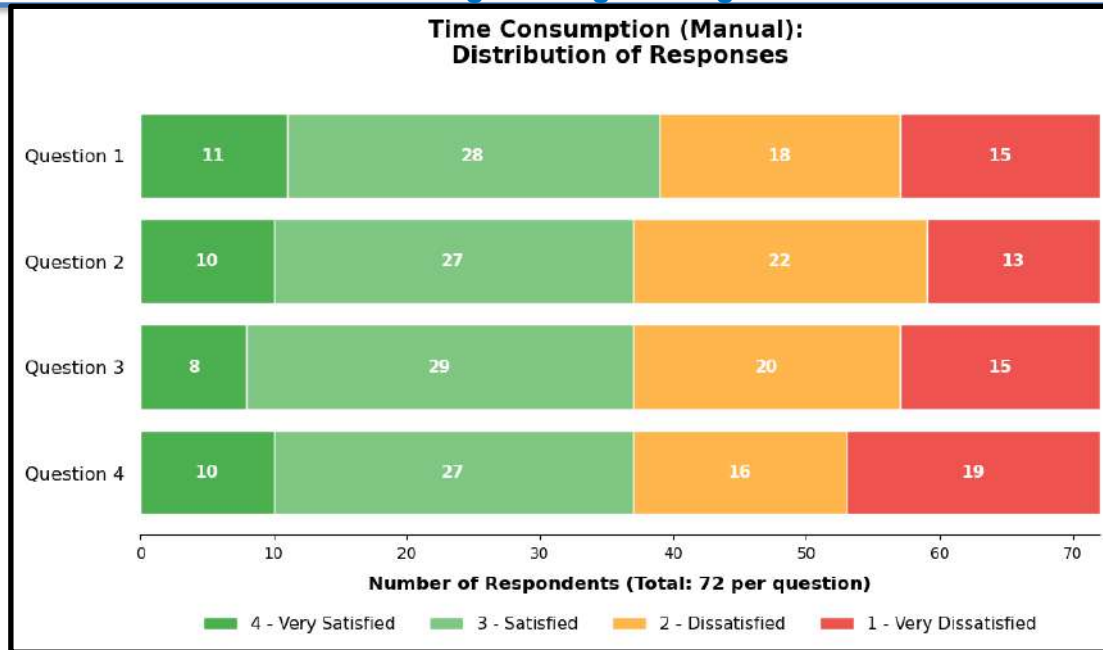
QR Code Computation of Survey Data



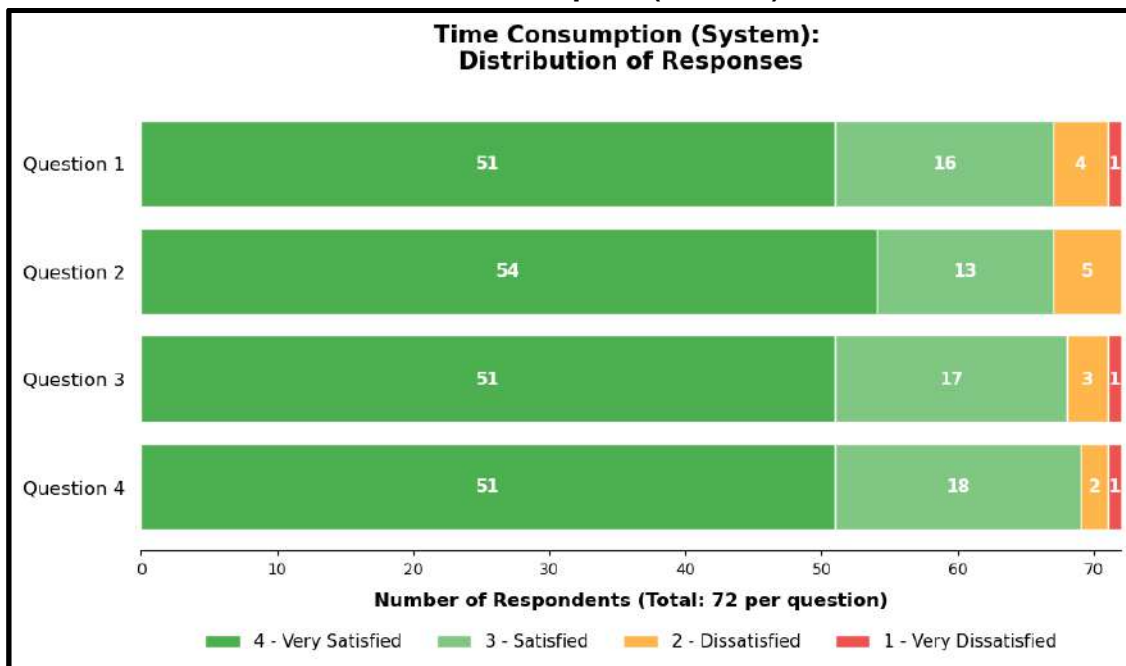
Student Distribution by Section



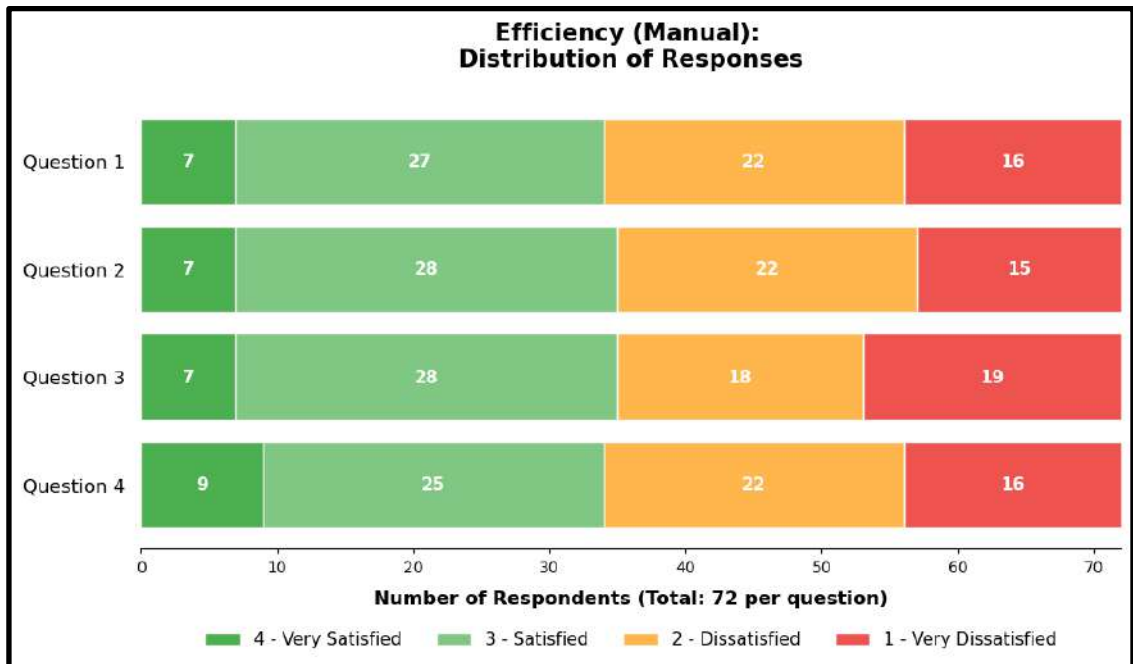
Comparison score of total respondents



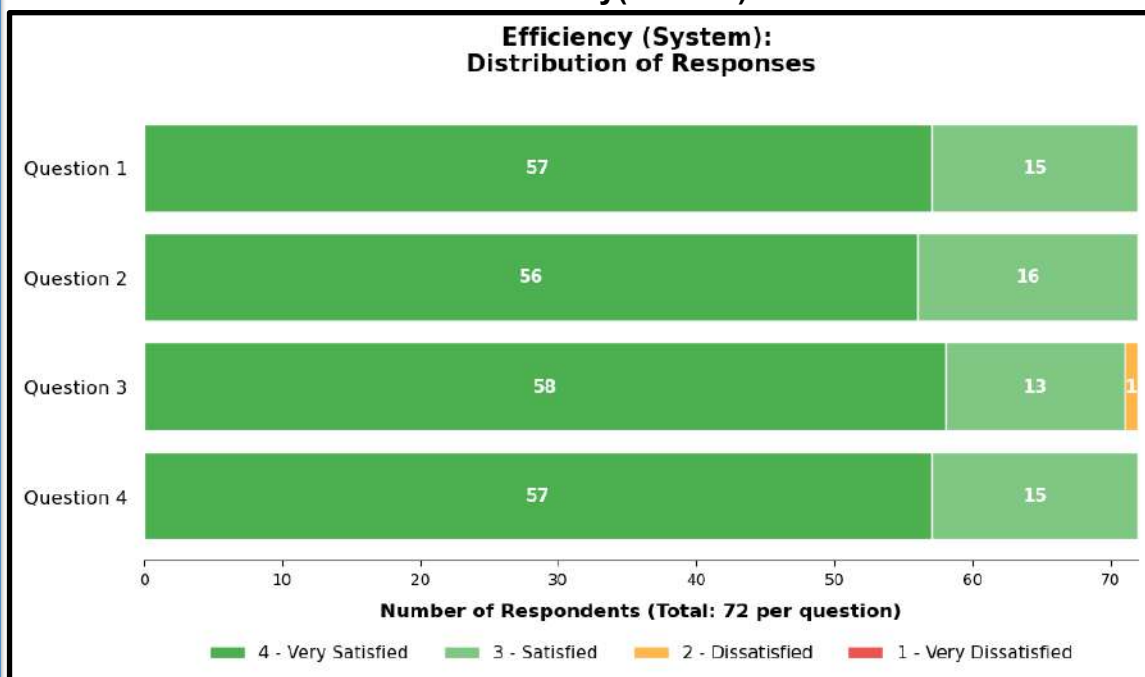
Time Consumption(Manual)



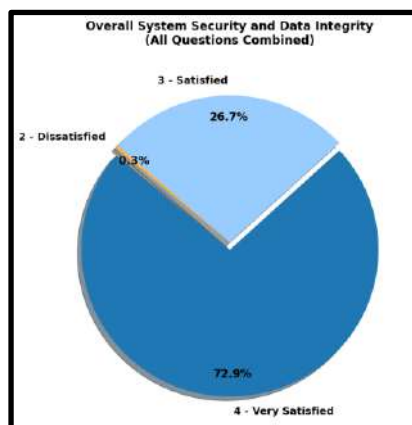
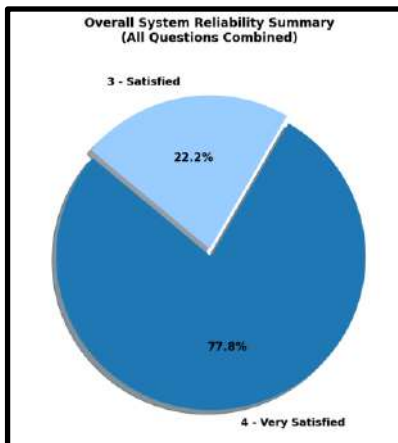
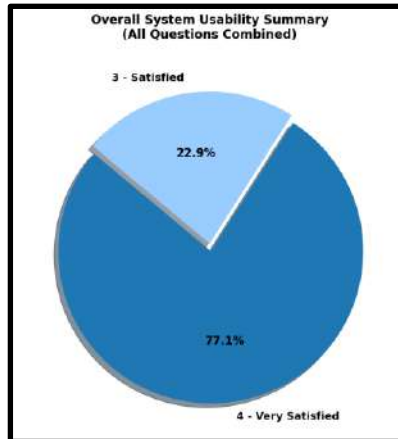
Time Consumption(4 in 1 kiosk)

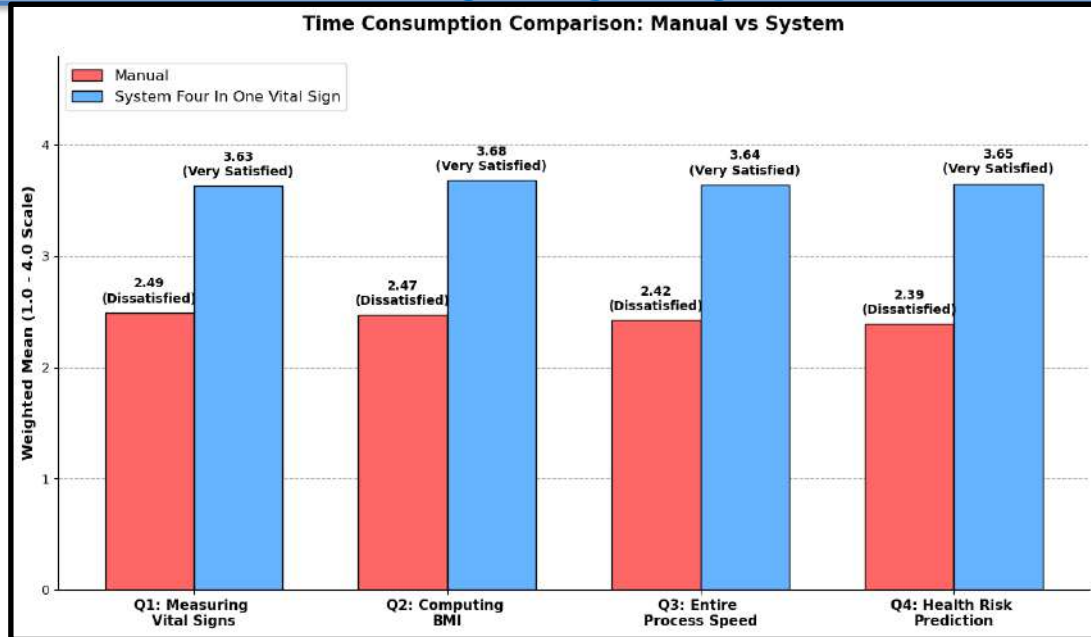


Efficiency(Manual)

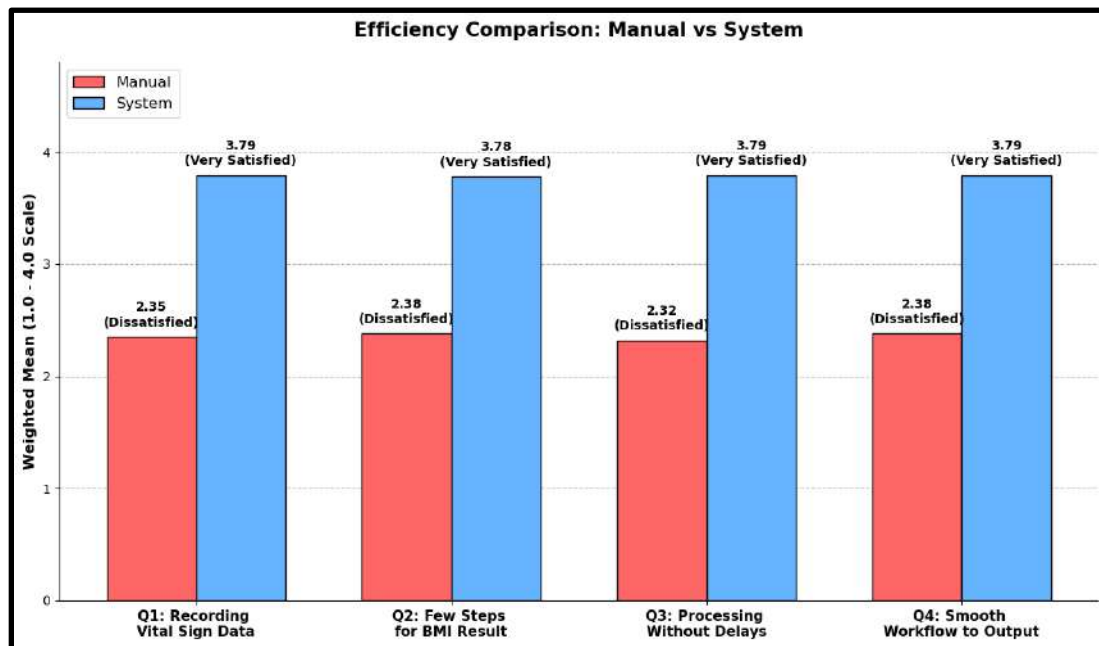


Efficiency(4 in 1 kiosk)





Mean Ratings of Time Consumption



Efficiency Ratings of Time Consumption

APPENDIX D

Supporting Documents

RIZAL TECHNOLOGICAL UNIVERSITY
College of Engineering
Clinical Validation and Accuracy Assessment

174



RIZAL TECHNOLOGICAL UNIVERSITY
College of Engineering, Architecture and Technology
M. Eusebio Avenue, Maybunga, Pasig City



March 03, 2026

Dr. Lusieto B. Oporto, M.D.
University Physician
RTU Pasig Clinic
Rizal Technological University
Pasig City

Subject: Request for Clinical Validation and Accuracy Assessment

Good day Dr. Oporto,

We are fourth-year Computer Engineering students from Rizal Technological University currently conducting our undergraduate thesis entitled: **"FOVB-AIoT: Four-in-One Vital Sign Sensor with BMI Calculation using Artificial Intelligence and IoT for Health Risk Prediction in RTU Pasig Clinic"**

As part of our research, we developed a prototype system capable of measuring heart rate, oxygen saturation (SpO₂), body temperature, blood pressure, height, and weight. The system computes Body Mass Index (BMI) and utilizes an Artificial Intelligence (AI) model to generate a health risk score and corresponding risk interpretation based on the collected vital signs.

The AI component functions strictly as a clinical decision-support tool. It analyzes patterns in the patient's vital signs and provides preliminary health risk suggestions (e.g., possible physiological stress, fatigue indicators, elevated cardiovascular risk, or other abnormal vital sign patterns). These outputs are not intended to serve as medical diagnoses but rather as supportive assessments that require professional medical evaluation.

In this regard, we respectfully request your professional expertise to



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- Evaluate whether the AI-generated health risk interpretations are clinically reasonable and appropriate based on the presented vital signs.
- Provide your professional review if the system operates within acceptable preliminary screening standards.
- Offer recommendations, corrections, or adjustments to improve the clinical reliability of the system.

Your professional validation and endorsement will significantly strengthen the medical credibility of our study and ensure that our system aligns with proper clinical practice.

We would be deeply grateful for your evaluation and, if deemed appropriate, your written approval supporting the clinical soundness of our methodology and AI-based decision-support system.

Thank you very much for your time, guidance, and continued support of student research.

Respectfully yours,

System Chapers

Alcantara, Mar Kevin P.

Berongoy, Bernie C.

Garcia, Khin Andrei R.

Lloria, Reynart G.

Olivarez, Erick John A.

Releva, Paul Andrew A.

Sagadillo, Ian Lorenz C.



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College of Engineering, Architecture and Technology
M. Eusebio Avenue, Maybunga, Pasig City



March 03, 2026

This letter serves to formally confirm that I have reviewed and evaluated the undergraduate thesis project entitled:

"FOVB-AIoT: Four-in-One Vital Sign Sensor with BMI Calculation using Artificial Intelligence and IoT for Health Risk Prediction in RTU Pasig Clinic," developed by fourth-year Computer Engineering students of Rizal Technological University.

I have examined the proposed comparative testing procedure, wherein the prototype's sensor readings are validated against standard medical-grade equipment used in the clinical setting. Based on the documentation and evaluation presented, the methodology for assessing sensor accuracy is appropriate and aligned with acceptable preliminary screening practices.

Furthermore, I have reviewed the AI-based health risk prediction component. The AI model functions as a clinical decision-support tool, providing preliminary health risk suggestions derived from measured vital signs. It is clearly stated that the system does not replace professional medical diagnosis and is intended solely to assist in early risk identification and monitoring. The interpretation framework and risk classification approach are clinically reasonable within the scope of non-diagnostic screening applications.

In addition, the Body Mass Index (BMI) classification used in the system is based on Asia-Pacific clinical guidelines. These standards are appropriate for the intended users of the system, as the population being assessed primarily belongs to the Asia-Pacific region. The blood pressure categories are therefore clinical practice guidelines based on the American Heart Association.

Based on my professional assessment, the overall process including sensor validation, data interpretation, regional standard alignment, and AI-generated health risk scoring has undergone professional review and is deemed appropriate for preliminary health screening purposes under proper supervision. The system demonstrates alignment with basic clinical standards for non-invasive vital sign monitoring and decision-support implementation.



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This approval is granted for academic and research purposes in support of the students' thesis study and validation activities within the RTU Pasig Clinic.

Issued this 3 day of March, 2026.

Sincerely,

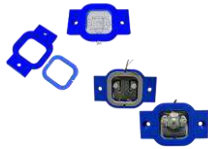
Dr. Lusieto B. Oporto, M.D.
University Physician

APPENDIX E

System Documentation

Appendix E Table 1 Hardware Requirements

Materials	Description
 Mini PC (Dell i5 10th Gen)	1. Dell Mini PC (Intel i5 10th Gen) handles data processing, storage, and AI-based health risk analysis. It communicates with the Arduino and database for real-time monitoring, record-keeping, and web-based reporting.
 Fortress CF 160TM	2. Fortress CF160TM 15.6" Capacitive Touchscreen Monitor functions as the primary user interface, allowing users to interact with the system, display real-time readings, and view AI-generated results conveniently.
 Arduino Mega	3. Arduino Mega serves as the main microcontroller that receives and processes data from all sensors, transmitting results to the connected mini PC and display unit. Its multiple input/output pins support simultaneous integration of components.



Load Cell Case

4. **Load Cell with HX711 Module** measures weight by converting mechanical force into an electrical signal. The amplified signal is sent to the microcontroller for BMI computation.



TF-Luna TOF Lidar Sensor

5. **TF Luna LiDAR Sensor** measures height using LiDAR (Light Detection and Ranging) technology by emitting laser pulses and calculating the reflected light's travel time, offering fast and precise results.



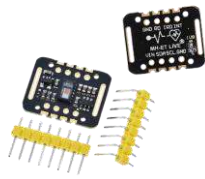
DFRobot MLX90614 Temp
Sensor

6. **MLX90614 Infrared Sensor** measures body temperature without physical contact by detecting infrared radiation emitted by the skin, ensuring quick and hygienic readings.



RFID Card Reader

7. **RFID** module checks if a user is registered when the card is tapped. Unregistered users are asked to enter basic info (name, age, sex, student/employee number, and school ID) which is then saved to the database.



**MAX30102 Pulse Oximeter
Sensor**

8. **MAX30102 Sensor** reads heart rate, respiratory rate, and oxygen saturation (SpO_2) using photoplethysmography (PPG) technology for real-time, non-invasive measurement.



**Digital Blood Pressure
Monitor**

9. **Blood pressure** is used to measure both systolic and diastolic pressure by detecting changes in arterial pressure during cuff inflation and deflation. In the system, the module generates blood pressure readings that are displayed on the digital BP device, which are then captured by a camera and processed using object detection techniques. This approach enables the extraction of essential cardiovascular data for further processing and display within the customized health assessment system.



Thermal Receipt Printer

10. **Thermal Printer** generates printed copies of the user's health results and recommendations after each session for personal reference or submission to the clinic.



Webcam 1080P Full HD

11. **Webcam** to accurately capture real-time video input for object detection. It serves as the primary visual sensor, allowing the AI model to identify and monitor any items or objects that users may bring during the health check process. The webcam ensures consistent image quality and stable frame rates, enabling reliable detection and preventing misuse or interference with the device.



Arduino NANO

12. **Arduino NANO** serves as the primary microcontroller for the digital BP device. It controls and interfaces with the LCD screen, LED, and user input button within the BP device. The Nano receives processed measurement data from the mini PC, which uses a camera and object detection algorithms to extract blood pressure readings from the digital BP display. Upon receiving this data, the Arduino Nano updates the LCD screen to display the detected results, lights the LED for camera visibility, and handles button inputs for system operation.



LCD Screen (16x2)

13. **LCD Screen** is used in the customized digital blood pressure (BP) device as an external indicator display. It provides real-time readings of the blood pressure measurement and status, allowing users to easily view measurement results during the BP measurement process.



LED

14. **LEDs** are installed inside the digital BP device to have enough lighting for the camera to capture and interpret blood pressure readings displayed on the digital BP screen. The LEDs ensure proper visibility, reduce shadows, and improve image quality, thereby increasing the accuracy and reliability of the detection process.



Button

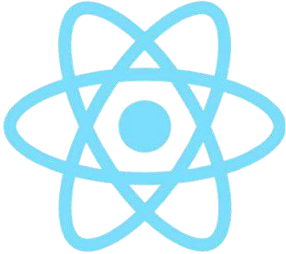
15. **Button** serves as the user input control for initiating the blood pressure (BP) measurement to allow users to start the BP measurement process. When pressed, the button sends a signal to the Arduino Nano activating the digital BP device.



Speaker (Voice Assistance)

16. **Speaker** - an output component of the FOVB-AIoT system that provides **voice assistance and audio feedback** to users during system operation. It guides users through the registration and measurement process by spoken instructions, prompts, and notifications. The speaker enhances accessibility by supporting users and ensures smooth interaction with the kiosk.

Appendix E Table 2 Software Requirements

Software	Description
 React JS Framework	1. Flask Framework will be used as the backend of our system to manage communication between the hardware, database, and web interface. It will handle sensor data sent from the Arduino, process it in real time, and connect it to the AI model for HRP.



Flask Framework

2. **React JS Framework** will be used to develop the system's modern user interface, allowing users and clinic staff to view real-time vital sign readings and AI-based predictions. It will make the web application dynamic and responsive, updating displayed values instantly without refreshing the page.



HTML (HyperText Markup Language)

3. **HTML (HyperText Markup Language)** will be used to structure the web interface of the system. It will organize the layout for displaying vital signs, BMI results, and system alerts in a clean and readable format.



CSS (Cascading Style Sheets)

4. **CSS (Cascading Style Sheets)** will be used to design and style the system's interface. It will ensure that the web application looks modern, easy to use, and visually consistent with the clinic's health monitoring theme.



Javascript

5. **JavaScript** will be used to make the frontend interactive and responsive. It will handle user actions such as starting measurements, displaying updated data, and validating inputs in real time.



Python

6. **Python** will be used to process sensor readings and perform calculations like BMI and average health metrics. It will also act as the bridge between the sensors, AI model, and Flask server for smooth data processing.



XGBoost

7. **XGBoost** will be used to build and integrate the AI model into the system. It will analyze the collected vital sign data and BMI to predict possible health risks, helping the system provide early alerts and recommendations.



8. **GPT OSS 120B** will be used to provide an independent, secondary analysis of the collected vital signs and the initial risk score. It will act as a validation layer to cross-reference the data and generate supplementary health suggestions.



9. **Gemini 2.0** will be used as the primary semantic reasoning engine to finalize the health assessment. It will compare its own analysis against XGBoost and GPT OSS to establish a consensus risk score, and then generate clear, user-friendly preventative health recommendations.



YOLO (You Only Look Once)

8. **YOLO** AI model for object detection to automatically identify and filter out unnecessary objects in the measurement area. This ensures that only relevant items are considered, improving the accuracy and reliability of our data collection.



Roboflow

9. **Roboflow** is used to manage datasets and train the YOLOv8 object detection model. It supports image annotation, preprocessing, and augmentation to prepare training data, enabling the system to accurately detect blood pressure readings from the digital BP device.



Google Colab

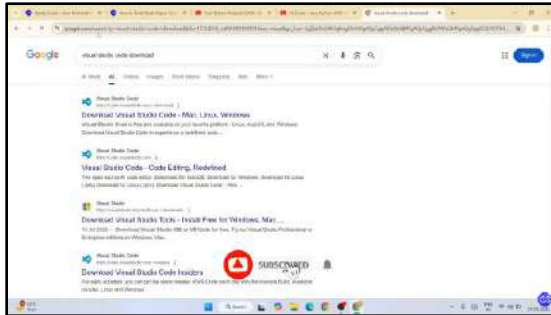
10. **Google Colab** is utilized as a cloud-based development environment for training and testing the object detection model. It provides access to GPU acceleration, for faster training of the YOLOv8 model. Google Colab is used in conjunction with Roboflow to train, evaluate, and refine the AI model for accurate detection of blood pressure values from captured images.



Tailscale

11. **Tailscale** is used in the system to create a secure private network that connects the IoT device and the server over the internet. It allows safe and encrypted transmission of vital sign data without requiring complex network configuration.

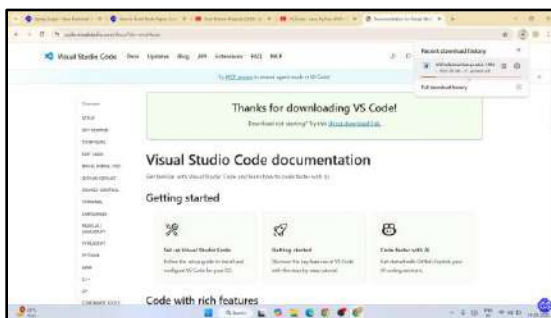
Installation of Visual Studio



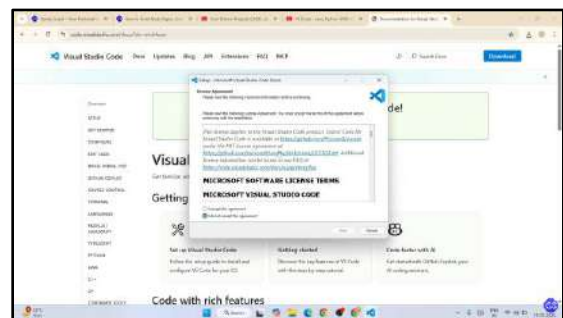
Open a web browser, search for “Visual Studio Code download,” and select the official Visual Studio Code website from the search results.



Click “MySQL Installer for Windows” from the MySQL Community Downloads page.



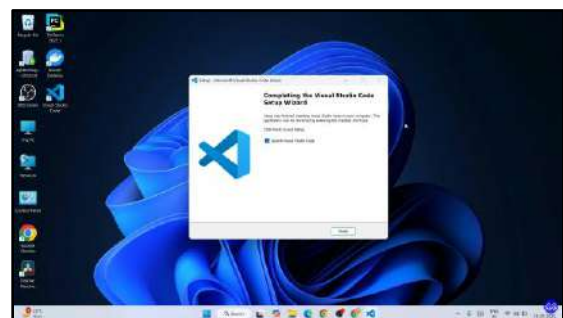
Once the download starts, wait for the installer file to finish downloading before proceeding with the installation process.



Run the installer and review the License Agreement, then select “I accept the agreement” to continue with the setup.

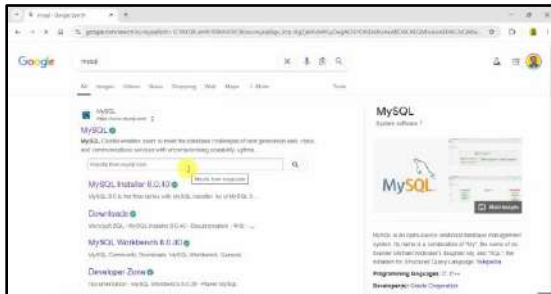


In the Select Additional Tasks window, choose the preferred options such as creating a desktop icon and adding Visual Studio Code to the system PATH, then click Next to proceed.

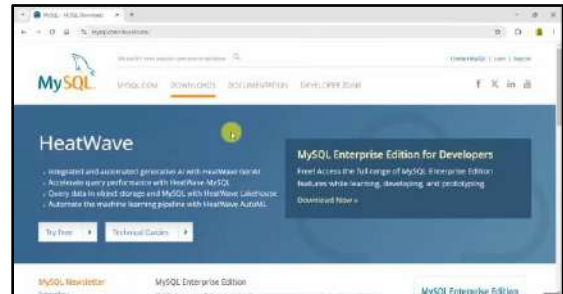


After the installation process is completed, click Finish to exit the Setup Wizard and launch Visual Studio Code successfully.

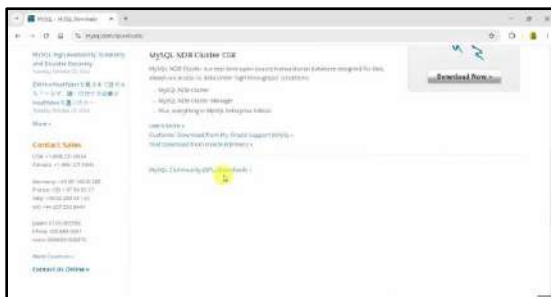
Installation of MySQL



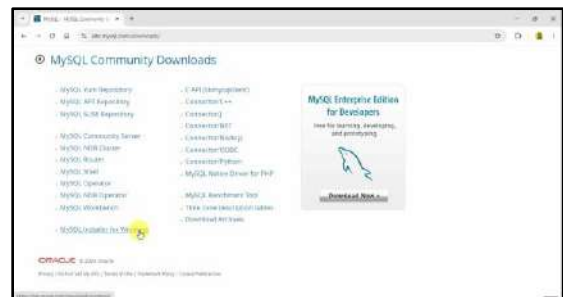
Search **“MySQL”** in Google to display the official MySQL website in the search results.



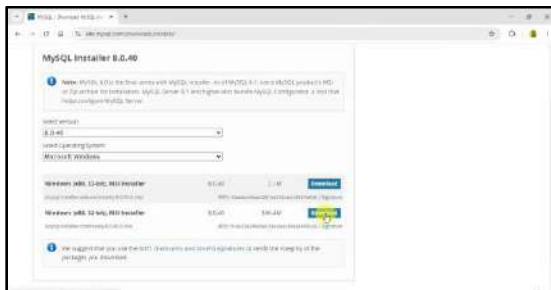
From the search results, select the official MySQL website (www.mysql.com).



Scroll down and click **MySQL Community (GPL) Downloads**.



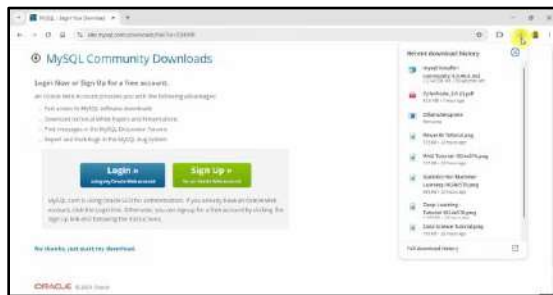
Select **MySQL Installer for Windows** from the list of available downloads.



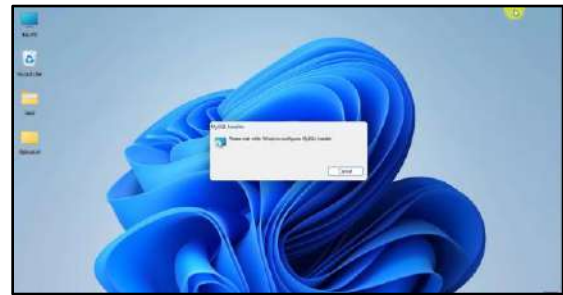
On the Community Downloads page, click **MySQL Installer for Windows**.



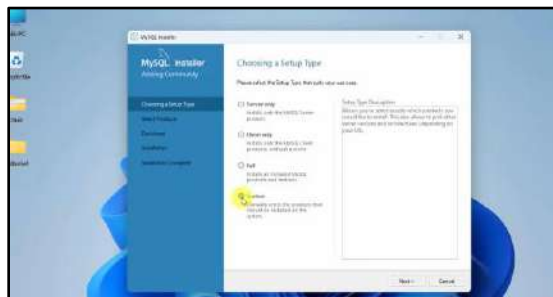
Click **“No thanks, just start my download”** to continue.



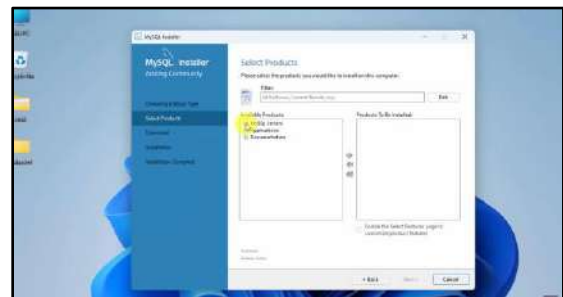
You can see the MySQL application downloading in the browser's download history



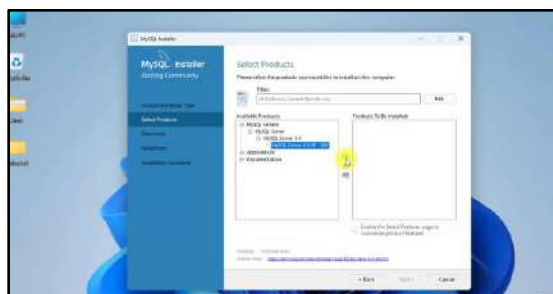
You can now install the application using the MySQL Installer.



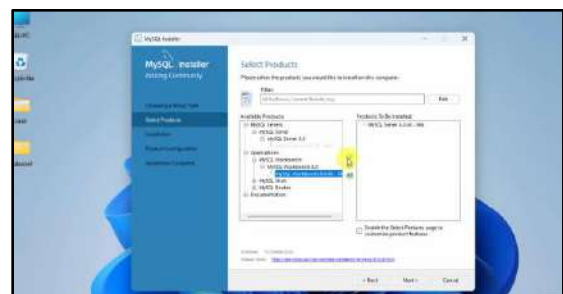
In the Setup Type window, select **Custom** installation.



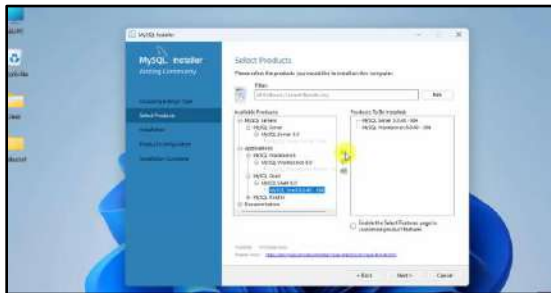
In the Select Products window, choose **MySQL Server 8.0.x (x64)** and add it.



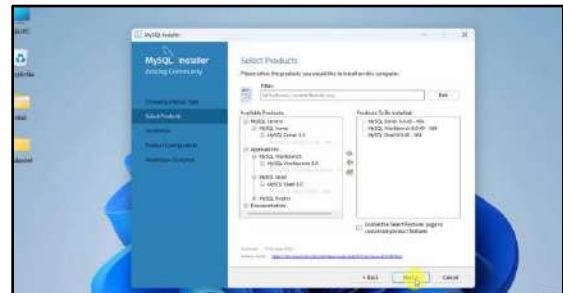
Add **MySQL Workbench 8.0.x (x64)** to the selected products



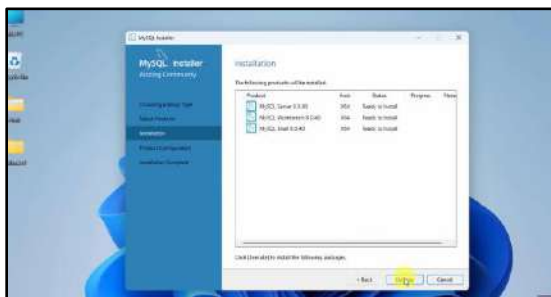
Add **MySQL Workbench 8.0.x (x64)** and click **Next**.



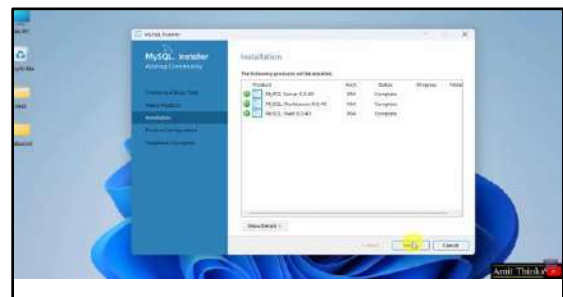
Add **MySQL Shell 8.0.x (x64)** and click **Next**.



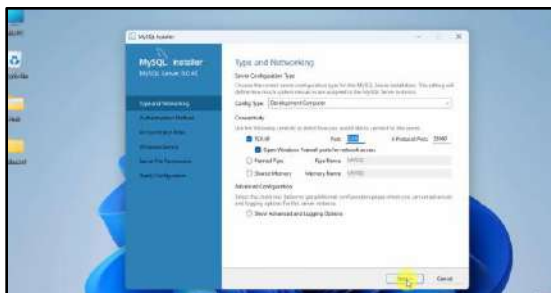
Click **Next** for continuation



Click **Execute** to begin installing the selected components.



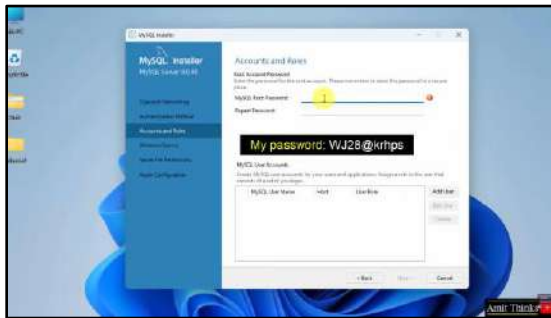
Wait until all products are successfully installed, then click **Next**.



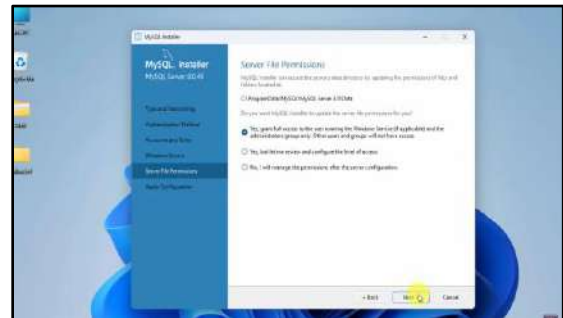
In the *Type and Networking* section, select **Development Computer** and keep the default port (3306).



In the *Authentication Method* section, select **Use Strong Password Encryption (Recommended)**.



In the *Accounts and Roles* section, create and confirm the **root** password.



In the *Server File Permissions* section, select the recommended option.



Click "Execute" to apply the configuration settings and wait for the process to complete.



After the configuration is successfully completed, click "Finish" to finalize the installation.

Source Code (Arduino)

```
#include <Wire.h>
#include <HX711_ADC.h>
#include <TFLI2C.h>
#include <Adafruit_MLX90614.h>
#include "MAX30105.h"
#include "heartRate.h"
#include "spo2_algorithm.h"

HX711_ADC LoadCell(4, 5); TFLI2C heightSensor; Adafruit_MLX90614 mlx;
MAX30105 particleSensor;

const          float          WEIGHT_CALIBRATION_FACTOR=21333.55,
SENSOR_MOUNT_HEIGHT_CM=213.36,      HUMAN_MIN_VALID_TEMP=35.0,
HUMAN_MAX_VALID_TEMP=43.0, CALIBRATION_BIAS=3.5;
const  int    TEMP_SAMPLES_PER_READING=10,    BPM_DEDUCTION=25,
DATA_PRINT_INTERVAL=200;
#define BUFFER_SIZE 50
#define HR_HISTORY 5
#define FINGER_THRESHOLD 30000
#define RR_DEDUCTION 4

bool  weightActive=0,  heightActive=0,  tempActive=0,  max30102Active=0,
weightSensorReady=0,      tempSensorInitialized=0,      max30102Initialized=0,
stringComplete=0, hrFilled=0, measurementStarted=0, fingerDetected=0;
unsigned long lastWeightPrint=0, lastHeightPrint=0, lastTempPrint=0;
int16_t tfDist=0, tfFlux=0, tfTemp=0;
String inputString="", signalQuality="WAITING";
```

```

uint32_t irBuffer[BUFFER_SIZE], redBuffer[BUFFER_SIZE];
int32_t spo2, heartRate; int8_t validSPO2, validHeartRate;
float respiratoryRate=0, perfusionIndex=0;
int hrHistory[HR_HISTORY], stableHR=0;
byte hrIndex=0, fingerDetectedCounter=0, fingerRemovedCounter=0;

int median(int *arr, int size) {
    int temp[HR_HISTORY]; for(int i=0;i<size;i++) temp[i]=arr[i];
    for(int i=0;i<size-1;i++) for(int j=i+1;j<size;j++) if(temp[j]<temp[i]) { int t=temp[i];
temp[i]=temp[j]; temp[j]=t; }
    return temp[size/2];
}

void updateStableHR(int newHR) {
    if(newHR<40||newHR>180 || (stableHR!=0 && abs(newHR-stableHR)>20)) return;
    hrHistory[hrIndex++]=newHR;                if(hrIndex>=HR_HISTORY){hrIndex=0;
hrFilled=true;}
    stableHR = hrFilled ? median(hrHistory, HR_HISTORY) : newHR;
}

int estimateRespiratoryRate(int bpm) { return constrain(bpm<40 ? 8 : (bpm<=100 ?
bpm/4.0 : (bpm<=140 ? bpm/4.5 : bpm/5.0)), 8, 40); }

float calculatePI(uint32_t* buf, int size) {
    if(size<2) return 0; uint32_t minV=buf[0], maxV=buf[0]; uint64_t sum=0;
    for(int i=0;i<size;i++){ if(buf[i]<minV) minV=buf[i]; if(buf[i]>maxV) maxV=buf[i];
sum+=buf[i]; }
    float dc=(float)sum/size; return dc<1 ? 0 : ((float)(maxV-minV)/dc)*100.0;
}

String          getSignalQuality(float          pi)          {          return
pi>=2.0?"EXCELLENT":pi>=1.0?"GOOD":pi>=0.5?"FAIR":pi>=0.2?"WEAK":"POOR"
; }

```

```
float calculateDynamicOffset(float amb) { return amb<=15.0?1.0:amb<=25.0?1.0-
((amb-15.0)*(0.5/10.0)):amb<=30.0?0.5-((amb-25.0)*(0.3/5.0)):0.1; }

void startAutoTare() {
    Serial.println("STATUS:TARE_STARTED"); LoadCell.start(2000, true);
    if(LoadCell.getTareTimeoutFlag())
        Serial.println("ERROR:WEIGHT_SENSOR_TIMEOUT");
    else { LoadCell.setCalFactor(WEIGHT_CALIBRATION_FACTOR);
        weightSensorReady=true;
        Serial.println("STATUS:AUTO_TARE_COMPLETE\nSTATUS:WEIGHT_SENSOR_
READY"); }
}

void startMaxMeasurement() {
    measurementStarted=true; hrIndex=0; hrFilled=false; stableHR=0;
    for(int i=0;i<HR_HISTORY;i++) hrHistory[i]=0;

    Serial.println("STATUS:MAX30102_MEASUREMENT_STARTED\nMAX30102_STA
TE:MEASURING");
}

void stopMaxMeasurement() { measurementStarted=fingerDetected=false;
    Serial.println("MAX30102_STATE:MEASUREMENT_STOPPED_FINGER_REMOV
ED\nFINGER_REMOVED"); }

void processCommand(String cmd) {
    cmd.trim();
    if(cmd=="AUTO_TARE"||cmd=="INITIALIZE_WEIGHT") startAutoTare();
    else if(cmd=="START_WEIGHT") {weightActive=true;
        Serial.println("STATUS:WEIGHT_MEASUREMENT_STARTED\nSTATUS:WEIGHT
_SENSOR_POWERED_UP");}
```

```

else if(cmd=="POWER_DOWN_WEIGHT"||cmd=="STOP_WEIGHT")
{weightActive=false;
Serial.println("STATUS:WEIGHT_MEASUREMENT_COMPLETE\nSTATUS:WEIGH
T_SENSOR_POWERED_DOWN");}
else if(cmd=="TARE_WEIGHT") {LoadCell.tareNoDelay();
Serial.println("STATUS:TARE_COMPLETE");}
else if(cmd=="START_HEIGHT") {heightActive=true;
Serial.println("STATUS:HEIGHT_MEASUREMENT_STARTED\nSTATUS:HEIGHT_
SENSOR_POWERED_UP");}
else if(cmd=="POWER_DOWN_HEIGHT"||cmd=="STOP_HEIGHT")
{heightActive=false;
Serial.println("STATUS:HEIGHT_MEASUREMENT_COMPLETE\nSTATUS:HEIGHT
_SENSOR_POWERED_DOWN");}
else if(cmd=="START_TEMPERATURE") {
if(tempSensorInitialized||mlx.begin()){tempSensorInitialized=tempActive=true;
Serial.println("STATUS:TEMPERATURE_MEASUREMENT_STARTED\nSTATUS:T
EMPERATURE_SENSOR_POWERED_UP");}
else Serial.println("ERROR:TEMPERATURE_SENSOR_NOT_INITIALIZED");
}
else
if(cmd=="POWER_DOWN_TEMPERATURE"||cmd=="STOP_TEMPERATURE")
{tempActive=false;
Serial.println("STATUS:TEMPERATURE_MEASUREMENT_COMPLETE\nSTATUS:
TEMPERATURE_SENSOR_POWERED_DOWN");}
else if(cmd=="POWER_UP_MAX30102"||cmd=="START_MAX30102") {
if(max30102Initialized||particleSensor.begin(Wire, I2C_SPEED_STANDARD)){
max30102Initialized=max30102Active=true; particleSensor.setup();
particleSensor.setPulseAmplitudeRed(0x32);
particleSensor.setPulseAmplitudeGreen(0);

```

```

fingerDetected=measurementStarted=fingerDetectedCounter=fingerRemovedCounter=0; delay(200);
    if((particleSensor.getIR()>FINGER_THRESHOLD && (delay(50),
particleSensor.getIR()>FINGER_THRESHOLD)) {
        fingerDetected=true;                fingerDetectedCounter=2;
Serial.println("FINGER_DETECTED"); startMaxMeasurement();
    }

Serial.println("STATUS:MAX30102_SENSOR_POWERED_UP\nSTATUS:MAX30102_MEASUREMENT_STARTED");
    } else Serial.println("ERROR:MAX30102_NOT_FOUND");
    }
    else if(cmd=="POWER_DOWN_MAX30102"||cmd=="STOP_MAX30102") {
        max30102Active=fingerDetected=measurementStarted=false;
particleSensor.shutDown();
Serial.println("STATUS:MAX30102_SENSOR_POWERED_DOWN\nSTATUS:MAX30102_MEASUREMENT_COMPLETE");
    }
    else if(cmd=="FULL_INITIALIZE") startAutoTare();
}

void monitorFingerPresence() {
    static unsigned long last=0; static byte logCtr=0;
    if(millis()-last>50) {
        long                ir=particleSensor.getIR();                if(++logCtr>=4){
Serial.print("MAX30102_IR_VALUE:"); Serial.println(ir); logCtr=0; }

```

```

    if(ir>FINGER_THRESHOLD){fingerDetectedCounter++;
fingerRemovedCounter=0;}                                else{fingerRemovedCounter++;
fingerDetectedCounter=0;}
    if(fingerDetectedCounter>=1      &&      !fingerDetected){fingerDetected=true;
fingerDetectedCounter=0;                Serial.println("FINGER_DETECTED");
startMaxMeasurement();}
    else if(fingerRemovedCounter>=1 && fingerDetected){fingerRemovedCounter=0;
stopMaxMeasurement();}
    last=millis();
}
}

void runMeasurementPhase() {
    if(!measurementStarted) return; static byte bufIdx=0; particleSensor.check();
    while(particleSensor.available()) {
        redBuffer[bufIdx]=particleSensor.getRed(); irBuffer[bufIdx]=particleSensor.getIR();
particleSensor.nextSample(); bufIdx++;
        if(bufIdx==BUFFER_SIZE) {
            perfusionIndex=calculatePI(irBuffer,                BUFFER_SIZE);
signalQuality=getSignalQuality(perfusionIndex);
            maxim_heart_rate_and_oxygen_saturation(irBuffer, BUFFER_SIZE, redBuffer,
&spo2, &validSPO2, &heartRate, &validHeartRate);
            if(heartRate>0) { updateStableHR(constrain(heartRate-BPM_DEDUCTION, 40,
180));                respiratoryRate=constrain(estimateRespiratoryRate(stableHR)-
RR_DEDUCTION, 8, 40); }
            if(spo2>0 && stableHR>0) {
                spo2 = spo2>=95 ? random(96,100) : (spo2>=90 ? 95 : random(90,96));
                stableHR = stableHR<60 ? random(60,66) : (stableHR>115 ? random(110,116)
: constrain(stableHR+random(-3,4), 60, 115));

```

```

        int fRR=respiratoryRate,
ded=fRR>=30?random(8,12):fRR>=24?random(5,9):fRR>=20?random(3,6):fRR>=1
6?random(1,4):fRR>=12?random(0,2):random(-2,1);
        respiratoryRate=constrain(fRR-ded, 12, 30);
        Serial.print("MAX30102_LIVE_DATA:HR=");        Serial.print(stableHR);
Serial.print(",SPO2=");        Serial.print(spo2);        Serial.print(",RR=");
Serial.print((int)respiratoryRate); Serial.print(",PI="); Serial.print(perfusionIndex, 2);
Serial.print(",QUALITY="); Serial.println(signalQuality);
    }
    bufIdx=0;
}
}
}

void setup() {
    Serial.begin(115200);        delay(100);
    Serial.println("=====\\nSYSTEM:BO
OTING_UP");
    Wire.begin(); Wire.setWireTimeout(3000, true); LoadCell.begin();
    tempSensorInitialized = mlx.begin(); Serial.println(tempSensorInitialized ?
"STATUS:TEMPERATURE_SENSOR_INITIALIZED" :
"ERROR:MLX90614_NOT_FOUND");
    if(!(max30102Initialized=particleSensor.begin(Wire, I2C_SPEED_STANDARD)))
Serial.println("ERROR:MAX30102_NOT_FOUND");
    else { Serial.println("STATUS:MAX30102_INITIALIZED"); particleSensor.setup(); }
    Serial.println("SYSTEM:READY_FOR_COMMANDS");        startAutoTare();
    Serial.println("=====");
    inputString.reserve(200);
}

```

```

void loop() {
  if(stringComplete) {      processCommand(inputString);      inputString="";
stringComplete=false; }
  if(LoadCell.update()      &&      weightActive      &&      millis()-
lastWeightPrint>DATA_PRINT_INTERVAL) { Serial.print("DEBUG:Weight reading: ");
Serial.println(LoadCell.getData(), 2); lastWeightPrint=millis(); }
  if(heightActive && millis()-lastHeightPrint>DATA_PRINT_INTERVAL) {
    if(heightSensor.getData(tfDist,  tfFlux,  tfTemp,  0x10)  &&  tfDist>0)  {
Serial.print("DEBUG:Height              reading:              ");
Serial.println(SENSOR_MOUNT_HEIGHT_CM-tfDist, 1); }
    lastHeightPrint=millis();
  }
  if(tempActive      &&      tempSensorInitialized      &&      millis()-
lastTempPrint>DATA_PRINT_INTERVAL) {
    float sumObj=0, sumAmb=0; int val=0;
    for(int      i=0;      i<TEMP_SAMPLES_PER_READING+5      &&
val<TEMP_SAMPLES_PER_READING; i++) {
      float rO=mlx.readObjectTempC(), rA=mlx.readAmbientTempC();
      if(rO>0&&rO<100&&rA>0&&rA<50){sumObj+=rO;      sumAmb+=rA;      val++;}
delay(20);
    }
    if(val>0) {
      float aO=sumObj/val, aA=sumAmb/val;
      if(!isnan(aO)&&aO>0) {
        float      off=constrain(calculateDynamicOffset(aA),0.0,1.0),
bT=aO+CALIBRATION_BIAS+off;

```

```
Serial.print("Amb: "); Serial.print(aA, 1); Serial.print("C | Raw: "); Serial.print(aO,
1); Serial.print("C | Bias: +"); Serial.print(CALIBRATION_BIAS, 1); Serial.print(" |
Offset: +"); Serial.println(off, 2);

    if(bT<HUMAN_MIN_VALID_TEMP) { Serial.print(" -> READ: "); Serial.print(bT,
1); Serial.println("C [IGNORED: NO HUMAN DETECTED]"); }

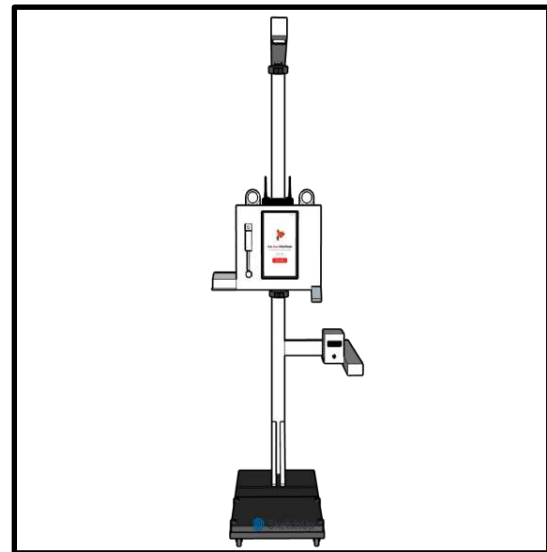
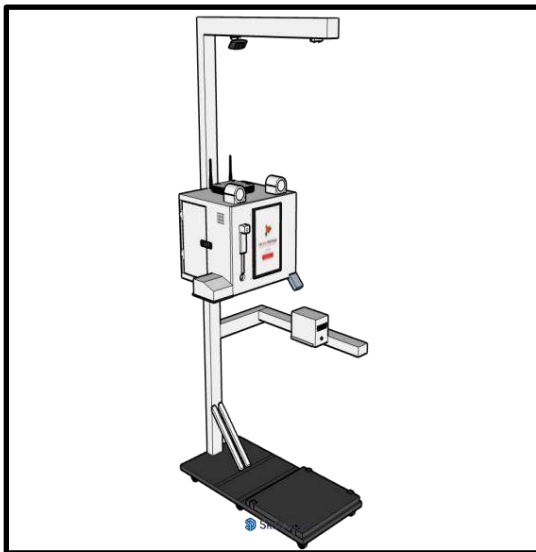
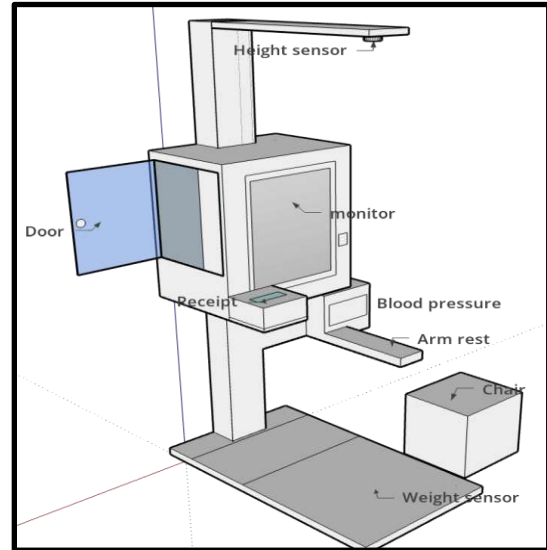
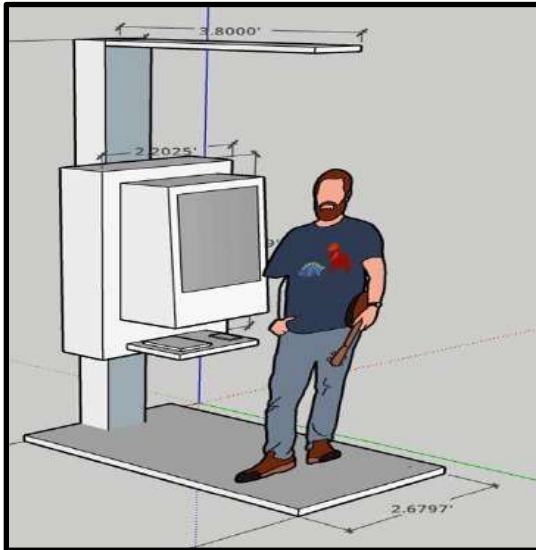
    else if(bT>HUMAN_MAX_VALID_TEMP) { Serial.print(" -> READ: ");
Serial.print(bT, 1); Serial.println("C [IGNORED: INVALID HIGH]"); }

    else {
        Serial.print(" -> BODY: "); Serial.print(bT, 1); Serial.print(" C [");
Serial.print(bT<=37.2?"Normal":bT<=38.0?"Slight fever":"Critical"); Serial.println("]");
        Serial.print("DEBUG:Temperature reading: "); Serial.println(bT, 2);
    }
}
}
}

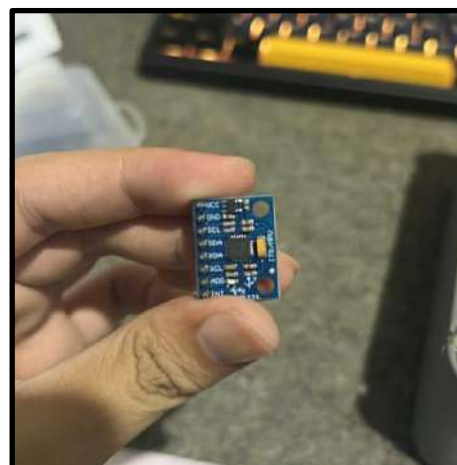
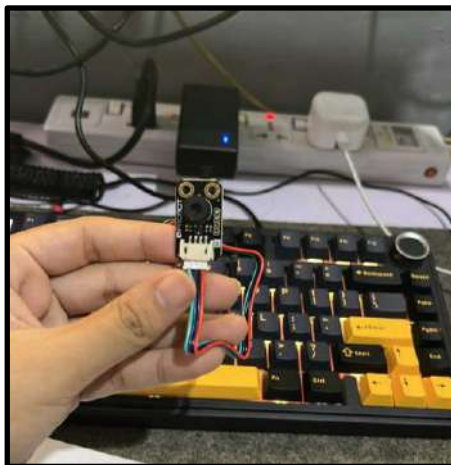
lastTempPrint=millis();
}

if(max30102Active && max30102Initialized) { monitorFingerPresence();
if(measurementStarted && fingerDetected) runMeasurementPhase(); }
}

void serialEvent() { while(Serial.available()){ char c=(char)Serial.read(); if(c=='\n')
stringComplete=true; else inputString+=c; } }
```



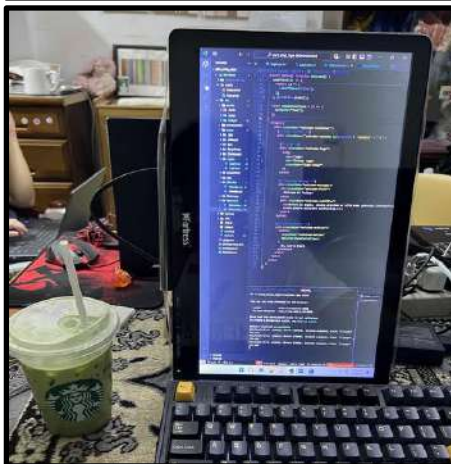
Designing Prototype



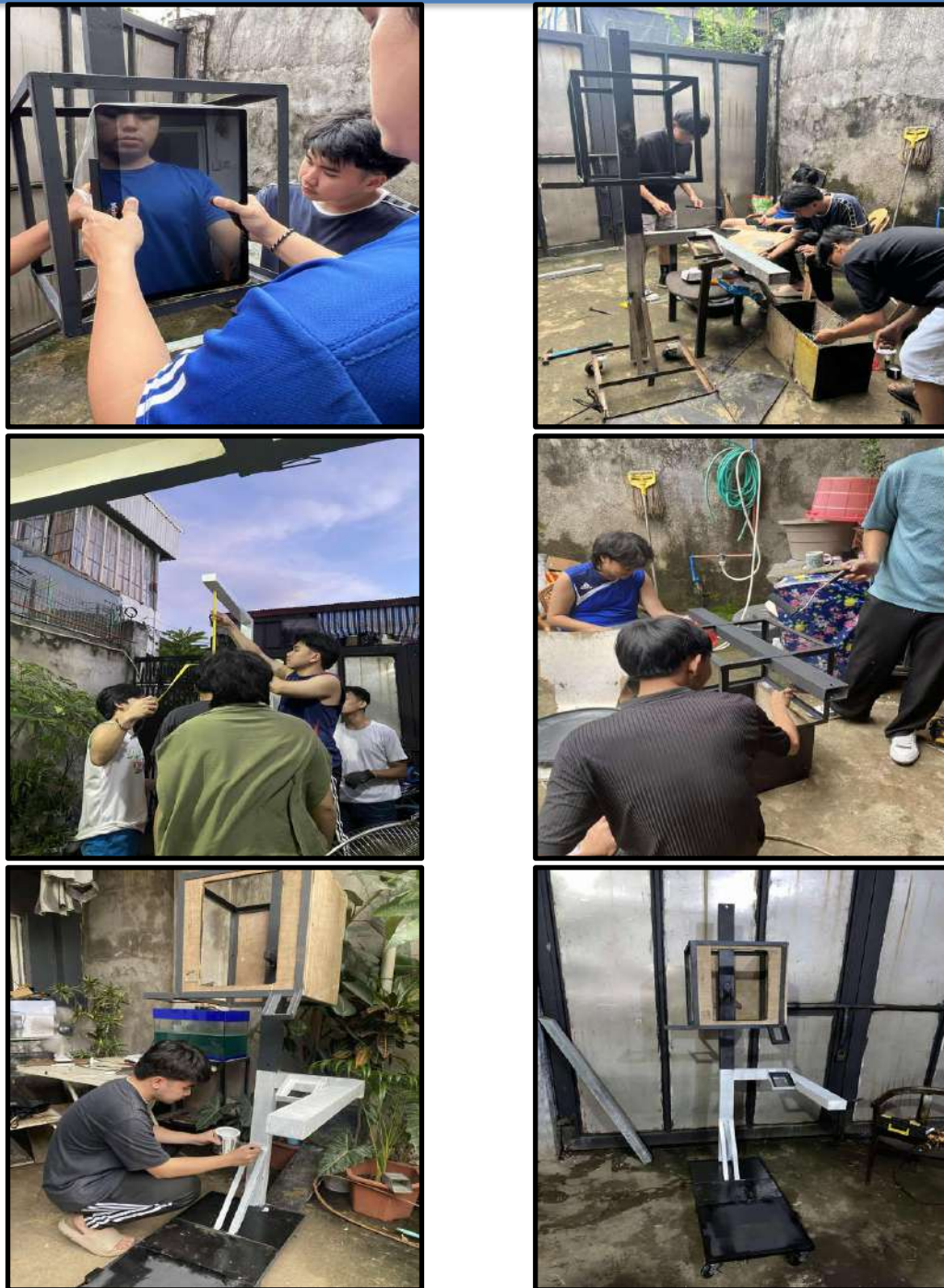
Canvassing and Gathering Materials



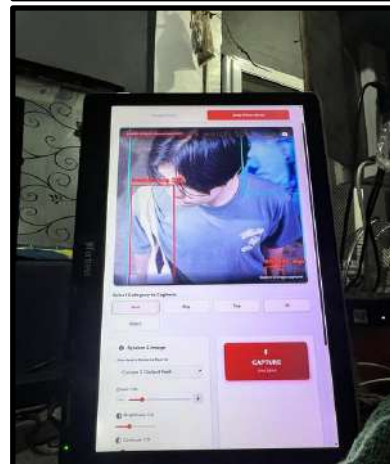
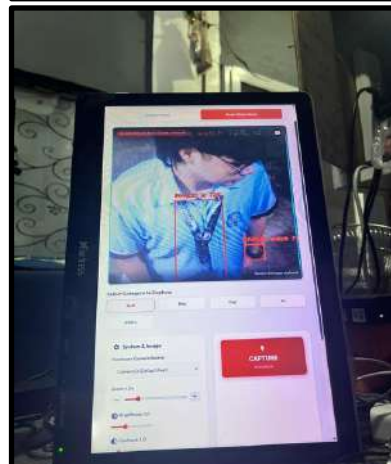
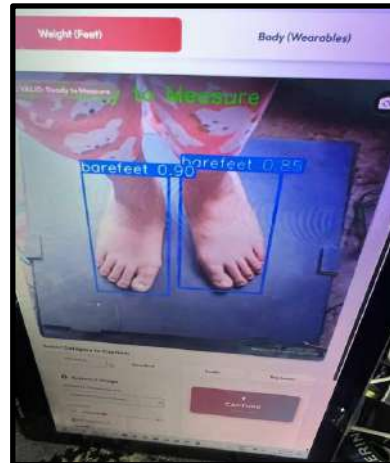
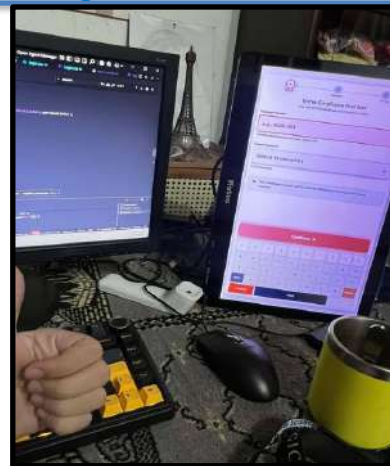
Calibrating Load Cells, TF Luna LiDAR and MLX90614 IR Temp sensor

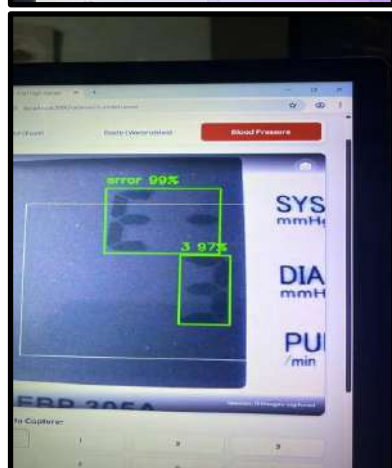
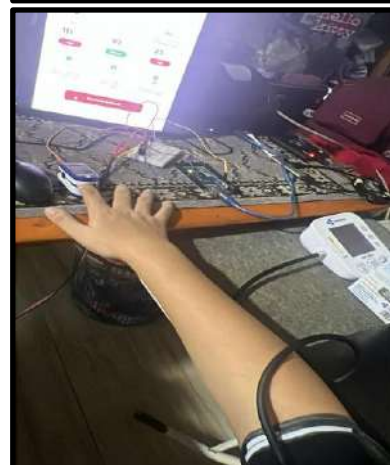
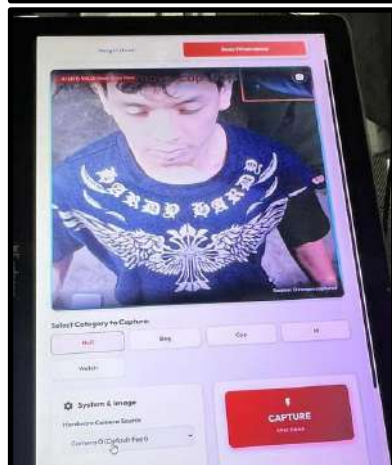
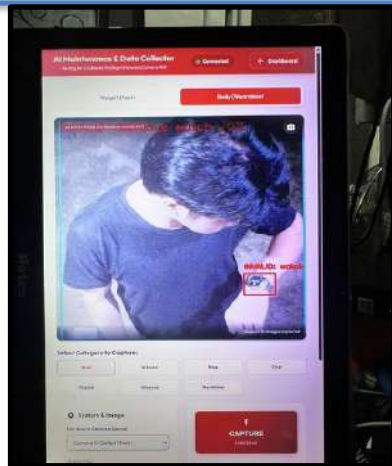


Building Structural Base of the Prototype & Software Display Designing



Polishing & Refining the Prototype





AI Detection Making and Testing



System Testing and Wiring



Deploying and Testing

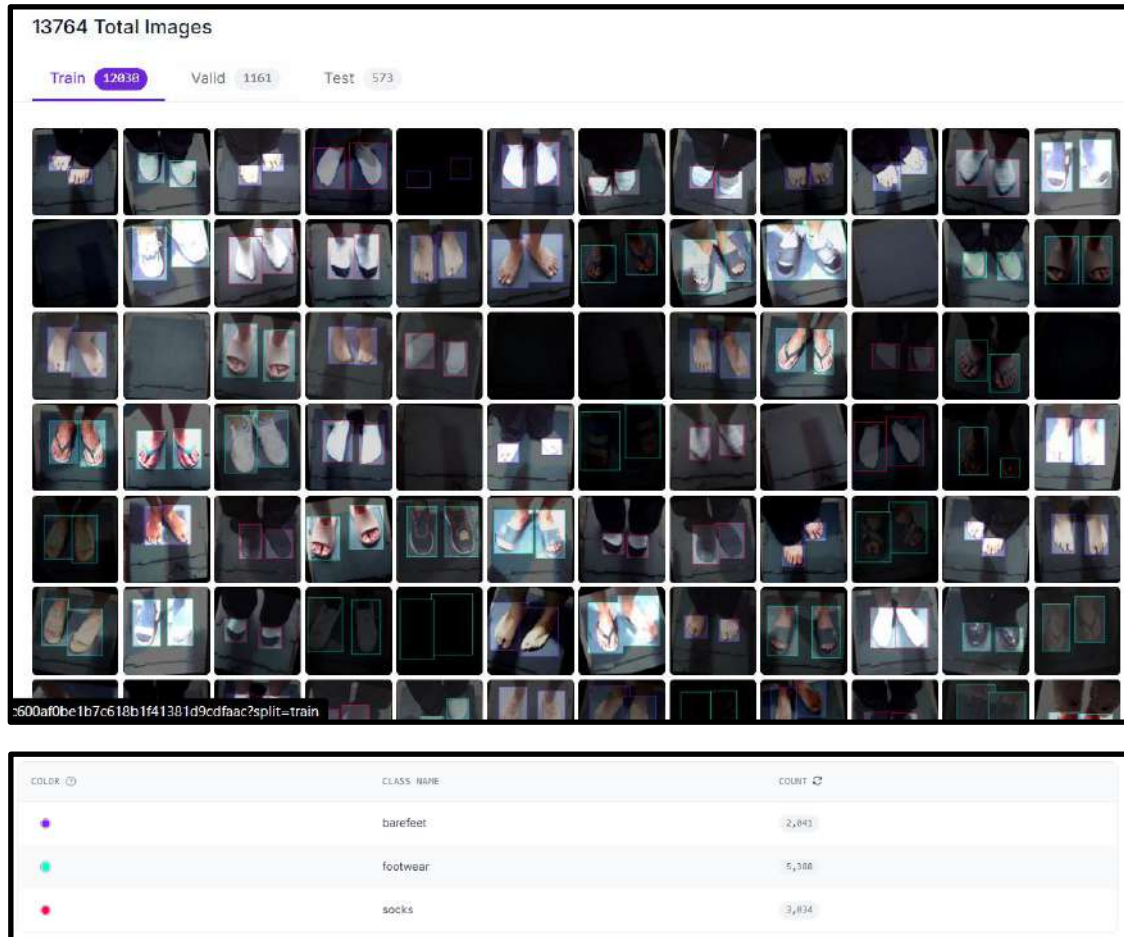


Deployment of Prototype and Data Gathering

APPENDIX F

Testing and Validation

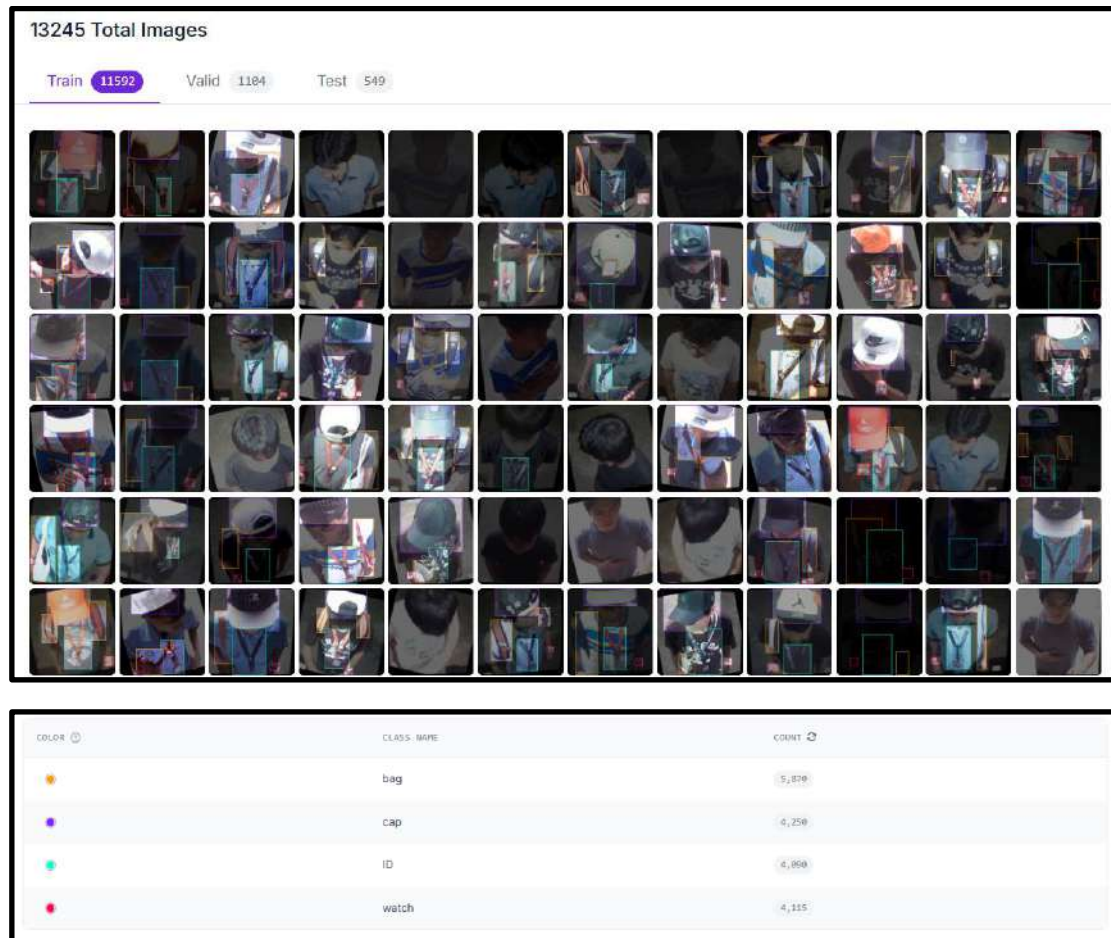
Footwear Detection Using AI Object Detection for Accurate Weight Acquisition



A total of 13,764 images were collected and annotated in Roboflow, categorized into three classes: barefoot, footwear, and socks, to support footwear detection prior to weight measurement. The dataset was divided into training, validation, and testing sets, then exported and trained in Google Colab using an object detection model.

The trained AI model verifies whether the user has removed footwear before body weight acquisition, ensuring accurate measurements by preventing unnecessary weight discrepancies caused by shoes or other foot wearables.

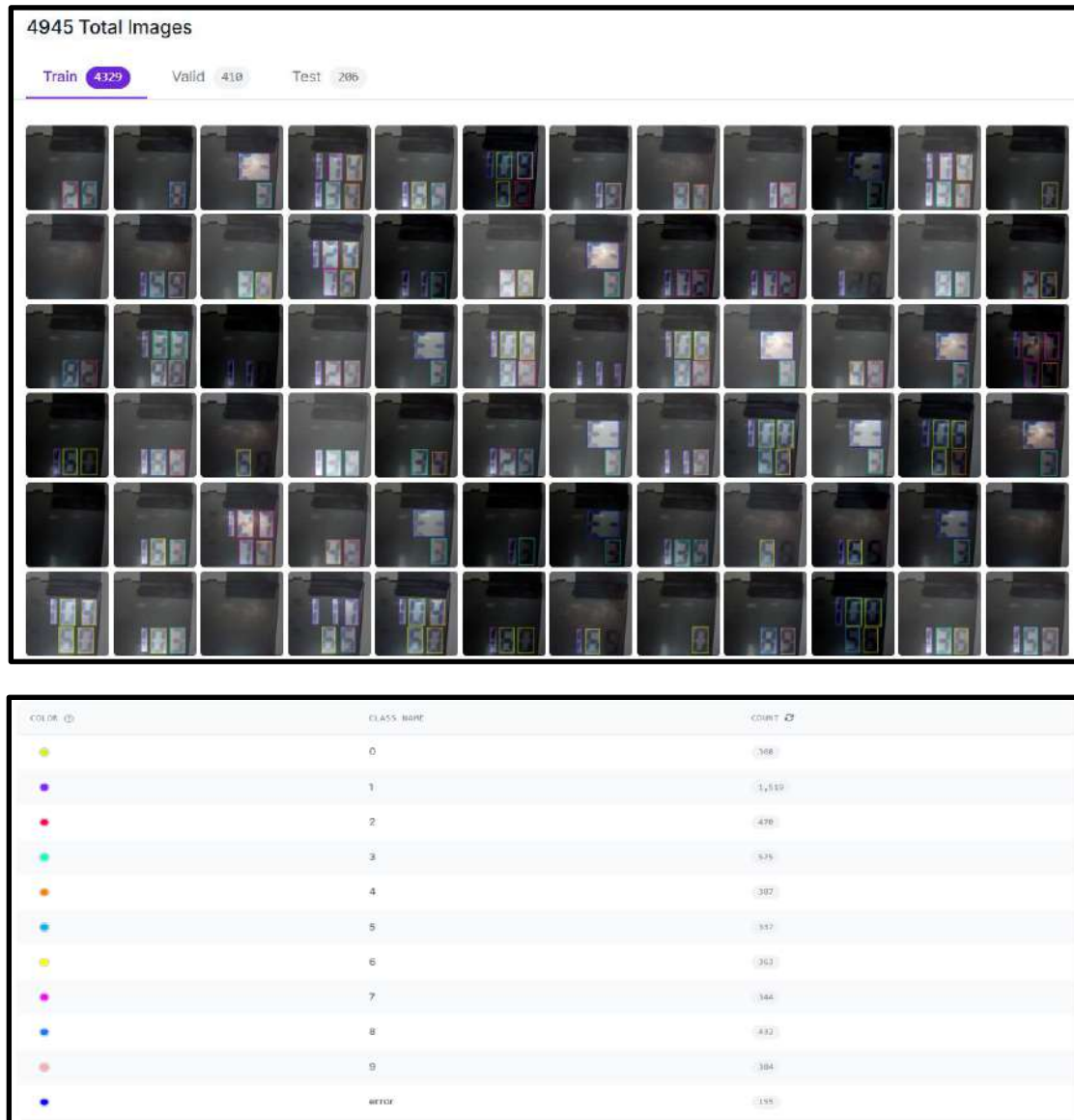
Training an Object Detection Model for Identifying Body Wearables Prior to Measurement



A total of 13,245 images were collected and annotated in Roboflow to develop an object detection model for identifying body wearables such as bags, caps, identification cards (ID), and watches. The dataset was divided into training, validation, and testing sets, and the labeled images were used to train the model in Google Colab.

The trained AI model automatically detects external body wearables prior to measurement to ensure users remove unnecessary items that may affect body weight accuracy and overall system assessment results.

Training an AI-Based Object Detection Model for Integrating Digital Blood Pressure Readings into the System



A total of 4,945 images of the digital blood pressure monitor display were annotated in Roboflow, where bounding boxes were assigned to each digit (0–9), and the dataset was divided into training, validation, and testing sets. The annotated dataset was trained in Google Colab using an object detection model to accurately detect and extract systolic, diastolic, and pulse readings for integration into the system.

Appendix Table 1 System Raw Data (First 10 Clients)

Created at	User Code	Temp (°C)	Sys (mm Hg)	Dia (mm Hg)	HR (bpm)	Sp O ₂ (%)	RR (cpm)	Weight (kg)	Height (cm)	BMI	Risk Level	Risk Category	Email	Print Receipt
2026-01-09 15:19:37	USR-GKM1 - CQBG	35.7	129	56	97	98	24	87.1	170.4	30	52	Moderate Risk	1	0
2026-01-09 15:35:50	USR-B9O9-0AFV	35.3	111	68	102	95	24	54.3	109.4	45.4	100	Critical Risk	1	0
2026-01-09 14:36:29	USR-GKM1 - CQBG	36.3	111	56	82	97	20	61.6	164.4	22.8	4	Low Risk	0	0
2026-01-09 12:57:34	USR-IOJI-ZT2E	36.4	105	59	103	97	24	61.9	164.4	22.9	57	Moderate Risk	0	0
2026-01-09 12:48:09	USR-IOJI-ZT2E	36.3	100	52	100	98	24	61.7	164.4	22.8	8	Low Risk	0	0
2026-01-09 13:11:59	USR-IOJI-ZT2E	35.8	105	59	80	98	20	61.9	164.4	22.9	38	Mild Risk	0	0
2026-01-09 15:12:12	USR-B9O9-0AFV	35.9	110	70	104	97	25	54.3	176.4	17.5	57	Moderate Risk	0	0
2026-01-09 0:20:10	USR-IOJI-ZT2E	35.4	113	59	74	98	18	61.7	165.4	22.6	34	Mild Risk	0	0
2026-01-09 16:36:40	USR-B9O9-0AFV	35.6	109	62	110	97	28	54.6	177.4	17.3	64	Moderate Risk	1	0

QR Code for Whole System Raw Data



APPENDIX G

Certificate of Originality or Anti-Plagiarism Report

APPENDIX H

AI Declaration

APPENDIX I

Certificate of Proofreading



APPENDIX J

Curriculum Vitae

Alcantara, Mar Kevin P.

Address: Block 14 Lot 6 Samasipat St.

Pulong Kendi 2, Sta. Ana, Taguig City

Contact: 09920030148

RTU Email: 2022-200901@rtu.edu.ph



PERSONAL INFORMATION

Age: 21

Gender: Male

Birthday: September 4, 2004

Civil Status: Single

Citizenship: Filipino

EDUCATIONAL BACKGROUND

Primary School :St. Joe Development Academy

Year: 2010-2016

Secondary School :Arellano University- Bonifacio Campus

Year: 2020-2022

Tertiary School :Rizal Technological University

Year: 2022-Present

Berongoy, Bernie C.

Address: Purok 2 Corbada,

La Hacienda, Alicia, Bohol

Contact: 09922714005

RTU Email: 2022-103187@rtu.edu.ph



PERSONAL INFORMATION

Age: 23

Gender: Male

Birthday: November 23, 2002

Civil Status: Single

Citizenship: Filipino

EDUCATIONAL BACKGROUND

Primary School :La Hacienda Elementary School

Year: 2010-2016

Secondary School :Eusebio High School

Year: 2016-2021

Tertiary School :Rizal Technological University

Year: 2022-Present

Gamboa, Khin Andrei R.

Address: Purok 5, Brgy Putlod,

Jaen, Nueva Ecija

Contact: 09924215130

RTU Email: 2022-200950@rtu.edu.ph



PERSONAL INFORMATION

Age: 21

Gender: Male

Birthday: October 21, 2004

Civil Status: Single

Citizenship: Filipino

EDUCATIONAL BACKGROUND

Primary School :Putlod Elementary School

Year: 2010-2016

Secondary School :Rizal High School

Year: 2016-2022

Tertiary School :Rizal Technological University

Year: 2022-Present

Llona, Reymart G.

Address: Purok 6c South

Daanghari Taguig City

Contact: 09304883774

RTU Email: 2022-201011@rtu.edu.ph



PERSONAL INFORMATION

Age: 22

Gender: Male

Birthday: August 29, 2003

Civil Status: Single

Citizenship: Filipino

EDUCATIONAL BACKGROUND

Primary School :Bagong Tanyag Elementary School Annex-A

Year: 2010-2016

Secondary School :Bagumbayan National High School

Year: 2016-2022

Tertiary School :Rizal Technological University

Year: 2022-Present

Relevo, Paul Andrew A.

Address: L.T Extension San

Gabriel Teresa Rizal

Contact: 09916903193

RTU Email: 2022-200980@rtu.edu.ph



PERSONAL INFORMATION

Age: 22

Gender: Male

Birthday: February 27, 2004

Civil Status: Single

Citizenship: Filipino

EDUCATIONAL BACKGROUND

Primary School :Quiterio San Jose Elementary School

Year: 2010-2016

Secondary School :Sumulong College of Arts and Sciences

Year: 2016-2022

Tertiary School :Rizal Technological University

Year: 2022-Present

Oavenada, Erick John A.

Address: Block 1 Cesar Street

Barangay Sta Ana Taytay Rizal

Contact: 09940940053

RTU Email: 2022-103187@rtu.edu.ph



PERSONAL INFORMATION

Age: 24

Gender: Male

Birthday: June 5, 2001

Civil Status: Single

Citizenship: Filipino

EDUCATIONAL BACKGROUND

Primary School :Exodus Elementary School

Year: 2008-2014

Secondary School :Manuel I Santos Memorial National High School

Year:2014-2018

Tertiary School :Rizal Technological University

Year:2022-Present

Sagadraca, Yuri Lorenz

Address: Blk 9 Lot 11 Roseville Subd.,

Sta Ana, Taytay, Rizal

Contact: 09270626367

RTU Email: 2022-200889@rtu.edu.ph



PERSONAL INFORMATION

Age: 21

Gender: Male

Birthday: October 28, 2004

Civil Status: Single

Citizenship: Filipino

EDUCATIONAL BACKGROUND

Primary School :San Miguel Elementary School

Year: 2010-2016

Secondary School :APEC SCHOOLS - C.RAYMUNDO

Year: 2016-2022

Tertiary School :Rizal Technological University

Year: 2022-Present